LibMuse Android 8.0.9

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Chapter 1

Muse SDK Documentation

This library enables communication with and control of Muse headbands from client software.

1.0.1 Quick start

1.0.1.1 Android

Start with com.choosemuse.libmuse.MuseManager and go from there, or use the Classes tab.

1.0.2 Threading

Most LibMuse methods are thread safe, except execute() and runAsynchronously() . Each method describes its threading concerns if there are any.

1.0.3 Exceptions

The LibMuse native library catches and handles every exception that it can reasonably be expected to do so. There are, however, some situations in which native code can still throw an exception. Any API method that can throw an exception documents this behaviour.

Chapter 2

Deprecated List

Member com.choosemuse.libmuse.MuseDataPacket.values ()

Use MuseDataPacket.getAccelerometerValue() , MuseDataPacket.getBatteryValue() , MuseDataPacket.getDrlRefValue() , MuseDataPacket.getEegChannelValue() , MuseDataPacket.getPpgChannelValue() , MuseDataPacket.getGyroValue() instead.

Member com.choosemuse.libmuse.MuseDataPacketType.DROPPED_ACCELEROMETER

This is never emitted in an actual Muse session; instead, NaN-filled packets of the basic type (EEG or ACC↩ ELEROMETER) are emitted to stand in for dropped packets. This can only appear when reading Muse files written with older versions of the library.

Member com.choosemuse.libmuse.MuseDataPacketType.DROPPED_EEG

This is never emitted in an actual Muse session; instead, NaN-filled packets of the basic type (EEG or ACC↩ ELEROMETER) are emitted to stand in for dropped packets. This can only appear when reading Muse files written with older versions of the library.

Member com.choosemuse.libmuse.ReaderMuse.run ()

Use playback() with the playback setting ReaderMusePlaybackSettings.AS_FAST_AS_POSSIBLE_WITH_SAVED_TIMESTAMP to replicate this behaviour.

Member com.choosemuse.libmuse.ReaderMuse.runInRealTimespan ()

Use playback()with the playback setting ReaderMusePlaybackSettings.SIMULATED_WITH_SAVED_TIMESTAMP to replicate this behaviour.

Member com.choosemuse.libmuse.ReaderMuseBuilder.buildWithAsync (MuseFileReader reader, EventLoop asyncLoop)

Set the EventLoop with ReaderMuseBuilder.withEventLoop() and then call ReaderMuseBuilder.build() instead.

Chapter 3

Namespace Index

3.1 Namespace List

Here is a list of all namespaces with brief descriptions:

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Chapter 5

Class Index

5.1 Class List

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Chapter 7

Namespace Documentation

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Packages

- package choosemuse

7.2 Package com.choosemuse

Packages

- package libmuse

7.3 Package com.choosemuse.libmuse

Classes

- enum Accelerometer
- class Action
- class AdvertisingStats
- class AndroidLogListener
- class AnnotationData
- enum AnnotationFormat
- class ApiVersion
- enum Battery
- class ComputingDeviceConfiguration
- class ComputingDeviceConfigurationFactory
- enum ConnectionState
- enum DrlRef
- class DspData
- enum Eeg
- class Error
- enum ErrorType

- class EventLoop
- enum Gyro
- class HandlerEventLoop
- class LibmuseVersion
- class LogListener
- class LogManager
- class LogPacket
- enum Magnetometer
- enum MessageType
- class Muse
- class MuseArtifactPacket
- class MuseConfiguration
- class MuseConnectionListener
- class MuseConnectionPacket
- class MuseDataListener
- class MuseDataPacket
- enum MuseDataPacketType
- class MuseErrorListener
- class MuseFile
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- class MuseFileReader
- class MuseFileWriter
- class MuseListener
- class MuseLog
- class MuseManager
- class MuseManagerAndroid
- enum MuseModel
- enum MusePreset
- class MuseVersion
- enum NotchFrequency
- enum Optics
- enum Ppg
- enum Pressure
- class ReaderListener
- class ReaderMuse
- class ReaderMuseBuilder
- enum ReaderMusePlaybackSettings
- class ReaderPlaybackListener
- class Result
- enum ResultLevel
- enum Severity
- class Stringify
- enum TimestampMode
- enum UltraViolet

7.3.1 Detailed Description

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Chapter 8

Class Documentation

8.1 com.choosemuse.libmuse.Accelerometer Enum Reference

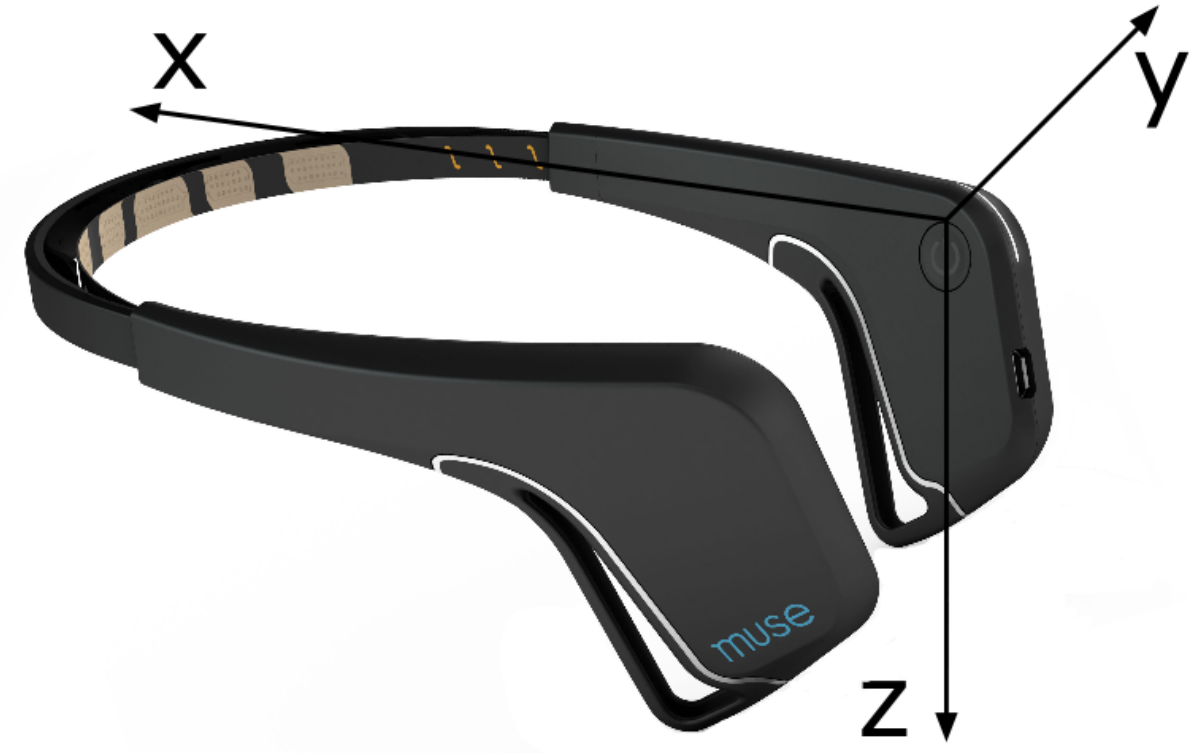
Public Attributes

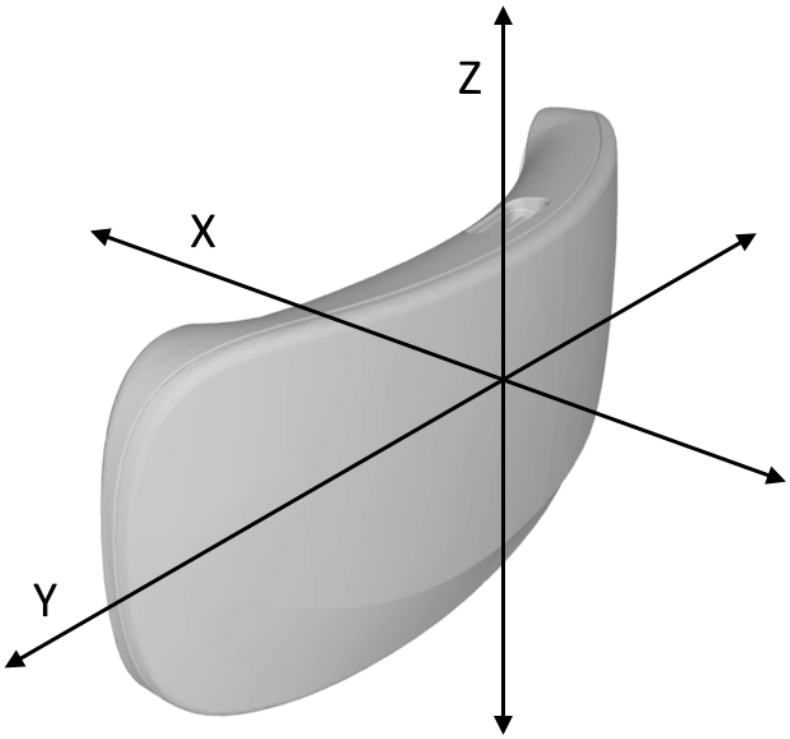
- X
- Y
- Z

8.1.1 Detailed Description

Represents the data mapping in an Accelerometer data packet.

The accelerometer data is measured on 3 axes as shown in the pictures below:





The axes are oriented to a Right Hand Coordinate System along the headband axes. Values are given in g (9.81 m/s^2) and are negated to align with the orientation of the headband in Earth's gravitational field. This convention is described in more detail in the following application note.

https://www.nxp.com/docs/en/application-note/AN3461.pdf

Values along the X axis increase as the head tilts down aligning the X axis with the downward force of gravity. Negative values indicate a tilt upwards.

Values along the Y axis increase as the head tilts to the right. Negative values indicate a tilt to the left.

When worn on a level head, or held in the level position shown in the figure above, the net acceleration of the device will only be caused from gravity. It will be in the direction of the ground aligned with the Z axis. This will give a_x =0, a_y = 0 and a_z = 1 g. As the headband tilts out of this orientation, the value of Z will decrease. -1 in Z represents a headband that is upside down.

See also

MuseDataPacketType.ACCELEROMETER

MuseDataPacket.getAccelerometerValue()

8.1.2 Member Data Documentation

8.1.2.1 X

`com.choosemuse.libmuse.Accelerometer.X`

Orientation of the X axis relative to gravity in g. Values along the X axis increase as the head tilts down. Negative values indicate a tilt up.

8.1.2.2 Y

`com.choosemuse.libmuse.Accelerometer.Y`

Orientation of the Y axis relative to gravity in g. Values along the Y axis increase as the head tilts to the right. Negative values indicate a tilt to the left.

8.1.2.3 Z

`com.choosemuse.libmuse.Accelerometer.Z`

Orientation of the Z axis relative to gravity in g.

The documentation for this enum was generated from the following file:

- Accelerometer.java

8.2 com.choosemuse.libmuse.Action Class Reference

Public Member Functions

- abstract void run ()

8.2.1 Detailed Description

A wrapper that represents a block or runnable action.

See also

EventLoop

8.2.2 Member Function Documentation

8.2.2.1 run()

```
abstract void com.choosemuse.libmuse.Action.run ( )  [abstract]
```

Performs the action.

The documentation for this class was generated from the following file:

- Action.java

8.3 com.choosemuse.libmuse.AdvertisingStats Class Reference

Public Member Functions

- AdvertisingStats (int numAdvertisingPackets, double avgAdvertisingInterval, double sigmaAdvertising↩ Interval, double maxAdvertisingInterval, boolean isLost, boolean hasBadMac)
- int getNumAdvertisingPackets ()
- double getAvgAdvertisingInterval ()
- double getSigmaAdvertisingInterval ()
- double getMaxAdvertisingInterval ()
- boolean getIsLost ()
- boolean getHasBadMac ()
- String toString ()

8.3.1 Detailed Description

Statistics about the advertising packets that LibMuse sees from Muse 2016 (MU_02) or later headbands.

8.3.2 Constructor & Destructor Documentation

8.3.2.1 AdvertisingStats()

```
com.choosemuse.libmuse.AdvertisingStats.AdvertisingStats (
            int numAdvertisingPackets,
            double avgAdvertisingInterval,
            double sigmaAdvertisingInterval,
            double maxAdvertisingInterval,
            boolean isLost,
            boolean hasBadMac )  [inline]
```

8.3.3 Member Function Documentation

8.3.3.1 getAvgAdvertisingInterval()

```
double com.choosemuse.libmuse.AdvertisingStats.getAvgAdvertisingInterval ( )  [inline]
```

Returns the average interval in seconds between advertising packets. This is a running average since the time the statistics were created or reset.

Returns

the average interval between advertising packets.

8.3.3.2 getHasBadMac()

```
boolean com.choosemuse.libmuse.AdvertisingStats.getHasBadMac ( )  [inline]
```

Indicates an issue with the MAC address of the headband.

Returns

`true` if for some reason the MAC address of a headband has changed since initial detection. `false` under normal circumstances.

8.3.3.3 getIsLost()

```
boolean com.choosemuse.libmuse.AdvertisingStats.getIsLost ( )  [inline]
```

Indicates if the phone has lost contact with the Muse headband. This is a subjective measurement. Even if this is true, the phone may still be able to connect with the headband if a connection is requested.

Returns

`true` if the phone has lost contact with the headband. `false` otherwise.

8.3.3.4 getMaxAdvertisingInterval()

```
double com.choosemuse.libmuse.AdvertisingStats.getMaxAdvertisingInterval ( )  [inline]
```

Returns the maximum interval in seconds between receiving 2 advertising packets.

Returns

the maximum interval between advertising packets.

8.3.3.5 getNumAdvertisingPackets()

```
int com.choosemuse.libmuse.AdvertisingStats.getNumAdvertisingPackets ( )  [inline]
```

Returns the number of advertising packets seen.

Returns

the number of advertising packets seen.

8.3.3.6 getSigmaAdvertisingInterval()

```
double com.choosemuse.libmuse.AdvertisingStats.getSigmaAdvertisingInterval ( )  [inline]
```

Returns the standard deviation of the average advertising interval.

Returns

the standard deviation of the average advertising interval.

8.3.3.7 toString()

```
String com.choosemuse.libmuse.AdvertisingStats.toString ( )  [inline]
```

Returns a string representation of this object.

Returns

a string representation of this object.

The documentation for this class was generated from the following file:

- AdvertisingStats.java

8.4 com.choosemuse.libmuse.AndroidLogListener Class Reference

Inheritance diagram for com.choosemuse.libmuse.AndroidLogListener:

8.5 com.choosemuse.libmuse.AnnotationData Class Reference

Collaboration diagram for com.choosemuse.libmuse.AnnotationData:

Public Member Functions

- AnnotationData (String data, AnnotationFormat format, String eventType, String eventId, String parentId)
- String getData ()
- AnnotationFormat getFormat ()
- String getEventType ()
- String getEventId ()
- String getParentId ()
- String toString ()

8.5.1 Detailed Description

A data structure that corresponds to an Annotation message in the .muse file format.

8.5.2 Constructor & Destructor Documentation

8.5.2.1 AnnotationData()

```
com.choosemuse.libmuse.AnnotationData.AnnotationData (
            String data,
            AnnotationFormat format,
            String eventType,
            String eventId,
            String parentId )  [inline]
```

8.5.3 Member Function Documentation

8.5.3.1 getData()

```
String com.choosemuse.libmuse.AnnotationData.getData ( )  [inline]
```

This is additional data for the event, can be in any format specified by AnnotationFormat

8.5.3.2 getEventId()

```
String com.choosemuse.libmuse.AnnotationData.getEventId ( )  [inline]
```

This can be used to pair events together or give them unique IDs. If you do not need this value, set it to the empty string "".

8.5.3.3 getEventType()

```
String com.choosemuse.libmuse.AnnotationData.getEventType ( )  [inline]
```

This can be used to classify events, for example "blink" or "error". If you do not need this value, set it to the empty string "".

8.5.3.4 getFormat()

```
AnnotationFormat com.choosemuse.libmuse.AnnotationData.getFormat ( )  [inline]
```

This specifies the string format of the event data.

8.5.3.5 getParentId()

```
String com.choosemuse.libmuse.AnnotationData.getParentId ( )  [inline]
```

This can be used to pair events together or give them unique IDs. If you do not need this value, set it to the empty string "".

8.5.3.6 toString()

```
String com.choosemuse.libmuse.AnnotationData.toString ( )  [inline]
```

Returns a string representation of this object.

Returns

a string representation of this object.

The documentation for this class was generated from the following file:

- AnnotationData.java

8.6 com.choosemuse.libmuse.AnnotationFormat Enum Reference

Public Attributes

- PLAIN_STRING
- JSON
- OSC

8.6.1 Detailed Description

Represents all possible data types in the annotation data. These fields correspond to our Muse protobuf schema.

8.6.2 Member Data Documentation

8.6.2.1 JSON

`com.choosemuse.libmuse.AnnotationFormat.JSON`

The data is a string of JSON name, value pairs.

8.6.2.2 OSC

`com.choosemuse.libmuse.AnnotationFormat.OSC`

The data is formatted as OSC (open sound control) data.

8.6.2.3 PLAIN_STRING

`com.choosemuse.libmuse.AnnotationFormat.PLAIN_STRING`

The data is a string with no inherrent format.

The documentation for this enum was generated from the following file:

- AnnotationFormat.java

8.7 com.choosemuse.libmuse.ApiVersion Class Reference

Public Member Functions

- abstract long getMonotonic ()
- abstract long getMajor ()
- abstract long getMinor ()
- abstract long getPatch ()
- abstract long getApi ()
- abstract String getString ()

8.7.1 Detailed Description

Library version interface.

Describes the version of LibMuse.

8.7.2 Member Function Documentation

8.7.2.1 getApi()

```
abstract long com.choosemuse.libmuse.ApiVersion.getApi ( )  [abstract]
```

API

This number is incremented when changes are introduced to the API that are not backwards compatible.

Returns

The API version.

8.7.2.2 getMajor()

```
abstract long com.choosemuse.libmuse.ApiVersion.getMajor ( )  [abstract]
```

Major release.

This number is incremented on a major release of new functionality.

Returns

The major version.

8.7.2.3 getMinor()

```
abstract long com.choosemuse.libmuse.ApiVersion.getMinor ( )  [abstract]
```

Minor release.

This number is incremented when minor changes are made that are backwards compatible.

Returns

The minor version.

8.7.2.4 getMonotonic()

```
abstract long com.choosemuse.libmuse.ApiVersion.getMonotonic ( )  [abstract]
```

Returns a monotonically increasing version number.

This number is guaranteed to increase from release to release. Currently it is equivalent to:

```
get_major() * 1000000 + get_minor() * 10000 + get_patch() * 100 + get_api()
```

So for major = 5, minor = 1, patch = 3, api = 2, it would be 5010302.

8.7.2.5 getPatch()

```
abstract long com.choosemuse.libmuse.ApiVersion.getPatch ( )  [abstract]
```

Patch release.

This number is incremented when backwards-compatible bugfixes are made.

Returns

The patch version.

8.7.2.6 getString()

```
abstract String com.choosemuse.libmuse.ApiVersion.getString ( )  [abstract]
```

String representation of the version.

This is a string like "x.y.z API Version j" where x is get_major(), y is get_minor(), z is get_patch() and j is get_api(). The format may change from release to release. In particular, don't make assumptions about it or try to parse it. Use the other API calls on this interface to get the numerical versions.

Returns

The version string.

The documentation for this class was generated from the following file:

- ApiVersion.java

8.8 com.choosemuse.libmuse.Battery Enum Reference

Public Attributes

- CHARGE_PERCENTAGE_REMAINING
- MILLIVOLTS
- TEMPERATURE_CELSIUS
- AVERAGE_CURRENT
- TIME_TO_EMPTY
- TIME_TO_FULL
- BATTERY_CAPACITY
- REMAINING_CAPACITY
- BATTERY_AGE
- TOTAL_CYCLES

8.8.1 Detailed Description

Represents the data mapping in a Battery packet.

See also

MuseDataPacketType.BATTERY

MuseDataPacket.getBatteryValue()

8.8.2 Member Data Documentation

8.8.2.1 AVERAGE_CURRENT

`com.choosemuse.libmuse.Battery.AVERAGE_CURRENT`

Average current in microamps.

8.8.2.2 BATTERY_AGE

`com.choosemuse.libmuse.Battery.BATTERY_AGE`

Percentage battery age calculated from capacity.

8.8.2.3 BATTERY_CAPACITY

`com.choosemuse.libmuse.Battery.BATTERY_CAPACITY`

Full capacity of battery in mAh.

8.8.2.4 CHARGE_PERCENTAGE_REMAINING

`com.choosemuse.libmuse.Battery.CHARGE_PERCENTAGE_REMAINING`

Charge percentage remaining of battery.

8.8.2.5 MILLIVOLTS

`com.choosemuse.libmuse.Battery.MILLIVOLTS`

Millivolts of battery from the view of the fuel gauge.

8.8.2.6 REMAINING_CAPACITY

`com.choosemuse.libmuse.Battery.REMAINING_CAPACITY`

Remaining capacity of battery in mAh.

8.8.2.7 TEMPERATURE_CELSIUS

`com.choosemuse.libmuse.Battery.TEMPERATURE_CELSIUS`

Temperature in degrees Celsius.

8.8.2.8 TIME_TO_EMPTY

`com.choosemuse.libmuse.Battery.TIME_TO_EMPTY`

Remaining seconds until charge depleted (while discharging).

8.8.2.9 TIME_TO_FULL

`com.choosemuse.libmuse.Battery.TIME_TO_FULL`

Remaining seconds until fully charged (while charging).

8.8.2.10 TOTAL_CYCLES

`com.choosemuse.libmuse.Battery.TOTAL_CYCLES`

Total number of charge/discharge cycles.

The documentation for this enum was generated from the following file:

- Battery.java

8.9 com.choosemuse.libmuse.ComputingDeviceConfiguration Class Reference

Public Member Functions

- ComputingDeviceConfiguration (String osType, String osVersion, String hardwareModelName, String hardwareModelId, String processorName, String processorSpeed, int numberOfProcessors, String memorySize, String bluetoothVersion, String timeZone, int timeZoneOffsetSeconds)
- String getOsType ()
- String getOsVersion ()
- String getHardwareModelName ()
- String getHardwareModelId ()
- String getProcessorName ()
- String getProcessorSpeed ()
- int getNumberOfProcessors ()
- String getMemorySize ()
- String getBluetoothVersion ()
- String getTimeZone ()
- int getTimeZoneOffsetSeconds ()
- String toString ()

8.9.1 Detailed Description

This data structure contains fields, which describe the running machine. To prevent confusion, field names should correspond to names in the protobuf schema of the .muse file format. If the computing device configuration cannot be obtained for any field, they will be empty strings.

8.9.2 Constructor & Destructor Documentation

8.9.2.1 ComputingDeviceConfiguration()

```
com.choosemuse.libmuse.ComputingDeviceConfiguration.ComputingDeviceConfiguration (
            String osType,
            String osVersion,
            String hardwareModelName,
            String hardwareModelId,
            String processorName,
            String processorSpeed,
            int numberOfProcessors,
            String memorySize,
            String bluetoothVersion,
            String timeZone,
            int timeZoneOffsetSeconds )  [inline]
```

8.9.3 Member Function Documentation

8.9.3.1 getBluetoothVersion()

```
String com.choosemuse.libmuse.ComputingDeviceConfiguration.getBluetoothVersion ( )  [inline]
```

Bluetooth version

8.9.3.2 getHardwareModelId()

```
String com.choosemuse.libmuse.ComputingDeviceConfiguration.getHardwareModelId ( )  [inline]
```

Hardware model id. For ex.: "SM-N900W8", "MacBookPro-101"

8.9.3.3 getHardwareModelName()

```
String com.choosemuse.libmuse.ComputingDeviceConfiguration.getHardwareModelName ( )  [inline]
```

Hardware model. For ex.: "Samsung Galaxy Note 3", "Macbook Pro"

8.9.3.4 getMemorySize()

```
String com.choosemuse.libmuse.ComputingDeviceConfiguration.getMemorySize ( )  [inline]
```

Memory size. For ex.: "500MB", "2000MB"

8.9.3.5 getNumberOfProcessors()

```
int com.choosemuse.libmuse.ComputingDeviceConfiguration.getNumberOfProcessors ( )  [inline]
```

Number of cores

8.9.3.6 getOsType()

```
String com.choosemuse.libmuse.ComputingDeviceConfiguration.getOsType ( )  [inline]
```

Operation system type. For ex.: "Android", "iOS", "Windows"

8.9.3.7 getOsVersion()

```
String com.choosemuse.libmuse.ComputingDeviceConfiguration.getOsVersion ( )  [inline]
```

Operation system version. For ex.: "10.1", "4.4.2"

8.9.3.8 getProcessorName()

```
String com.choosemuse.libmuse.ComputingDeviceConfiguration.getProcessorName ( )  [inline]
```

Processor name. For ex.: "Intel", "ARM"

8.9.3.9 getProcessorSpeed()

```
String com.choosemuse.libmuse.ComputingDeviceConfiguration.getProcessorSpeed ( )  [inline]
```

Processor frequency in Hz

8.9.3.10 getTimeZone()

```
String com.choosemuse.libmuse.ComputingDeviceConfiguration.getTimeZone ( )  [inline]
```

time zone indicator

8.9.3.11 getTimeZoneOffsetSeconds()

```
int com.choosemuse.libmuse.ComputingDeviceConfiguration.getTimeZoneOffsetSeconds ( )  [inline]
```

time zone offset in seconds

8.9.3.12 toString()

```
String com.choosemuse.libmuse.ComputingDeviceConfiguration.toString ( )  [inline]
```

Returns a string representation of this object.

Returns

a string representation of this object.

The documentation for this class was generated from the following file:

- ComputingDeviceConfiguration.java

8.10 com.choosemuse.libmuse.ComputingDeviceConfigurationFactory Class Reference

Public Member Functions

- ComputingDeviceConfiguration getComputingDeviceConfiguration ()

Static Public Member Functions

- static ComputingDeviceConfigurationFactory getInstance ()

8.10.1 Detailed Description

Provides access to the computingDeviceConfiguration object containing information about the computing device.

8.10.2 Member Function Documentation

8.10.2.1 getComputingDeviceConfiguration()

```
ComputingDeviceConfiguration com.choosemuse.libmuse.ComputingDeviceConfigurationFactory.get↩
ComputingDeviceConfiguration ( )  [inline]
```

Retrieves the appropriate ComputingDeviceConfiguration structure for the current computing device.

Returns

A ComputingDeviceConfiguration structure for the current computing device.

8.10.2.2 getInstance()

```
static ComputingDeviceConfigurationFactory com.choosemuse.libmuse.ComputingDeviceConfiguration↩
Factory.getInstance ( )  [inline], [static]
```

Static constructor for the singleton object.

Returns

An instance of the ComputingDeviceConfigurationFactory object.

The documentation for this class was generated from the following file:

- ComputingDeviceConfigurationFactory.java

8.11 com.choosemuse.libmuse.ConnectionState Enum Reference

Public Attributes

- UNKNOWN
- CONNECTED
- CONNECTING
- DISCONNECTED
- NEEDS_UPDATE
- NEEDS_LICENSE

8.11.1 Detailed Description

Lists all possible connection states

8.11.2 Member Data Documentation

8.11.2.1 CONNECTED

```
com.choosemuse.libmuse.ConnectionState.CONNECTED
```

This state is set if the connection was correctly established.

8.11.2.2 CONNECTING

```
com.choosemuse.libmuse.ConnectionState.CONNECTING
```

This state is set while trying to establish a connection.

8.11.2.3 DISCONNECTED

`com.choosemuse.libmuse.ConnectionState.DISCONNECTED`

This state is set in case of an unsuccessful connection operation or after execution of disconnect method.

8.11.2.4 NEEDS_LICENSE

`com.choosemuse.libmuse.ConnectionState.NEEDS_LICENSE`

This state is set when the connection succeeded but the headband's license is invalid.

8.11.2.5 NEEDS_UPDATE

`com.choosemuse.libmuse.ConnectionState.NEEDS_UPDATE`

This state is set when the connection succeeded but the headband's firmware is out of date – if this occurs, you should instruct your users to use the Muse app to upgrade their firmware.

8.11.2.6 UNKNOWN

`com.choosemuse.libmuse.ConnectionState.UNKNOWN`

Initial state

The documentation for this enum was generated from the following file:

- ConnectionState.java

8.12 com.choosemuse.libmuse.DrlRef Enum Reference

Public Attributes

- DRL
- REF

8.12.1 Detailed Description

Represents the data mapping in a DRL_REF packet.

See also

MuseDataPacketType.DRL_REF

MuseDataPacket.getDrlRefValue()

8.12.2 Member Data Documentation

8.12.2.1 DRL

`com.choosemuse.libmuse.DrlRef.DRL`

DRL sensor

8.12.2.2 REF

`com.choosemuse.libmuse.DrlRef.REF`

REF sensor

The documentation for this enum was generated from the following file:

- DrlRef.java

8.13 com.choosemuse.libmuse.DspData Class Reference

Collaboration diagram for com.choosemuse.libmuse.DspData:

Public Member Functions

- DspData (String type, ArrayList< Double > floatArray, ArrayList< Long > intArray, String version)
- String getType ()
- ArrayList< Double > getFloatArray ()
- ArrayList< Long > getIntArray ()
- String getVersion ()
- String toString ()

8.13.1 Detailed Description

Data struct corresponds to a DSP message in the .muse file format. This message is very generic and allows you to save custom messages.

8.13.2 Constructor & Destructor Documentation

8.13.2.1 DspData()

```
com.choosemuse.libmuse.DspData.DspData (
            String type,
            ArrayList< Double > floatArray,
            ArrayList< Long > intArray,
            String version )  [inline]
```

8.13.3 Member Function Documentation

8.13.3.1 getFloatArray()

```
ArrayList<Double> com.choosemuse.libmuse.DspData.getFloatArray ( )  [inline]
```

array of floating point data

8.13.3.2 getIntArray()

```
ArrayList<Long> com.choosemuse.libmuse.DspData.getIntArray ( )  [inline]
```

array of integer data

8.13.3.3 getType()

```
String com.choosemuse.libmuse.DspData.getType ( )  [inline]
```

Type of your message

8.13.3.4 getVersion()

```
String com.choosemuse.libmuse.DspData.getVersion ( )  [inline]
```

version of your message format

8.13.3.5 toString()

```
String com.choosemuse.libmuse.DspData.toString ( )  [inline]
```

Returns a string representation of this object.

Returns

a string representation of this object.

The documentation for this class was generated from the following file:

- DspData.java

8.14 com.choosemuse.libmuse.Eeg Enum Reference

Public Attributes

- EEG1
- EEG2
- EEG3
- EEG4
- AUX_LEFT
- AUX_RIGHT
- AUX1
- AUX2
- AUX3
- AUX4

8.14.1 Detailed Description

Represents the data mapping in an EEG packet. Enum values correspond to head locations.

Raw EEG values are given in microvolts. EEG derived values may have different units as defined in MuseDataPacketType .

See also

MuseDataPacketType.EEG

MuseDataPacket.getEegChannelValue()

8.14.2 Member Data Documentation

8.14.2.1 AUX1

`com.choosemuse.libmuse.Eeg.AUX1`

Auxiliary input 1 on MuseS 2025 (MS_03).

8.14.2.2 AUX2

`com.choosemuse.libmuse.Eeg.AUX2`

Auxiliary input 2 on MuseS 2025 (MS_03).

8.14.2.3 AUX3

`com.choosemuse.libmuse.Eeg.AUX3`

Auxiliary input 3 on MuseS 2025 (MS_03)

8.14.2.4 AUX4

`com.choosemuse.libmuse.Eeg.AUX4`

Auxiliary input 4 on MuseS 2025 (MS_03)

8.14.2.5 AUX_LEFT

`com.choosemuse.libmuse.Eeg.AUX_LEFT`

Left auxiliary (maps to AUX1 on MuseS 2025 (MS_03)).

8.14.2.6 AUX_RIGHT

`com.choosemuse.libmuse.Eeg.AUX_RIGHT`

Right auxiliary (maps to AUX2 on MuseS 2025 (MS_03)).

8.14.2.7 EEG1

`com.choosemuse.libmuse.Eeg.EEG1`

Left ear

8.14.2.8 EEG2

`com.choosemuse.libmuse.Eeg.EEG2`

Left forehead

8.14.2.9 EEG3

`com.choosemuse.libmuse.Eeg.EEG3`

Right forehead

8.14.2.10 EEG4

`com.choosemuse.libmuse.Eeg.EEG4`

Right ear

The documentation for this enum was generated from the following file:

- Eeg.java

8.15 com.choosemuse.libmuse.Error Class Reference

Collaboration diagram for com.choosemuse.libmuse.Error:

Public Member Functions

- Error (ErrorType type, int code, String info)
- ErrorType getType ()
- int getCode ()
- String getInfo ()
- String toString ()

8.15.1 Detailed Description

Represents an error from an operation (synchronous or asynchronous). These will be forwarded to any Error↩ Listeners that are registered.

See also

MuseErrorListener

Muse.registerErrorListener()

8.15.2 Constructor & Destructor Documentation

8.15.2.1 Error()

```
com.choosemuse.libmuse.Error.Error (
            ErrorType type,
            int code,
            String info )  [inline]
```

8.15.3 Member Function Documentation

8.15.3.1 getCode()

```
int com.choosemuse.libmuse.Error.getCode ( )  [inline]
```

Machine-parseable error code

Returns

a machine parseable error code.

8.15.3.2 getInfo()

```
String com.choosemuse.libmuse.Error.getInfo ( )  [inline]
```

Human-readable description.

Returns

the english error message.

8.15.3.3 getType()

```
ErrorType com.choosemuse.libmuse.Error.getType ( )  [inline]
```

Type of error

Returns

the type of error that has occurred.

8.15.3.4 toString()

```
String com.choosemuse.libmuse.Error.toString ( )  [inline]
```

Returns a string representation of this object.

Returns

a string representation of this object.

The documentation for this class was generated from the following file:

- Error.java

8.16 com.choosemuse.libmuse.ErrorType Enum Reference

Public Attributes

- FAILURE
- TIMEOUT
- OVERLOADED
- UNIMPLEMENTED
- HARDWARE_EVENT

8.16.1 Detailed Description

Error classifications.

These are designed to provide users with courses of action based on the code that's received: e.g. retry on timeout, try a different way of doing the same thing on unimplemented, show the user an error message on failure.

8.16.2 Member Data Documentation

8.16.2.1 FAILURE

`com.choosemuse.libmuse.ErrorType.FAILURE`

A generic failure occurred without any further information. Retrying the operation is unlikely to result in success.

8.16.2.2 HARDWARE_EVENT

`com.choosemuse.libmuse.ErrorType.HARDWARE_EVENT`

A hardware event occurred.

8.16.2.3 OVERLOADED

`com.choosemuse.libmuse.ErrorType.OVERLOADED`

Some resource (queue space, memory, bandwidth) was exhausted. Retry with backoff.

8.16.2.4 TIMEOUT

`com.choosemuse.libmuse.ErrorType.TIMEOUT`

Some timeout was exceeded; the operation might succeed if retried.

8.16.2.5 UNIMPLEMENTED

`com.choosemuse.libmuse.ErrorType.UNIMPLEMENTED`

Something was tried that isn't implemented.

The documentation for this enum was generated from the following file:

- ErrorType.java

8.17 com.choosemuse.libmuse.EventLoop Class Reference

Inheritance diagram for com.choosemuse.libmuse.EventLoop:

Public Member Functions

- abstract void post (Action action)
- abstract void postDelayed (Action action, long delayMilliseconds)
- abstract void cancel ()

8.17.1 Detailed Description

Wrapper around a platform-specific event loop.

8.17.2 Member Function Documentation

8.17.2.1 cancel()

```
abstract void com.choosemuse.libmuse.EventLoop.cancel ( )  [abstract]
```

Cancel all pending actions.

Reimplemented in com.choosemuse.libmuse.HandlerEventLoop.

8.17.2.2 post()

```
abstract void com.choosemuse.libmuse.EventLoop.post (
            Action action )  [abstract]
```

Posts an action to the event loop for processing on the next pass.

Parameters

action	The action to perform.

Reimplemented in com.choosemuse.libmuse.HandlerEventLoop.

8.17.2.3 postDelayed()

```
abstract void com.choosemuse.libmuse.EventLoop.postDelayed (
```

```
        Action action,
        long delayMilliseconds )  [abstract]
```

Posts an action to the event loop for processing after the specified delay in milliseconds has passed.

Parameters

action	The action to perform.
delayMilliseconds	The time to wait before posting the action to the event loop for processing.

Reimplemented in com.choosemuse.libmuse.HandlerEventLoop.

The documentation for this class was generated from the following file:

- EventLoop.java

8.18 com.choosemuse.libmuse.Gyro Enum Reference

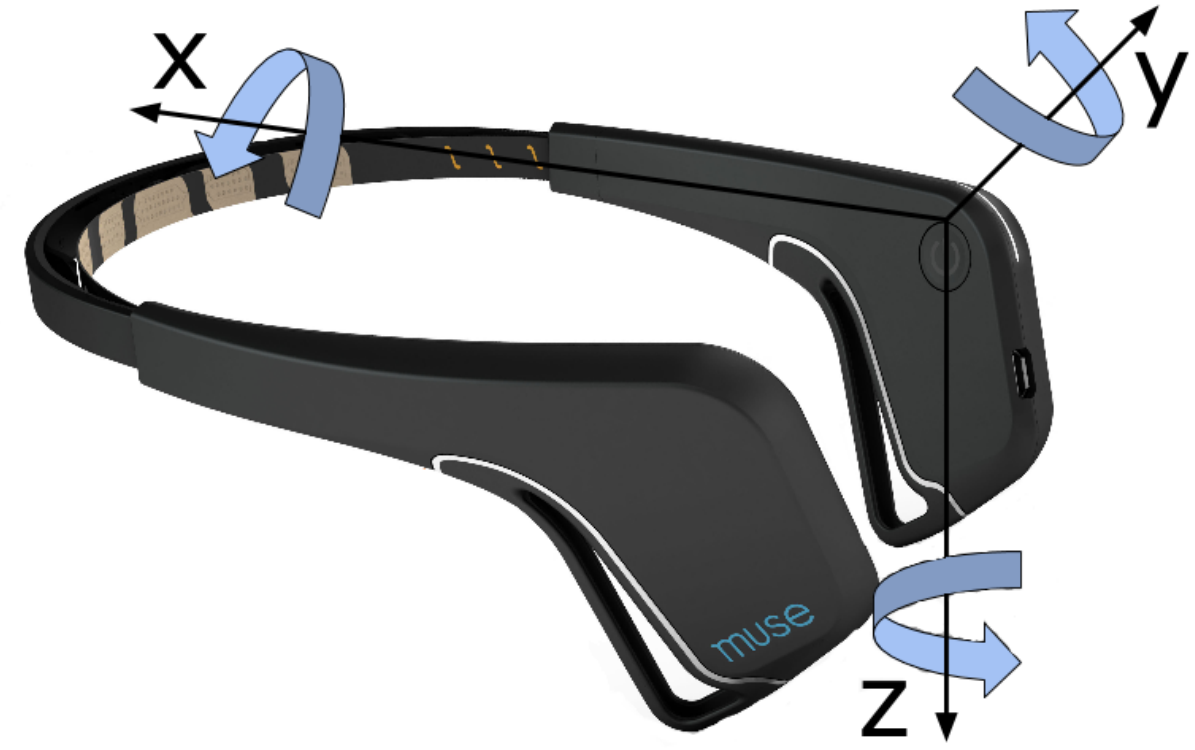
Public Attributes

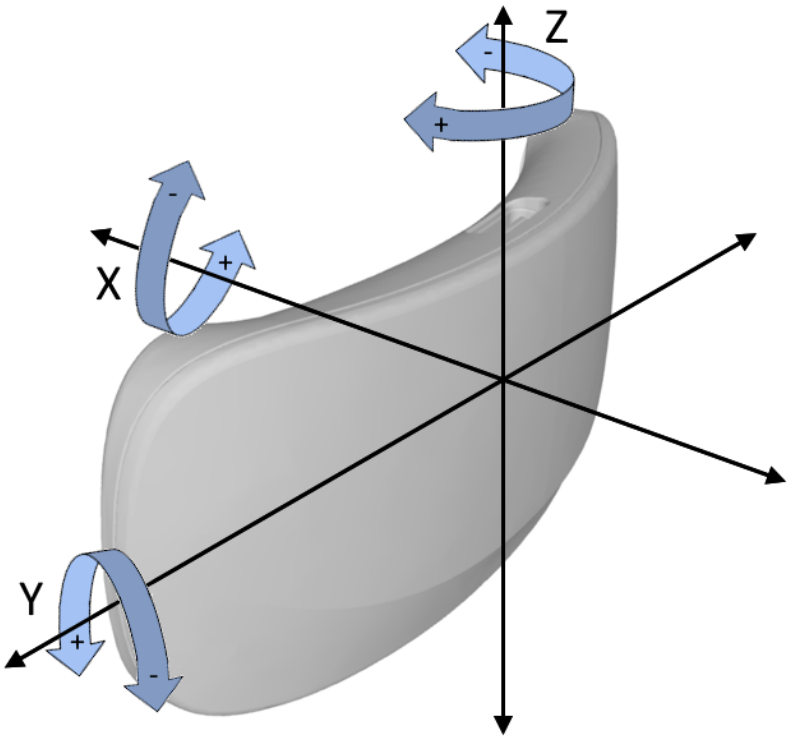
- X
- Y
- Z

8.18.1 Detailed Description

Represents the data mapping in a Gyro data packet

The gyro data is measured as the rotation about 3 axes which is shown in the pictures below:





The axes are oriented to a Right Hand Coordinate System.
Rotation about the X axis corresponds to tilting the head side to side. Positive values increase when tilting to the right. This is also known as roll.
Rotation about the Y axis corresponds to tilting the head up and down. Positive values increase when looking up. This is also known as pitch.
Rotation about the Z axis corresponds to looking left and right. Positive values increase when looking to the right. This is also known as yaw.

See also

MuseDataPacketType.GYRO

MuseDataPacket.getGyroValue()

8.18.2 Member Data Documentation

8.18.2.1 X

`com.choosemuse.libmuse.Gyro.X`

Rotation about the X axis in degrees per second. Rotation about the X axis corresponds to tilting the head side to side. Positive values increase when tilting to the right. This is also known as roll.

8.18.2.2 Y

`com.choosemuse.libmuse.Gyro.Y`

Rotation about the Y axis in degrees per second. Rotation about the Y axis corresponds to tilting the head up and down. Positive values increase when looking up. This is also known as pitch.

8.18.2.3 Z

`com.choosemuse.libmuse.Gyro.Z`

Rotation about the Z axis in degrees per second. Rotation about the Z axis corresponds to looking left and right. Positive values increase when looking to the right. This is also known as yaw.

The documentation for this enum was generated from the following file:

- Gyro.java

8.19 com.choosemuse.libmuse.HandlerEventLoop Class Reference

Inheritance diagram for com.choosemuse.libmuse.HandlerEventLoop:

Collaboration diagram for com.choosemuse.libmuse.HandlerEventLoop:

Public Member Functions

- HandlerEventLoop (final Handler handler)
- void post (Action action)
- void postDelayed (Action action, long delayMilliseconds)
- void cancel ()

8.19.1 Detailed Description

Android EventLoop for an existing Handler.

8.19.2 Constructor & Destructor Documentation

8.19.2.1 HandlerEventLoop()

```
com.choosemuse.libmuse.HandlerEventLoop.HandlerEventLoop (
            final Handler handler )  [inline]
```

Constructs a HandlerEventLoop using the provided Handler.

Parameters

handler	The handler to use for the event loop.

8.19.3 Member Function Documentation

8.19.3.1 cancel()

```
void com.choosemuse.libmuse.HandlerEventLoop.cancel ( )  [inline]
```

Cancel all pending actions.

Reimplemented from com.choosemuse.libmuse.EventLoop.

8.19.3.2 post()

```
void com.choosemuse.libmuse.HandlerEventLoop.post (
            Action action )  [inline]
```

Posts an action to the event loop for processing on the next pass.

Parameters

action	The action to perform.

Reimplemented from com.choosemuse.libmuse.EventLoop.

8.19.3.3 postDelayed()

```
void com.choosemuse.libmuse.HandlerEventLoop.postDelayed (
            Action action,
            long delayMilliseconds )  [inline]
```

Posts an action to the event loop for processing after the specified delay in milliseconds has passed.

Parameters

action	The action to perform.
delayMilliseconds	The time to wait before posting the action to the event loop for processing.

Reimplemented from com.choosemuse.libmuse.EventLoop.

The documentation for this class was generated from the following file:

- HandlerEventLoop.java

8.20 com.choosemuse.libmuse.LibmuseVersion Class Reference

Static Public Member Functions

- static native ApiVersion instance ()

8.20.1 Detailed Description

The version of the LibMuse library.

8.20.2 Member Function Documentation

8.20.2.1 instance()

```
static native ApiVersion com.choosemuse.libmuse.LibmuseVersion.instance ( )  [static]
```

Returns the LibMuse API version.

Returns

the LibMuse API version.

The documentation for this class was generated from the following file:

- LibmuseVersion.java

8.21 com.choosemuse.libmuse.LogListener Class Reference

Inheritance diagram for com.choosemuse.libmuse.LogListener:

Public Member Functions

- abstract void receiveLog (LogPacket log)

8.21.1 Detailed Description

User-supplied log message handler.

You can provide an instance of this class in order to integrate LibMuse logs into your application's logging system.

See also

LogManager

8.21.2 Member Function Documentation

8.21.2.1 receiveLog()

```
abstract void com.choosemuse.libmuse.LogListener.receiveLog (
            LogPacket log ) [abstract]
```

Handler method for log messages. These are sent when interesting events happen. A simple handler may just print each message to stderr or stdout separated by newlines. It's also possible to implement arbitrary log policies within this handler.

Warning

receiveLog may be entered concurrently from multiple threads; if your listener implementation requires synchronization, it should provide its own.

Your listener should not make any LibMuse API calls; otherwise, unbounded recursion may occur, leading to resource exhaustion.

Parameters

log	The log message.

Reimplemented in com.choosemuse.libmuse.AndroidLogListener.

The documentation for this class was generated from the following file:

- LogListener.java

8.22 com.choosemuse.libmuse.LogManager Class Reference

Public Member Functions

- abstract LogListener makeDefaultLogListener ()
- abstract void setLogListener (LogListener listener)
- abstract void setMinimumSeverity (Severity severity)
- abstract void writeLog (Severity severity, boolean raw, String tag, String message)
- abstract long getTimestamp ()
- abstract double timeSince (long timestamp)

Static Public Member Functions

- static native LogManager instance ()

8.22.1 Detailed Description

Manages the listener for any logging events generated by LibMuse.

To disable logging:

setLogListener(null);

To install a custom log listener for your application:

setLogListener(your_log_listener);

To restore default logging to standard error (if you have previously changed the log listener):

setLogListener(makeDefaultLogListener());

An instance of the default listener is installed unless another listener is supplied. The default log listener prints all messages to standard error as they show up, without any buffering, using the following formatting rules:

- Raw logs are printed directly without any formatting or newline.
- Other messages are formatted to pretty-print with the tag, timestamp, severity, and log message on a single line (prepending a newline if the last message printed was raw.)

For example, suppose that these packets showed up in order: {info, false, "A", 4, "thing A"}, {info, true, "FOO", 1, "+"}, {err, true, "BAR", 3, "-"}, {warn, false, "B", 2, "thing B"} where the second parameter idicates if this is a raw log message. The log output would be formatted roughly as such:

```
A [I 4.000] thing A
+-
B [W 2.000] thing B
```

In order to provide this formatting without buffering or losing messages, the default log listener uses a mutex, effectively making the logging subsystem single-threaded and IO-bound.

See also

LogListener

8.22.2 Member Function Documentation

8.22.2.1 getTimestamp()

```
abstract long com.choosemuse.libmuse.LogManager.getTimestamp ( )  [abstract]
```

Return the current timestamp.

The returned value is in microseconds since some epoch – January 1 1970 on UNIX, but possibly other values on other systems.

Returns

The current timestamp.

8.22.2.2 instance()

```
static native LogManager com.choosemuse.libmuse.LogManager.instance ( )  [static]
```

Returns the shared log manager instance.

Returns

The shared log manager instance.

8.22.2.3 makeDefaultLogListener()

```
abstract LogListener com.choosemuse.libmuse.LogManager.makeDefaultLogListener ( )  [abstract]
```

Constructs and returns a default log listener.

Returns

an instance of the default log listener.

8.22.2.4 setLogListener()

```
abstract void com.choosemuse.libmuse.LogManager.setLogListener (
            LogListener listener )  [abstract]
```

Sets the log listener.

There is only one log listener active at a time; hence this is just a plain setter rather than a register / unregister pair.

Parameters

listener	the LogListener to use.

8.22.2.5 setMinimumSeverity()

```
abstract void com.choosemuse.libmuse.LogManager.setMinimumSeverity (
            Severity severity )  [abstract]
```

Set the minimum log severity.

Messages at or above this severity will be passed to the configured log listener. Messages below this severity will be dropped.

Parameters

severity	the minimum log severity to log.

8.22.2.6 timeSince()

```
abstract double com.choosemuse.libmuse.LogManager.timeSince (
            long timestamp )  [abstract]
```

Return the time elapsed in seconds since the passed timestamp.

Parameters

timestamp	The prior timestamp.

8.22.2.7 writeLog()

```
abstract void com.choosemuse.libmuse.LogManager.writeLog (
            Severity severity,
            boolean raw,
            String tag,
            String message )  [abstract]
```

Write a log line.

This method sends messages to the configured log listener. It is exposed here primarily so that LibMuse can access it from its own platform-specific code, but there's nothing stopping you from using it and doing all your logging through LibMuse. Except please don't, because we may change the API at any time, and why would you?

The log's timestamp is the number of seconds since this log manager was created.

Parameters

severity	The severity of this log message.
raw	`true` if this is a raw log message (no formatting), `false` otherwise
tag	The tag for this message.
message	The log message.

The documentation for this class was generated from the following file:

- LogManager.java

8.23 com.choosemuse.libmuse.LogPacket Class Reference

Collaboration diagram for com.choosemuse.libmuse.LogPacket:

Public Member Functions

- LogPacket (Severity severity, boolean raw, String tag, double timestamp, String message)
- Severity getSeverity ()
- boolean getRaw ()
- String getTag ()
- double getTimestamp ()
- String getMessage ()
- String toString ()

8.23.1 Detailed Description

A log message.

8.23.2 Constructor & Destructor Documentation

8.23.2.1 LogPacket()

```
com.choosemuse.libmuse.LogPacket.LogPacket (
            Severity severity,
            boolean raw,
            String tag,
            double timestamp,
            String message )  [inline]
```

8.23.3 Member Function Documentation

8.23.3.1 getMessage()

```
String com.choosemuse.libmuse.LogPacket.getMessage ( )  [inline]
```

Content of this log message. Not newline-terminated; handlers should provide their own line-based formatting.

Returns

The log message content.

8.23.3.2 getRaw()

```
boolean com.choosemuse.libmuse.LogPacket.getRaw ( )  [inline]
```

Denotes raw messages.

Raw messages are generally short (one or two character) messages that can be passed straight to the screen without any formatting or added newlines. (We use this to reduce the noise level on some of our verbose log mesages.)

Returns

`true` if this is a raw log message, `false` otherwise.

8.23.3.3 getSeverity()

```
Severity com.choosemuse.libmuse.LogPacket.getSeverity ( )  [inline]
```

Severity of this message.

Returns

The severity of this message.

8.23.3.4 getTag()

```
String com.choosemuse.libmuse.LogPacket.getTag ( )  [inline]
```

Human-readable, usually all-caps, tag for this message, e.g. "CONNECTOR" or "DEVICE".

Returns

The message tag.

8.23.3.5 getTimestamp()

```
double com.choosemuse.libmuse.LogPacket.getTimestamp ( )  [inline]
```

Timestamp for this log, in seconds.

Returns

The timestamp of this log message.

8.23.3.6 toString()

```
String com.choosemuse.libmuse.LogPacket.toString ( )  [inline]
```

Returns a string representation of this object.

Returns

a string representation of this object.

The documentation for this class was generated from the following file:

- LogPacket.java

8.24 com.choosemuse.libmuse.Magnetometer Enum Reference

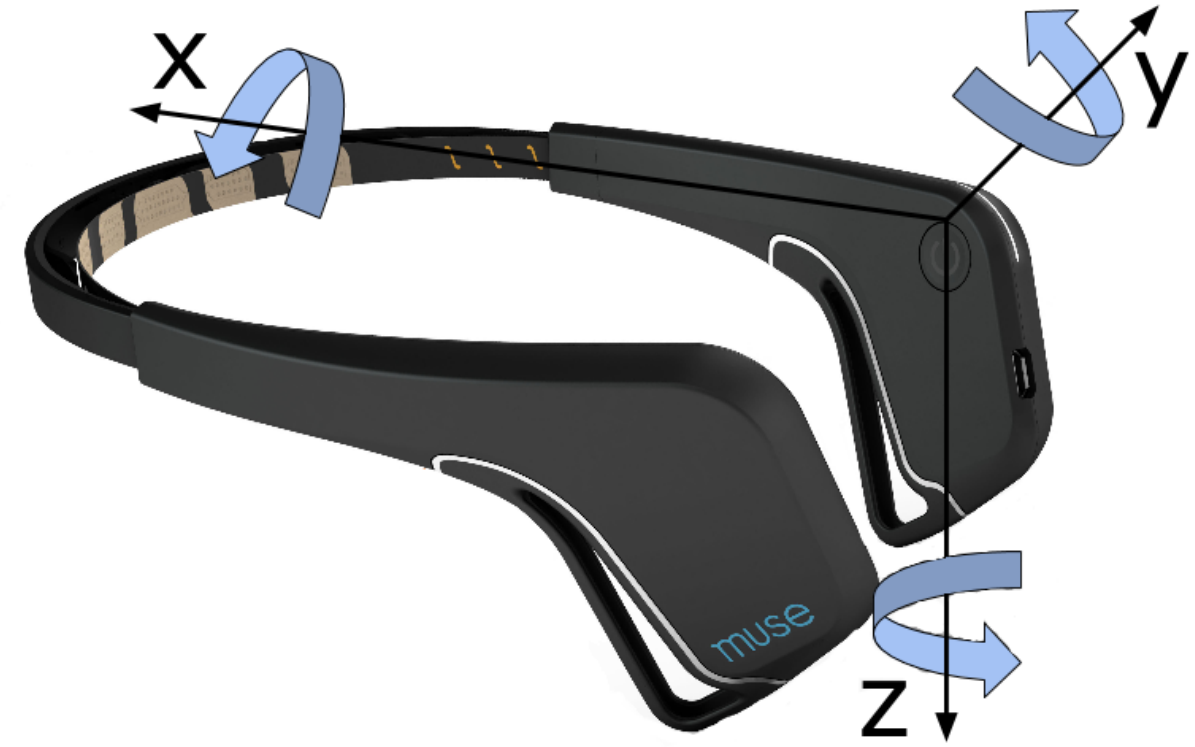
Public Attributes

- X
- Y
- Z

8.24.1 Detailed Description

Represents the data mapping in a Magnetometer data packet

The magnetometer data is measured on 3 axes as shown in the picture below:



The axes are oriented to a Right Hand Coordinate System.

See also

MuseDataPacketType.MAGNETOMETER

MuseDataPacket.getMagnetometerValue()

8.24.2 Member Data Documentation

8.24.2.1 X

`com.choosemuse.libmuse.Magnetometer.X`

8.24.2.2 Y

`com.choosemuse.libmuse.Magnetometer.Y`

8.24.2.3 Z

`com.choosemuse.libmuse.Magnetometer.Z`

The documentation for this enum was generated from the following file:

- Magnetometer.java

8.25 com.choosemuse.libmuse.MessageType Enum Reference

Public Attributes

- EEG
- QUANTIZATION
- ACCELEROMETER
- BATTERY
- VERSION
- CONFIGURATION
- ANNOTATION
- HISTOGRAM
- ALG_VALUE
- DSP
- COMPUTING_DEVICE
- EEG_DROPPED
- ACC_DROPPED
- CALM_APP
- CALM_ALG
- MUSE_ELEMENTS
- GYRO
- ARTIFACT
- PRESSURE
- TEMPERATURE
- ULTRA_VIOLET
- MAGNETOMETER
- PPG
- THERMISTOR
- OPTICS
- ALGORITHM
- DEEP_SLEEP_STIM

8.25.1 Detailed Description

Represents all possible MuseData message data types in a .muse file. This enum corresponds to the DataType enum in the protobuf schema.

8.25.2 Member Data Documentation

8.25.2.1 ACC_DROPPED

`com.choosemuse.libmuse.MessageType.ACC_DROPPED`

A message containing dropped acc data.

8.25.2.2 ACCELEROMETER

`com.choosemuse.libmuse.MessageType.ACCELEROMETER`

A message containing accelerometer data.

8.25.2.3 ALG_VALUE

`com.choosemuse.libmuse.MessageType.ALG_VALUE`

A message containing algorithm data.

8.25.2.4 ALGORITHM

`com.choosemuse.libmuse.MessageType.ALGORITHM`

A message containing algorithm data.

8.25.2.5 ANNOTATION

`com.choosemuse.libmuse.MessageType.ANNOTATION`

A message containing annotation data.

8.25.2.6 ARTIFACT

`com.choosemuse.libmuse.MessageType.ARTIFACT`

A message containing artifact packet.

8.25.2.7 BATTERY

`com.choosemuse.libmuse.MessageType.BATTERY`

A message containing battery data.

8.25.2.8 CALM_ALG

`com.choosemuse.libmuse.MessageType.CALM_ALG`

A message containing data from the calm algorithm.

8.25.2.9 CALM_APP

`com.choosemuse.libmuse.MessageType.CALM_APP`

A message containing data from the calm application.

8.25.2.10 COMPUTING_DEVICE

`com.choosemuse.libmuse.MessageType.COMPUTING_DEVICE`

A message containing device data.

8.25.2.11 CONFIGURATION

`com.choosemuse.libmuse.MessageType.CONFIGURATION`

A message containing configuration data.

8.25.2.12 DEEP_SLEEP_STIM

`com.choosemuse.libmuse.MessageType.DEEP_SLEEP_STIM`

A message containing deep sleep stimulation data.

8.25.2.13 DSP

`com.choosemuse.libmuse.MessageType.DSP`

A message containing dsp data.

8.25.2.14 EEG

`com.choosemuse.libmuse.MessageType.EEG`

A message containing eeg data.

8.25.2.15 EEG_DROPPED

`com.choosemuse.libmuse.MessageType.EEG_DROPPED`

A message containing dropped eeg data.

8.25.2.16 GYRO

`com.choosemuse.libmuse.MessageType.GYRO`

A message containing gyro data.

8.25.2.17 HISTOGRAM

`com.choosemuse.libmuse.MessageType.HISTOGRAM`

A message containing histogram data.

8.25.2.18 MAGNETOMETER

`com.choosemuse.libmuse.MessageType.MAGNETOMETER`

A message containing magnetometer data.

8.25.2.19 MUSE_ELEMENTS

`com.choosemuse.libmuse.MessageType.MUSE_ELEMENTS`

A message containing muse element data.

8.25.2.20 OPTICS

`com.choosemuse.libmuse.MessageType.OPTICS`

A message containing optics data.

8.25.2.21 PPG

`com.choosemuse.libmuse.MessageType.PPG`

A message containing ppg data.

8.25.2.22 PRESSURE

`com.choosemuse.libmuse.MessageType.PRESSURE`

A message containing pressure data.

8.25.2.23 QUANTIZATION

`com.choosemuse.libmuse.MessageType.QUANTIZATION`

A message containing quantization data.

8.25.2.24 TEMPERATURE

`com.choosemuse.libmuse.MessageType.TEMPERATURE`

A message containing temperature data.

8.25.2.25 THERMISTOR

`com.choosemuse.libmuse.MessageType.THERMISTOR`

A message containing thermistor data.

8.25.2.26 ULTRA_VIOLET

`com.choosemuse.libmuse.MessageType.ULTRA_VIOLET`

A message containing ultra violet data.

8.25.2.27 VERSION

`com.choosemuse.libmuse.MessageType.VERSION`

A message containing version data.

The documentation for this enum was generated from the following file:

- MessageType.java

8.26 com.choosemuse.libmuse.Muse Class Reference

Public Member Functions

- abstract void connect ()
- abstract void disconnect ()
- abstract void execute ()
- abstract void runAsynchronously ()
- abstract ConnectionState getConnectionState ()
- abstract String getMacAddress ()
- abstract String getName ()
- abstract double getRssi ()
- abstract MuseModel getModel ()
- abstract double getLastDiscoveredTime ()
- abstract void setNumConnectTries (int numTries)
- abstract MuseConfiguration getMuseConfiguration ()
- abstract MuseVersion getMuseVersion ()
- abstract void registerConnectionListener (MuseConnectionListener listener)
- abstract void unregisterConnectionListener (MuseConnectionListener listener)
- abstract void registerDataListener (MuseDataListener listener, MuseDataPacketType type)
- abstract void unregisterDataListener (MuseDataListener listener, MuseDataPacketType type)
- abstract void registerErrorListener (MuseErrorListener listener)
- abstract void unregisterErrorListener (MuseErrorListener listener)
- abstract void unregisterAllListeners ()
- abstract void setPreset (MusePreset preset)
- abstract void enableLedIndicator (boolean enable)
- abstract void enableDataTransmission (boolean enable)
- abstract void setNotchFrequency (NotchFrequency newFrequency)
- abstract boolean isLowEnergy ()
- abstract boolean isPaired ()
- abstract boolean isConnectable ()
- abstract void setLicenseData (byte[] data)
- abstract void enableException (boolean enable)
- abstract void setProperty (String name, String value)

8.26.1 Detailed Description

Provides the client interface to a specific Muse Headband.

This class maps to a single Muse Headband, providing an API for receiving connection events and data packets, as well as the ability to inspect various properties of the headband it corresponds to. Users can initiate and halt a connection to the device, enable data transmission, find out the human-readable name of the headband, etc.

This class should never be constructed directly by a user; rather, references to it can be retrieved from MuseManager.

There are two main ways of interacting with a Muse:

1. calling runAsynchronously()

2. calling connect() on your own and then repeatedly calling execute().

If you call runAsynchronously(), you do not need to call connect() or execute() on your own. Everything is taken care of by LibMuse for the duration of the connection (i.e. until you receive a disconnected event).

If you call connect() on your own, you are responsible for calling execute() on your own as well. Execute performs small, non-blocking steps of work such as initiating a connection, streaming data or disconnecting. If execute() is not called, no work is performed. The Muse will not transition its connection state and you will not receive any data. If you are using this method of interacting with a Muse you need to continue calling execute() until you receive a DISCONNECTED event through a registered connection listener. It may take multiple execute calls to reach the DISCONNECTED after issuing the disconnect request. Failure to wait for the DISCONNECTED event can result in unanticipated consequences.

8.26.2 Member Function Documentation

8.26.2.1 connect()

```
abstract void com.choosemuse.libmuse.Muse.connect ( )  [abstract]
```

Initiate a connection to a Muse headband.

This call is non-blocking and returns immediately. Since this returns before the connection is established, you should instead use a connection listener to implement any logic, such as updating a UI element, on successful connection. When the connection is successful you will receive a CONNECTED event. If the connection is unsuccessful you will receive a DISCONNECTED event.

Threading: method is thread-safe.

8.26.2.2 disconnect()

```
abstract void com.choosemuse.libmuse.Muse.disconnect ( )  [abstract]
```

Disconnects your mobile device from Muse Headband.

If you are calling execute() on your own, remember to continue calling execute() until you receive the DISCONN↩ ECTED event.

Threading: method is thread-safe.

If you don't want to receive disconnection event, unregister listeners manually first.

8.26.2.3 enableDataTransmission()

```
abstract void com.choosemuse.libmuse.Muse.enableDataTransmission (
            boolean enable )  [abstract]
```

Starts/stops data transmission (but keep-alive packets will be still sent). If you're using low-level interface (connect + execute), you should still call execute() to continue sending keep-alive messages. If you're using runAsynchronously(), then it will take care about sending keep-alive packets.
Threading: method is thread-safe. You can call it in the middle of an execute operation. If method is called before connection is established, the right setting will be passed to device during connection routine.

Parameters

enable	`true` to start streaming data. `false` to pause the data stream.

8.26.2.4 enableException()

```
abstract void com.choosemuse.libmuse.Muse.enableException (
            boolean enable )  [abstract]
```

Enable/disable libmuse to rethrow any exceptions caught in your app code from the various listeners. For example, if there is a bug in your data listener code that results in an exception. Libmuse will catch that and it can rethrow that to cause your app to crash if enabled. Or if disabled it will swallow that exception and prevent the app from crashing. It is useful to enable this in development to help track down any bugs in your code more easily. It is best to disable this in production code to prevent your app from crashing. If this method is not called, the default is disabled.

Parameters

enable	`true` to enable exception to be thrown. `false` to disable.

8.26.2.5 enableLedIndicator()

```
abstract void com.choosemuse.libmuse.Muse.enableLedIndicator (
            boolean enable )  [abstract]
```

Toggles the LED indicator state on Muse headbands supporting sleep.

Warning

The headband must be in the CONNECTED state before calling this method. **Threading:** method is thread-safe.

Parameters

enable	`true` to turn indicator LEDs on. `false` to turn them off.

8.26.2.6 execute()

```
abstract void com.choosemuse.libmuse.Muse.execute ( )  [abstract]
```

Runs a single, non-blocking step of processing.

Reads data from Bluetooth if there is any; updates the connection state; sends packets to listeners; etc. This should be called relatively frequently: max 250ms, ideally 20ms.

If you are listening for computed values (e.g. band powers) rather than just raw EEG samples, then be aware that this method may trigger some amount of processing in the foreground before it returns. If this has undesirable effects on app performance, then you may want to move it to a dedicated thread.

Also be aware that all packets will be received on the same thread as execute() is run.

Threading: This method is NOT thread safe. You must synchronize the Muse object on your own if you wish to call it concurrently from more than one thread.

8.26.2.7 getConnectionState()

```
abstract ConnectionState com.choosemuse.libmuse.Muse.getConnectionState ( )  [abstract]
```

Returns current connection state.

Threading: method is thread-safe.

Returns

The current connection state.

8.26.2.8 getLastDiscoveredTime()

```
abstract double com.choosemuse.libmuse.Muse.getLastDiscoveredTime ( )  [abstract]
```

Returns the time at which this device was most recently discovered.

Muse devices broadcast service information every few seconds, and this is the last time LibMuse has heard this broadcast info.

Only implemented on low-energy Muses. Returns NaN if called on non-low-energy Muses.

The value is in microseconds since some common start time (e.g. epoch, device start time, or app start time) that is guaranteed not to change within a running app process. The value may be used e.g. to determine which of two devices was discovered more recently than the other. It should not be used as an absolute time, nor should the common start time be assumed to be the same between any two running app instances.

Returns

The time at which this device was most recently discovered.

8.26.2.9 getMacAddress()

```
abstract String com.choosemuse.libmuse.Muse.getMacAddress ( )  [abstract]
```

Returns Bluetooth MAC address of the Muse Headband.

Threading: method is thread-safe.

Returns

The MAC address of the Muse Headband.

8.26.2.10 getModel()

```
abstract MuseModel com.choosemuse.libmuse.Muse.getModel ( )  [abstract]
```

Returns the model of the connected Muse Headband.

For Muse2 2018 (MU_03) and later, the Muse must have reached the CONNECTED state before this method is called. Otherwise it will return MU_02.

This should return a value for every Muse currently available.

Threading: method is thread-safe.

Returns

The model of the Muse Headband.

8.26.2.11 getMuseConfiguration()

```
abstract MuseConfiguration com.choosemuse.libmuse.Muse.getMuseConfiguration ( )  [abstract]
```

Returns struct which contains all information about Muse configuration.

Threading: method is thread-safe.

Warning

The Muse Configuration object is only populated during connection routine or after headband settings (like preset or notch frequency) are changed. If this is called before the Muse is connected, the configuration will be `null`.

Returns

The configuration information of this Muse or `null` if the configuration is unknown.

8.26.2.12 getMuseVersion()

```
abstract MuseVersion com.choosemuse.libmuse.Muse.getMuseVersion ( )  [abstract]
```

Returns all information about the version of the Muse.

Threading: method is thread-safe.

Warning

The Muse Version is populated during connection routine only. If this is called before the Muse is connected, the version will be `null`.

Returns

The version of this Muse or `null` if the version is unknown.

8.26.2.13 getName()

```
abstract String com.choosemuse.libmuse.Muse.getName ( )  [abstract]
```

Returns Bluetooth name of the Muse Headband.

Threading: method is thread-safe.

Returns

The name of the Muse Headband.

8.26.2.14 getRssi()

```
abstract double com.choosemuse.libmuse.Muse.getRssi ( )  [abstract]
```

Returns the RSSI of this device.

Only implemented on low-energy Muses. Returns NaN if called on non-low-energy Muses.

Threading: method is thread-safe.

Returns

The RSSI of a low-energy Muse or NaN if the Muse is not low-energy.

8.26.2.15 isConnectable()

```
abstract boolean com.choosemuse.libmuse.Muse.isConnectable ( )  [abstract]
```

Returns true if the Muse is connectable, false otherwise. This will always return true for Muse 2014 (MU_01). For Muse 2016 (MU_02) or later headbands this will return true on Android and iOS. For Muse 2016 (MU_02) or later headbands on Windows, this will return true if the headband is advertising it is connectable, false if it is not.

Returns

`true` if the Muse is connectable, `false` otherwise.

8.26.2.16 isLowEnergy()

```
abstract boolean com.choosemuse.libmuse.Muse.isLowEnergy ( )  [abstract]
```

True if this device supports Bluetooth Low-Energy.

Returns

`true` if this Muse supports Bluetooth Low-Energy. `false` if it does not.

8.26.2.17 isPaired()

```
abstract boolean com.choosemuse.libmuse.Muse.isPaired ( )  [abstract]
```

Returns true if the Muse is paired with the OS, false otherwise. This will always return true for Muse 2014 (MU_01). For Muse 2016 (MU_02) or later headbands this will return false on Android and iOS. For Muse 2016 (MU_02) or later headbands on Windows, this will return true if the headband is paired with the OS and false if it is not.

Returns

`true` if the Muse is paired with the OS, `false` otherwise.

8.26.2.18 registerConnectionListener()

```
abstract void com.choosemuse.libmuse.Muse.registerConnectionListener (
            MuseConnectionListener listener )  [abstract]
```

Registers a connection listener. The same listener cannot be registered twice. If the listener was already registered, then this method does nothing.

Threading: method is thread-safe.

Parameters

listener	The listener to register.

8.26.2.19 registerDataListener()

```
abstract void com.choosemuse.libmuse.Muse.registerDataListener (
            MuseDataListener listener,
            MuseDataPacketType type )  [abstract]
```

Registers a data listener. You can register the same listener to listen for different packet types - just call this method again. It's your responsibility to make sure that the listener handles all packet types correctly.

If this listener was already registered for this specific type then this method does nothing.

Threading: method is thread-safe.

Parameters

listener	The listener to register.
type	The type of data packet the listener will receive.

8.26.2.20 registerErrorListener()

```
abstract void com.choosemuse.libmuse.Muse.registerErrorListener (
            MuseErrorListener listener )  [abstract]
```

Registers an error listener.

Threading: method is thread-safe.

Parameters

listener	The listener to register.

8.26.2.21 runAsynchronously()

```
abstract void com.choosemuse.libmuse.Muse.runAsynchronously ( )  [abstract]
```

Connect to a headband out of the main logic flow.

This method handles all the connection logic for a single connection to a headband (i.e. starting with issuing a "connecting" event and ending with issuing a "disconnected" event). It may run in a separate thread or an event loop on the main thread; the details are platform-specific.

If this method is used, then all packets are delivered on the main thread.

The lifetime of effect of this method ends on disconnection; if you want to initiate a new connection to a headband that has disconnected, you should call runAsynchronously() on it again.

If this method is called multiple times within the lifetime of a single connection, any subsequent calls will do nothing other than print a log message.

Note that there is currently a race condition if runAsynchronously() is called

1. as a result of a DISCONNECTED packet, and
2. on a different thread from the one that delivered the notification.

This will be resolved in a future library release. For now, call runAsynchronously() on the same thread that sent the DISCONNECTED packet, or after some modest delay (say 1 second.)

Threading: This method is thread safe.

8.26.2.22 setLicenseData()

```
abstract void com.choosemuse.libmuse.Muse.setLicenseData (
            byte[] data )  [abstract]
```

Allows forwarding of license data from cloud.

Parameters

data	The encrypted license blob.

8.26.2.23 setNotchFrequency()

```
abstract void com.choosemuse.libmuse.Muse.setNotchFrequency (
            NotchFrequency newFrequency )  [abstract]
```

Changes notch frequency (power line frequency).

Threading: method is thread-safe. You can call it in the middle of execute operation, but in this case be aware that this operation will interrupt data streaming to set new notch frequency. Data streaming will be restored after that. If method is called before a connection is established, the right setting will be passed to device during connection routine.

When combining a call setNotchFrequency with Muse.setPreset() setNotchFrequency can be called before or after the call to Muse.setPreset() The notch frequency will be applied to the preset after the preset is set.

Warning

This is only supported on Muse 2014 (MU_01) headbands and then only with presets: MusePreset.PRESET_10 , MusePreset.PRESET_12 and MusePreset.PRESET_14 . Under those presets, the only valid frequencies are: NotchFrequency.NOTCH_50HZ NotchFrequency.NOTCH_60HZ

Calling this with NotchFrequency.NOTCH_NONE will do nothing.

This does nothing on Muse 2016 (MU_02) or later headbands and on Muse 2014 (MU_01) headbands with presets MusePreset.PRESET_AB or MusePreset.PRESET_AD

Parameters

newFrequency	The new notch frequency.

8.26.2.24 setNumConnectTries()

```
abstract void com.choosemuse.libmuse.Muse.setNumConnectTries (
            int numTries )  [abstract]
```

When LibMuse tries connect to a Muse, it will by default try 3 times before giving up. This function allows you to change the number of tries from 3 to anything you want. numTries must be a positive integer. This function only works for model MU-02 Muses, calling it on a MU-01 will do nothing. This function was added for internal testing purposes, and we do not anticipate that 3rd party clients of the LibMuse library will need to modify this value.

Threading: This method is thread safe.

Parameters

numTries	The number of times to try to connect before giving up.

8.26.2.25 setPreset()

```
abstract void com.choosemuse.libmuse.Muse.setPreset (
            MusePreset preset )  [abstract]
```

Changes Muse Headband settings. **Threading:** method is thread-safe. You can call it in the middle of execute operation, but in this case be aware that this operation will interrupt data streaming to set new preset. Data streaming will be restored after that. If method is called before connection is established, the right setting will be passed to device during connection routine.

Parameters

preset	The new preset.

8.26.2.26 setProperty()

```
abstract void com.choosemuse.libmuse.Muse.setProperty (
            String name,
            String value )  [abstract]
```

Factory use only

8.26.2.27 unregisterAllListeners()

```
abstract void com.choosemuse.libmuse.Muse.unregisterAllListeners ( )  [abstract]
```

Unregisters all registered connection listeners and data listeners.

Threading: method is thread-safe.

8.26.2.28 unregisterConnectionListener()

```
abstract void com.choosemuse.libmuse.Muse.unregisterConnectionListener (
            MuseConnectionListener listener )  [abstract]
```

Unregisters connection listeners.

Threading: method is thread-safe.

Parameters

listener	The listener to unregister.

8.26.2.29 unregisterDataListener()

```
abstract void com.choosemuse.libmuse.Muse.unregisterDataListener (
            MuseDataListener listener,
            MuseDataPacketType type )  [abstract]
```

Unregisters a data listener that was registered before.

If the listener was not registered before, then this method does nothing.

Threading: method is thread-safe.

Parameters

listener	The listener to unregister.
type	The type of data packet the listener will stop receiving.

8.26.2.30 unregisterErrorListener()

```
abstract void com.choosemuse.libmuse.Muse.unregisterErrorListener (
            MuseErrorListener listener )  [abstract]
```

Unregisters an info listener that was registered before.

If the listener was not registered before, then this method does nothing.

Threading: method is thread-safe.

Parameters

listener	The listener to unregister.

The documentation for this class was generated from the following file:

- Muse.java

8.27 com.choosemuse.libmuse.MuseArtifactPacket Class Reference

Public Member Functions

- MuseArtifactPacket (boolean headbandOn, boolean blink, boolean jawClench, long timestamp)
- boolean getHeadbandOn ()
- boolean getBlink ()
- boolean getJawClench ()
- long getTimestamp ()
- String toString ()

8.27.1 Detailed Description

Provides access to calculated artifacts.

8.27.2 Constructor & Destructor Documentation

8.27.2.1 MuseArtifactPacket()

```
com.choosemuse.libmuse.MuseArtifactPacket.MuseArtifactPacket (
            boolean headbandOn,
            boolean blink,
            boolean jawClench,
            long timestamp )  [inline]
```

8.27.3 Member Function Documentation

8.27.3.1 getBlink()

```
boolean com.choosemuse.libmuse.MuseArtifactPacket.getBlink ( )  [inline]
```

Eye blinking flag.

Returns

`true` if an eye blink was detected. `false` otherwise.

8.27.3.2 getHeadbandOn()

```
boolean com.choosemuse.libmuse.MuseArtifactPacket.getHeadbandOn ( )  [inline]
```

Flag which represents if the Muse Headband is on a person's head. Headband On detection is one of the first things muse elements algorithm does. If this flag is false you will not see other artifact or band power data.

Returns

`true` if the headband is on. `false` otherwise.

8.27.3.3 getJawClench()

```
boolean com.choosemuse.libmuse.MuseArtifactPacket.getJawClench ( )  [inline]
```

Jaw clench flag.

Returns

`true` if a jaw clench was detected. `false` otherwise.

8.27.3.4 getTimestamp()

```
long com.choosemuse.libmuse.MuseArtifactPacket.getTimestamp ( )  [inline]
```

Microseconds since epoch (usually Jan 1, 1970).

Returns

the timestamp of the packet expressed in microseconds since epoch.

8.27.3.5 toString()

```
String com.choosemuse.libmuse.MuseArtifactPacket.toString ( )  [inline]
```

Returns a string representation of this object.

Returns

a string representation of this object.

The documentation for this class was generated from the following file:

- MuseArtifactPacket.java

8.28 com.choosemuse.libmuse.MuseConfiguration Class Reference

Public Member Functions

- abstract MusePreset getPreset ()
- abstract String getHeadbandName ()
- abstract String getMicrocontrollerId ()
- abstract int getEegChannelCount ()
- abstract int getAfeGain ()
- abstract int getDownsampleRate ()
- abstract int getSeroutMode ()
- abstract int getOutputFrequency ()
- abstract int getAdcFrequency ()
- abstract boolean getNotchFilterEnabled ()
- abstract NotchFrequency getNotchFilter ()
- abstract int getAccelerometerSampleFrequency ()
- abstract boolean getBatteryDataEnabled ()
- abstract boolean getDrlRefEnabled ()
- abstract int getDrlRefFrequency ()
- abstract double getBatteryPercentRemaining ()
- abstract String getBluetoothMac ()
- abstract String getSerialNumber ()
- abstract String getHeadsetSerialNumber ()
- abstract MuseModel getModel ()
- abstract String getLicenseNonce ()
- abstract int getSwitch ()

8.28.1 Detailed Description

Encapsulates the configuration information of the Muse headband.

You must connect to the headband at least once to before this information is available. Once you have connected once the information will remain available, even after you disconnect.

8.28.2 Member Function Documentation

8.28.2.1 getAccelerometerSampleFrequency()

```
abstract int com.choosemuse.libmuse.MuseConfiguration.getAccelerometerSampleFrequency ( )
[abstract]
```

Returns accelerometer sample frequency in Hz.

Returns

The frequency of accelerometer samples.

8.28.2.2 getAdcFrequency()

```
abstract int com.choosemuse.libmuse.MuseConfiguration.getAdcFrequency ( )  [abstract]
```

The sample frequency of EEG packets (ADC) in Hz. A value of -1 means "unknown".

Returns

The ADC frequency of the headband.

8.28.2.3 getAfeGain()

```
abstract int com.choosemuse.libmuse.MuseConfiguration.getAfeGain ( )  [abstract]
```

Gain to apply to incoming EEG samples.

Returns

The gain that should be applied to incoming EEG samples.

8.28.2.4 getBatteryDataEnabled()

```
abstract boolean com.choosemuse.libmuse.MuseConfiguration.getBatteryDataEnabled ( )  [abstract]
```

Provides information about battery data transmission state (on/off).

Returns

`true` if the battery data transmission is enabled. `false` if it is not.

8.28.2.5 getBatteryPercentRemaining()

```
abstract double com.choosemuse.libmuse.MuseConfiguration.getBatteryPercentRemaining ( )  [abstract]
```

Returns battery charge remaining in percent.

Returns

The percentage of battery remaining.

8.28.2.6 getBluetoothMac()

```
abstract String com.choosemuse.libmuse.MuseConfiguration.getBluetoothMac ( )  [abstract]
```

Bluetooth MAC address reported by device.

Returns

The MAC address of the headband.

8.28.2.7 getDownsampleRate()

```
abstract int com.choosemuse.libmuse.MuseConfiguration.getDownsampleRate ( )  [abstract]
```

EEG downsampling rate. A value of -1 means "unknown".

Returns

The EEG downsampling rate.

8.28.2.8 getDrlRefEnabled()

```
abstract boolean com.choosemuse.libmuse.MuseConfiguration.getDrlRefEnabled ( )  [abstract]
```

Provides information about DRL/REF transmission state (on/off).

Returns

`true` if the DRL/REF data transmission is enabled. `false` if it is not.

8.28.2.9 getDrlRefFrequency()

```
abstract int com.choosemuse.libmuse.MuseConfiguration.getDrlRefFrequency ( )  [abstract]
```

DRL/REF data output frequency in Hz. 0 if DRL/REF is not enabled.

Returns

The frequency at which DRL/REF packets are transmitted.

8.28.2.10 getEegChannelCount()

```
abstract int com.choosemuse.libmuse.MuseConfiguration.getEegChannelCount ( )  [abstract]
```

Number of channels currently active.

Returns

The number of EEG channels currently active.

8.28.2.11 getHeadbandName()

```
abstract String com.choosemuse.libmuse.MuseConfiguration.getHeadbandName ( )  [abstract]
```

The name of the headband. This is only available on Muse 2016 (MU_02) or later.

Returns

The name of the headband.

8.28.2.12 getHeadsetSerialNumber()

```
abstract String com.choosemuse.libmuse.MuseConfiguration.getHeadsetSerialNumber ( )  [abstract]
```

Headset serial number reported by device. Only applicable to Aster.

Returns

The headset serial number.

8.28.2.13 getLicenseNonce()

```
abstract String com.choosemuse.libmuse.MuseConfiguration.getLicenseNonce ( )  [abstract]
```

The licensing nonce

Returns

The licensing nonce.

8.28.2.14 getMicrocontrollerId()

```
abstract String com.choosemuse.libmuse.MuseConfiguration.getMicrocontrollerId ( )  [abstract]
```

Unique 96 bit ID identifying the microcontroller on the headband. This is only available on Muse 2016 (MU_02) or later.

Returns

The id of the microcontroller.

8.28.2.15 getModel()

```
abstract MuseModel com.choosemuse.libmuse.MuseConfiguration.getModel ( )  [abstract]
```

The headband model

Returns

The model of the headband.

8.28.2.16 getNotchFilter()

```
abstract NotchFrequency com.choosemuse.libmuse.MuseConfiguration.getNotchFilter ( )  [abstract]
```

Provides information about notch filter frequency (50 Hz/60 Hz).

Returns

The frequency of the notch filter.

8.28.2.17 getNotchFilterEnabled()

```
abstract boolean com.choosemuse.libmuse.MuseConfiguration.getNotchFilterEnabled ( )  [abstract]
```

Provides information about notch filter state (on/off).

Returns

`true` if the notch filter is enabled. `false` if it is not.

8.28.2.18 getOutputFrequency()

```
abstract int com.choosemuse.libmuse.MuseConfiguration.getOutputFrequency ( )  [abstract]
```

The output frequency of EEG packets in Hz.

Returns

The output frequency of EEG packets.

8.28.2.19 getPreset()

```
abstract MusePreset com.choosemuse.libmuse.MuseConfiguration.getPreset ( )  [abstract]
```

Provides access to current preset.

8.28.2.20 getSerialNumber()

```
abstract String com.choosemuse.libmuse.MuseConfiguration.getSerialNumber ( )  [abstract]
```

Serial number reported by device.

Returns

The device serial number.

8.28.2.21 getSeroutMode()

```
abstract int com.choosemuse.libmuse.MuseConfiguration.getSeroutMode ( )  [abstract]
```

Serial output mode. The relates to the size of the EEG packets transmitted from the headband. A value of 2 indicates that the packet data is compressed (10 bits). A value of 3 indicates that the packet data is uncompressed (16 bits). A value of -1 means "unknown".

Returns

`2` if the data is compressed, `3` if the data is uncompressed or `-1` if this value is unknown.

8.28.2.22 getSwitch()

```
abstract int com.choosemuse.libmuse.MuseConfiguration.getSwitch ( )  [abstract]
```

Factory use only

The documentation for this class was generated from the following file:

- MuseConfiguration.java

8.29 com.choosemuse.libmuse.MuseConnectionListener Class Reference

Public Member Functions

- abstract void receiveMuseConnectionPacket (MuseConnectionPacket packet, Muse muse)

8.29.1 Detailed Description

Implement this interface to listen for changes to the current connection status (like connect/disconnect).

Important: DO NOT CALL Muse.connect()/ Muse.disconnect() directly from a connection listener handler. Either create a new thread and run it from there or use Muse.runAsynchronously() which creates a new thread implicitly.

8.29.2 Member Function Documentation

8.29.2.1 receiveMuseConnectionPacket()

```
abstract void com.choosemuse.libmuse.MuseConnectionListener.receiveMuseConnectionPacket (
            MuseConnectionPacket packet,
            Muse muse )  [abstract]
```

Called from the Muse connection thread whenever there is a change in the current connection status of the Muse

Warning

> It is important that you do not perform any computation intensive tasks in this callback. This would result in significant delays in all the listener callbacks from being called. You should delegate any intensive tasks to another thread or schedule it to run with a delay through handler/scheduler for the platform.

However, you can register/unregister listeners in this callback. All previously registered listeners would still receive callbacks for this current event. On subsequent events, the newly registered listeners will be called. For example, if you had 2 listeners 'A' and 'B' for this event. If, on the callback for listener A, listener A unregisters all listeners and registers a new listener 'C' and then in the callback for listener 'B', you unregister all listeners again and register a new listener 'D'. Then on the subsequent event callback, only listener D's callback will be invoked.

Parameters

packet	The connection packet
muse	The Muse that sent the connection packet.

The documentation for this class was generated from the following file:

- MuseConnectionListener.java

8.30 com.choosemuse.libmuse.MuseConnectionPacket Class Reference

Collaboration diagram for com.choosemuse.libmuse.MuseConnectionPacket:

Public Member Functions

- MuseConnectionPacket (ConnectionState previousConnectionState, ConnectionState currentConnection↩ State)
- ConnectionState getPreviousConnectionState ()
- ConnectionState getCurrentConnectionState ()
- String toString ()

8.30.1 Detailed Description

Packet provides information about the connection state.

8.30.2 Constructor & Destructor Documentation

8.30.2.1 MuseConnectionPacket()

```
com.choosemuse.libmuse.MuseConnectionPacket.MuseConnectionPacket (
            ConnectionState previousConnectionState,
            ConnectionState currentConnectionState )  [inline]
```

8.30.3 Member Function Documentation

8.30.3.1 getCurrentConnectionState()

```
ConnectionState com.choosemuse.libmuse.MuseConnectionPacket.getCurrentConnectionState ( )
[inline]
```

Provides access to the current connection status.

Returns

The current (new) connection state.

8.30.3.2 getPreviousConnectionState()

```
ConnectionState com.choosemuse.libmuse.MuseConnectionPacket.getPreviousConnectionState ( )
[inline]
```

Provides access to the previous connection status.

Returns

The prior connection state.

8.30.3.3 toString()

```
String com.choosemuse.libmuse.MuseConnectionPacket.toString ( )  [inline]
```

Returns a string representation of this object.

Returns

a string representation of this object.

The documentation for this class was generated from the following file:

- MuseConnectionPacket.java

8.31 com.choosemuse.libmuse.MuseDataListener Class Reference

Public Member Functions

- abstract void receiveMuseDataPacket (MuseDataPacket packet, Muse muse)
- abstract void receiveMuseArtifactPacket (MuseArtifactPacket packet, Muse muse)

8.31.1 Detailed Description

Implement this interface to receive data from the headband. The data may be either data packets or artifact packets. Check out MuseDataPacketType for information about all acceptable data packet types. All packets can be handled in one listener or you can specify a separate listener for every specific packet type.

A packet is only sent if a listener was registered to listen for this packet type.

8.31.2 Member Function Documentation

8.31.2.1 receiveMuseArtifactPacket()

```
abstract void com.choosemuse.libmuse.MuseDataListener.receiveMuseArtifactPacket (
            MuseArtifactPacket packet,
            Muse muse )  [abstract]
```

Handler method for Muse artifact packets. Packet is sent only if listener was registered to listen for such packets.

Warning

It is important that you do not perform any computation intensive tasks in this callback. This would result in significant delays in all the listener callbacks from being called. You should delegate any intensive tasks to another thread or schedule it to run with a delay through handler/scheduler for the platform.

However, you can register/unregister listeners in this callback. All previously registered listeners would still receive callbacks for this current event. On subsequent events, the newly registered listeners will be called. For example, if you had 2 listeners 'A' and 'B' for this event. If, on the callback for listener A, listener A unregisters all listeners and registers a new listener 'C' and then in the callback for listener 'B', you unregister all listeners again and register a new listener 'D'. Then on the subsequent event callback, only listener D's callback will be invoked.

Parameters

packet	The artifact packet
muse	The Muse that sent the artifact packet.

8.31.2.2 receiveMuseDataPacket()

```
abstract void com.choosemuse.libmuse.MuseDataListener.receiveMuseDataPacket (
            MuseDataPacket packet,
            Muse muse )  [abstract]
```

Handler method for Muse data packets

Warning

It is important that you do not perform any computation intensive tasks in this callback. This would result in significant delays in all the listener callbacks from being called. You should delegate any intensive tasks to another thread or schedule it to run with a delay through handler/scheduler for the platform.

However, you can register/unregister listeners in this callback. All previously registered listeners would still receive callbacks for this current event. On subsequent events, the newly registered listeners will be called. For example, if you had 2 listeners 'A' and 'B' for this event. If, on the callback for listener A, listener A unregisters all listeners and registers a new listener 'C' and then in the callback for listener 'B', you unregister all listeners again and register a new listener 'D'. Then on the subsequent event callback, only listener D's callback will be invoked.

Parameters

packet	The data packet
muse	The Muse that sent the data packet.

The documentation for this class was generated from the following file:

- MuseDataListener.java

8.32 com.choosemuse.libmuse.MuseDataPacket Class Reference

Public Member Functions

- abstract MuseDataPacketType packetType ()
- abstract long timestamp ()
- abstract ArrayList< Double > values ()
- abstract long valuesSize ()
- abstract double getEegChannelValue (Eeg channelNum)
- abstract double getPpgChannelValue (Ppg channelNum)
- abstract double getPpgMicroamps (MuseModel model, double ppgValue)
- abstract double getOpticsChannelValue (Optics channelNum)
- abstract double getBatteryValue (Battery b)
- abstract double getAccelerometerValue (Accelerometer a)
- abstract double getGyroValue (Gyro g)
- abstract double getMagnetometerValue (Magnetometer m)
- abstract double getDrlRefValue (DrlRef drl)
- abstract double getPressureValue (Pressure pressure)
- abstract double getTemperatureValue ()
- abstract double getUvValue (UltraViolet v)

Static Public Member Functions

- static native MuseDataPacket makeUninitializedPacket (long capacity)
- static native MuseDataPacket makePacket (MuseDataPacketType type, long timestamp, ArrayList< Double > values)

8.32.1 Detailed Description

Contains information received from the headband. Each packet can represent different data: eeg, accelerometer, quantization, etc. Take a look at MuseDataPacketType enum for all possible types

8.32.2 Member Function Documentation

8.32.2.1 getAccelerometerValue()

```
abstract double com.choosemuse.libmuse.MuseDataPacket.getAccelerometerValue (
            Accelerometer a )  [abstract]
```

Get the Accelerometer value from the packet.

Calling this function does not perform additional allocations and is preferable to using values().

Parameters

a	the Accelerometer value to retrieve (ie. Accelerometer.X)

Returns

the value requested.

Exceptions

SIGABRT	If this function is called on a packet type that is not type MuseDataPacketType.ACCELEROMETER , LibMuse will throw an exception. Use packetType() to check the type before calling this function.

8.32.2.2 getBatteryValue()

```
abstract double com.choosemuse.libmuse.MuseDataPacket.getBatteryValue (
            Battery b )  [abstract]
```

Get the Battery value from the packet.

Calling this function does not perform additional allocations and is preferable to using values().

Parameters

b	the Battery value to retrieve (ie. Battery.MILLIVOLTS)

Returns

the value requested.

Exceptions

SIGABRT	If this function is called on a packet type that is not type MuseDataPacketType.BATTERY , LibMuse will throw an exception. Use packetType() to check the type before calling this function.

8.32.2.3 getDrlRefValue()

```
abstract double com.choosemuse.libmuse.MuseDataPacket.getDrlRefValue (
            DrlRef drl )  [abstract]
```

Get the DrlRef value from the packet.

Calling this function does not perform additional allocations and is preferable to using values().

Parameters

drl	the DrlRef value to retrieve (ie. DrlRef.DRL)

Returns

the value requested.

Exceptions

SIGABRT	If this function is called on a packet type that is not type MuseDataPacketType.DRL_REF , LibMuse will throw an exception. Use packetType() to check the type before calling this function.

8.32.2.4 getEegChannelValue()

```
abstract double com.choosemuse.libmuse.MuseDataPacket.getEegChannelValue (
            Eeg channelNum )  [abstract]
```

Get the raw EEG or EEG derived value from the packet. EEG derived value are data that is calculated based on the raw EEG values. Take a look at enum in MuseDataPacketType for values derived from EEG channel. Calling this function does not perform additional allocations and is preferable to using values().

Parameters

channelNum	the EEG channel to retrieve. (ie. Eeg.EEG1)

Returns

the value requested.

Exceptions

SIGABRT	If this function is called on a packet type that is not type MuseDataPacketType.EEG , LibMuse will throw an exception. Use packetType() to check the type before calling this function.

8.32.2.5 getGyroValue()

```
abstract double com.choosemuse.libmuse.MuseDataPacket.getGyroValue (
            Gyro g )  [abstract]
```

Get the Gyro value from the packet.

Calling this function does not perform additional allocations and is preferable to using values().

Parameters

g	the Gyro value to retrieve (ie. Gyro.X)

Returns

the value requested.

Exceptions

SIGABRT	If this function is called on a packet type that is not type MuseDataPacketType.GYRO , LibMuse will throw an exception. Use packetType() to check the type before calling this function.

8.32.2.6 getMagnetometerValue()

```
abstract double com.choosemuse.libmuse.MuseDataPacket.getMagnetometerValue (
            Magnetometer m )  [abstract]
```

Get the Magnetometer value from the packet.

Calling this function does not perform additional allocations and is preferable to using values().

Parameters

m	the Magnetometer value to retrieve (ie. Magnetometer.X)

Returns

the value requested.

Exceptions

SIGABRT	If this function is called on a packet type that is not type MuseDataPacketType.MAGNETOMETER , LibMuse will throw an exception. Use packetType() to check the type before calling this function.

8.32.2.7 getOpticsChannelValue()

```
abstract double com.choosemuse.libmuse.MuseDataPacket.getOpticsChannelValue (
            Optics channelNum )  [abstract]
```

Get the optics value from the packet. Calling this function does not perform additional allocations and is preferable to using values().

Parameters

channelNum	the optics channel to retrieve. (ie.) the value requested. SIGABRT If this function is called on a packet type that is not type MuseDataPacketType.OP , LibMuse will throw an exception. Use packetType() to check the type before calling this function.

8.32.2.8 getPpgChannelValue()

```
abstract double com.choosemuse.libmuse.MuseDataPacket.getPpgChannelValue (
            Ppg channelNum )  [abstract]
```

Get the PPG value from the packet. Calling this function does not perform additional allocations and is preferable to using values().

Parameters

channelNum	the PPG channel to retrieve. (ie. Ppg.IR Ppg.RED)

Returns

the value requested.

Exceptions

SIGABRT	If this function is called on a packet type that is not type MuseDataPacketType.PPG , LibMuse will throw an exception. Use packetType() to check the type before calling this function.

8.32.2.9 getPpgMicroamps()

```
abstract double com.choosemuse.libmuse.MuseDataPacket.getPpgMicroamps (
            MuseModel model,
            double ppgValue )  [abstract]
```

Get PPG microamps from the raw PPG value.

Parameters

model	the Muse model to which the conversion is applied.
ppgValue	the raw PPG channel value.

Returns

the PPG value in microamps.

Exceptions

SIGABRT	

8.32.2.10 getPressureValue()

```
abstract double com.choosemuse.libmuse.MuseDataPacket.getPressureValue (
            Pressure pressure )  [abstract]
```

Get the Pressure value from the packet.

Calling this function does not perform additional allocations and is preferable to using values().

Parameters

pressure	the Pressure value to retrieve (ie. Pressure.AVERAGED)

Returns

the value requested.

Exceptions

SIGABRT	If this function is called on a packet type that is not type MuseDataPacketType.PRESSURE , LibMuse will throw an exception. Use packetType() to check the type before calling this function.

8.32.2.11 getTemperatureValue()

```
abstract double com.choosemuse.libmuse.MuseDataPacket.getTemperatureValue ( )  [abstract]
```

Get the temperature value from the packet.

Calling this function does not perform additional allocations and is preferable to using values().

Returns

the value requested.

Exceptions

SIGABRT	If this function is called on a packet type that is not type MuseDataPacketType.TEMPERATURE , LibMuse will throw an exception. Use packetType() to check the type before calling this function.

8.32.2.12 getUvValue()

```
abstract double com.choosemuse.libmuse.MuseDataPacket.getUvValue (
            UltraViolet v )  [abstract]
```

Get the UltraViolet value from the packet.

Calling this function does not perform additional allocations and is preferable to using values().

Parameters

v	the UltraViolet value to retrieve (ie. UltraViolet.UV_A)

Returns

the value requested.

Exceptions

SIGABRT	If this function is called on a packet type that is not type MuseDataPacketType.ULTRA_VIOLET , LibMuse will throw an exception. Use packetType() to check the type before calling this function.

8.32.2.13 makePacket()

```
static native MuseDataPacket com.choosemuse.libmuse.MuseDataPacket.makePacket (
            MuseDataPacketType type,
            long timestamp,
            ArrayList< Double > values )  [static]
```

Create a new packet with the given contents.

Parameters

type	the type of packet to create
timestamp	the timestamp of the packet
values	the data the packet contains.

8.32.2.14 makeUninitializedPacket()

```
static native MuseDataPacket com.choosemuse.libmuse.MuseDataPacket.makeUninitializedPacket (
            long capacity )  [static]
```

Create a new packet with reserved capacity but unspecified contents.

Parameters

capacity	the number of data entries to reserve.

8.32.2.15 packetType()

```
abstract MuseDataPacketType com.choosemuse.libmuse.MuseDataPacket.packetType ( )  [abstract]
```

Specifies what kind of values are stored in the packet.

Returns

the type of packet.

8.32.2.16 timestamp()

```
abstract long com.choosemuse.libmuse.MuseDataPacket.timestamp ( )  [abstract]
```

Microseconds since epoch (usually Jan 1, 1970).

Returns

the timestamp of the packet expressed in microseconds since epoch.

8.32.2.17 values()

```
abstract ArrayList<Double> com.choosemuse.libmuse.MuseDataPacket.values ( )  [abstract]
```

Raw packet data as an array. The size of this array and the meaning of its elements depend on the packet type. See the documentation for MuseDataPacketType for details.

Note that this method causes new memory to be allocated for an array of boxed double values. If this impacts your application's performance, you should use get_xxx_value() methods instead.

Deprecated Use MuseDataPacket.getAccelerometerValue() , MuseDataPacket.getBatteryValue() , MuseDataPacket.getDrlRefValue(, MuseDataPacket.getEegChannelValue() , MuseDataPacket.getPpgChannelValue() , MuseDataPacket.getGyroValue() instead.

8.32.2.18 valuesSize()

```
abstract long com.choosemuse.libmuse.MuseDataPacket.valuesSize ( )  [abstract]
```

Get the number of values in this packet.

Returns

the number of data values in this packet.

The documentation for this class was generated from the following file:

- MuseDataPacket.java

8.33 com.choosemuse.libmuse.MuseDataPacketType Enum Reference

Public Attributes

- ACCELEROMETER
- GYRO
- EEG
- DROPPED_ACCELEROMETER
- DROPPED_EEG
- QUANTIZATION
- BATTERY
- DRL_REF
- ALPHA_ABSOLUTE
- BETA_ABSOLUTE
- DELTA_ABSOLUTE
- THETA_ABSOLUTE
- GAMMA_ABSOLUTE
- ALPHA_RELATIVE
- BETA_RELATIVE
- DELTA_RELATIVE
- THETA_RELATIVE
- GAMMA_RELATIVE
- ALPHA_SCORE
- BETA_SCORE
- DELTA_SCORE
- THETA_SCORE
- GAMMA_SCORE
- IS_GOOD
- HSI
- HSI_PRECISION
- ARTIFACTS
- MAGNETOMETER
- PRESSURE
- TEMPERATURE
- ULTRA_VIOLET
- NOTCH_FILTERED_EEG
- VARIANCE_EEG
- VARIANCE_NOTCH_FILTERED_EEG
- PPG
- IS_PPG_GOOD
- IS_HEART_GOOD
- THERMISTOR
- IS_THERMISTOR_GOOD
- AVG_BODY_TEMPERATURE
- CLOUD_COMPUTED
- OPTICS
- TOTAL

8.33.1 Detailed Description

This Enum represents all possible packet types. The type of the packet tells you information about what data it contains. When you know the packet type, look at the corresponding enum for information about data mapping (e.g.: Accelerometer enum, EEG enum, etc).

EEG derived signals

Absolute band powers The absolute band power for a given frequency range (for instance, alpha, i.e. 7.5-13Hz) is the logarithm of the sum of the Power Spectral Density of the EEG data over that frequency range. They are provided for each of the channels/electrode sites on Muse. Since it is a logarithm, some of the values will be negative (i.e. when the absolute power is less than 1). They are given on a log scale, units are Bels.

Relative band powers The relative band powers are calculated by dividing the absolute linear-scale power in one band over the sum of the absolute linear-scale powers in all bands. The linear-scale band power can be calculated from the log-scale band power.

linear-scale band power = 10^ (log-scale band power)

Therefore, the relative band powers can be calculated as percentages of linear-scale band powers in each band. For example:

alpha_relative = (10^alpha_absolute / (10^alpha_absolute + 10^beta_absolute + 10^delta_absolute + 10^gamma_absolute + 10^theta_absolute))

Band power scores The band session score is computed by comparing the current value of a band power to its history. This current value is mapped to a score between 0 and 1 using a linear function that returns 0 if the current value is equal to or below the 10th percentile of the distribution of band powers, and returns 1 if it's equal to or above the 90th percentile. Linear scoring between 0 and 1 is done for any value between these two percentiles.

Be advised that these scores are based on recent history and it will take a few seconds before having a stable distribution to score the power against. The estimated distribution is continuously updated as long as the headband is on the head. However, every time it's updated, the newest values are weighted to have more importance than the historical values. This means that eventually old values will not be present anymore in the estimated distribution. The half-life of the estimated distribution at any given point is around 10 seconds.

The score will start being calculated as soon as the headband has established a good connection with the skin. Whenever the headset loses connection with the head (as determined by the DRL/REF contact quality) the estimated distributions are reset. This means that when the headband is removed, the session data from any previous user will be cleared.

8.33.2 Member Data Documentation

8.33.2.1 ACCELEROMETER

`com.choosemuse.libmuse.MuseDataPacketType.ACCELEROMETER`

3-axis accelerometer data packet An Accelerometer packet provides 3 pieces of data.

See also

- Accelerometer for mapping details.

8.33.2.2 ALPHA_ABSOLUTE

`com.choosemuse.libmuse.MuseDataPacketType.ALPHA_ABSOLUTE`

EEG derived value. Absolute alpha band powers for each channel. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.3 ALPHA_RELATIVE

`com.choosemuse.libmuse.MuseDataPacketType.ALPHA_RELATIVE`

EEG derived value. Relative alpha band powers for each channel. Values in this packet are in range [0; 1]. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.4 ALPHA_SCORE

`com.choosemuse.libmuse.MuseDataPacketType.ALPHA_SCORE`

EEG derived value. Alpha band power scores for each channel. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.5 ARTIFACTS

`com.choosemuse.libmuse.MuseDataPacketType.ARTIFACTS`

Artifacts packet type will be sent.

Note that this will result in your listener receiving MuseArtifactPacket . We never emit a MuseDataPacket with MuseDataPacketType.ARTIFACTS ; this is only here for use in register / unregister methods.

8.33.2.6 AVG_BODY_TEMPERATURE

`com.choosemuse.libmuse.MuseDataPacketType.AVG_BODY_TEMPERATURE`

Thermistor derived value.

8.33.2.7 BATTERY

`com.choosemuse.libmuse.MuseDataPacketType.BATTERY`

Muse headband battery data packet. This packet provides 3 pieces of data.

See also

- Battery for mapping details.

8.33.2.8 BETA_ABSOLUTE

`com.choosemuse.libmuse.MuseDataPacketType.BETA_ABSOLUTE`

EEG derived value. Absolute beta band powers for each channel. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.9 BETA_RELATIVE

`com.choosemuse.libmuse.MuseDataPacketType.BETA_RELATIVE`

EEG derived value. Relative beta band powers for each channel. Values in this packet are in range [0; 1]. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.10 BETA_SCORE

`com.choosemuse.libmuse.MuseDataPacketType.BETA_SCORE`

EEG derived value. Beta band power scores for each channel. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.11 CLOUD_COMPUTED

`com.choosemuse.libmuse.MuseDataPacketType.CLOUD_COMPUTED`

Specifies a cloud computed value.

Values in this packet have been computed on the cloud remotely. This packet type allows the libmuse library to expose new packet type that has been generated remotely on the cloud.

8.33.2.12 DELTA_ABSOLUTE

`com.choosemuse.libmuse.MuseDataPacketType.DELTA_ABSOLUTE`

EEG derived value. Absolute delta band powers for each channel. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.13 DELTA_RELATIVE

`com.choosemuse.libmuse.MuseDataPacketType.DELTA_RELATIVE`

EEG derived value. Relative delta band powers for each channel. Values in this packet are in range [0; 1]. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.14 DELTA_SCORE

`com.choosemuse.libmuse.MuseDataPacketType.DELTA_SCORE`

EEG derived value. Delta band power scores for each channel. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.15 DRL_REF

`com.choosemuse.libmuse.MuseDataPacketType.DRL_REF`

Packet contains raw data from Muse DRL and REF sensors. This packet provides 2 pieces of data. The units of both values are in microvolts.

See also

- DrlRef for mapping details.

8.33.2.16 DROPPED_ACCELEROMETER

`com.choosemuse.libmuse.MuseDataPacketType.DROPPED_ACCELEROMETER`

Packet stands in for n dropped samples of the accelerometer type. Size of the values array for this packet is always 1.

Deprecated This is never emitted in an actual Muse session; instead, NaN-filled packets of the basic type (EEG or ACCELEROMETER) are emitted to stand in for dropped packets. This can only appear when reading Muse files written with older versions of the library.

8.33.2.17 DROPPED_EEG

`com.choosemuse.libmuse.MuseDataPacketType.DROPPED_EEG`

Packet stands in for n dropped samples of the eeg type. Size of the values array for this packet is always 1.

Deprecated This is never emitted in an actual Muse session; instead, NaN-filled packets of the basic type (EEG or ACCELEROMETER) are emitted to stand in for dropped packets. This can only appear when reading Muse files written with older versions of the library.

8.33.2.18 EEG

`com.choosemuse.libmuse.MuseDataPacketType.EEG`

Specifies raw EEG samples.

Values in this packet correspond to EEG data read from the different sensor locations on the headband. The accessors in the Eeg enum define the mapping from packet values to sensor locations. The units of EEG values are microvolts.

The size of the data is unspecified, but it is large enough to hold all the EEG channels emitted by the current preset.

In the future new Muse Presets may be added, which will have extra values.

See also

- Eeg for mapping details.

8.33.2.19 GAMMA_ABSOLUTE

`com.choosemuse.libmuse.MuseDataPacketType.GAMMA_ABSOLUTE`

EEG derived value. Absolute gamma band powers for each channel. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.20 GAMMA_RELATIVE

`com.choosemuse.libmuse.MuseDataPacketType.GAMMA_RELATIVE`

EEG derived value. Relative band powers for each channel. Values in this packet are in range [0; 1]. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.21 GAMMA_SCORE

`com.choosemuse.libmuse.MuseDataPacketType.GAMMA_SCORE`

EEG derived value. Gamma band power scores for each channel. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.22 GYRO

`com.choosemuse.libmuse.MuseDataPacketType.GYRO`

3-axis gyro data packet A Gyro packet provides 3 pieces of data.

See also

- Gyro for mapping details.

8.33.2.23 HSI

`com.choosemuse.libmuse.MuseDataPacketType.HSI`

EEG derived value. HSI values represent the fit of the headband. (known as headband status indicator). This value is not emitted by the LibMuse SDK.

8.33.2.24 HSI_PRECISION

`com.choosemuse.libmuse.MuseDataPacketType.HSI_PRECISION`

EEG derived value. HSI precision values represent the fit of the headband. This packet contains 4 values corresponding to Eeg.EEG1 , Eeg.EEG2 , Eeg.EEG3 and Eeg.EEG4 . There are no MuseDataPacketType.HSI_PRECISION values for the auxillary channels.

Each channel represents the fit at that location. A value of `1` represents a good fit, `2` represents a mediocre fit, and a value or `4` represents a poor fit.

See also

- Eeg for mapping details.

8.33.2.25 IS_GOOD

`com.choosemuse.libmuse.MuseDataPacketType.IS_GOOD`

EEG derived value. Is Good indicates whether or not the last 1 second of raw EEG data on each channel was good or not. Eye blinks or muscle movement can interfere with EEG data and cause Is Good to report that the data is not good. This is emitted every 1/10 of a second to represent the rolling window of the last second of EEG data. This is only useful for real time EEG analysis. This packet only contains 4 values for the 4 sensors on the headband, there is no support for the auxillary channels.

See also

- Eeg for mapping details.

8.33.2.26 IS_HEART_GOOD

`com.choosemuse.libmuse.MuseDataPacketType.IS_HEART_GOOD`

PPG derived value.

See also

- Ppg for mapping details.

8.33.2.27 IS_PPG_GOOD

`com.choosemuse.libmuse.MuseDataPacketType.IS_PPG_GOOD`

PPG derived value.

See also

- Ppg for mapping details.

8.33.2.28 IS_THERMISTOR_GOOD

`com.choosemuse.libmuse.MuseDataPacketType.IS_THERMISTOR_GOOD`

Thermistor derived value.

8.33.2.29 MAGNETOMETER

`com.choosemuse.libmuse.MuseDataPacketType.MAGNETOMETER`

3-axis magnetometer data packet A Magnetometer packet provides 3 pieces of data.

See also

- Magnetometer for mapping details.

8.33.2.30 NOTCH_FILTERED_EEG

`com.choosemuse.libmuse.MuseDataPacketType.NOTCH_FILTERED_EEG`

EEG derived value. Notch filtered EEG is the raw EEG passed through a band stop filter to remove frequencies between 45 and 65 Hz inclusive. This packet contains 4 values corresponding to Eeg.EEG1 , Eeg.EEG2 , Eeg.EEG3 and Eeg.EEG4 .

See also

- Eeg for mapping details.

8.33.2.31 OPTICS

`com.choosemuse.libmuse.MuseDataPacketType.OPTICS`

Specifies OPTICS samples.

Values in this packet correspond to optics data read from supported hardware. The accessors in the Optics enum define the mapping from packet data to 16 different values. The units of OPTICS values are microamps.

See also

- Optics for mapping details.

8.33.2.32 PPG

`com.choosemuse.libmuse.MuseDataPacketType.PPG`

Specifies PPG samples.

Values in this packet correspond to PPG data read from supported hardware. The accessors in the Ppg enum define the mapping from packet data to 3 different values Ambient, IR, and RED. The units of PPG values are arbitrary.

See also

- Ppg for mapping details.

8.33.2.33 PRESSURE

`com.choosemuse.libmuse.MuseDataPacketType.PRESSURE`

Pressure packet provides both a raw and averaged ambient pressure value.

See also

- Pressure for mapping details.

8.33.2.34 QUANTIZATION

`com.choosemuse.libmuse.MuseDataPacketType.QUANTIZATION`

Packet contains information about signal quantization. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

Each index in this packet corresponds to the same index in an EEG packet. Quantization occurs when there is a particularly noisy signal, which generally happens when there is not a good contact between the headband and the skin.

Higher numbers are worse; 1 is no quantization, and 16 is maximum quantization.

These values are used under the hood by the library and by Muse Elements in reconstructing the EEG signal and contributing to an overall measure of noise; it is extremely unlikely that you will be interested in them. For measuring noise, it is recommended to instead use the more useful computed values like 'headband_on' or 'hsi'.

Each quantization packet applies to the next 16 EEG packets.

See also

- Eeg for mapping details.

8.33.2.35 TEMPERATURE

`com.choosemuse.libmuse.MuseDataPacketType.TEMPERATURE`

Temperature packet provides ambient temperature value.

8.33.2.36 THERMISTOR

`com.choosemuse.libmuse.MuseDataPacketType.THERMISTOR`

Provides temperature values from the thermistor that is in contact with the user's skin.

8.33.2.37 THETA_ABSOLUTE

`com.choosemuse.libmuse.MuseDataPacketType.THETA_ABSOLUTE`

EEG derived value. Absolute theta band powers for each channel. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.38 THETA_RELATIVE

`com.choosemuse.libmuse.MuseDataPacketType.THETA_RELATIVE`

EEG derived value. Relative band powers for each channel. Values in this packet are in range [0; 1]. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.39 THETA_SCORE

`com.choosemuse.libmuse.MuseDataPacketType.THETA_SCORE`

EEG derived value. Theta band power scores for each channel. This packet contains the same amount of data as an EEG packet and has the same channel mapping.

See also

- Eeg for mapping details.

8.33.2.40 TOTAL

`com.choosemuse.libmuse.MuseDataPacketType.TOTAL`

The total number of possible data packet types

8.33.2.41 ULTRA_VIOLET

`com.choosemuse.libmuse.MuseDataPacketType.ULTRA_VIOLET`

UltraViolet packet provides both UVA and UVB index value and the average of the 2.

See also

- UltraViolet for mapping details.

8.33.2.42 VARIANCE_EEG

`com.choosemuse.libmuse.MuseDataPacketType.VARIANCE_EEG`

EEG derived value. Signal variance for raw EEG values. Variance is the numerical value that measures how widely a set of numbers within the interval are spread out from the average value. This packet contains the variance value of raw EEG over the last second.

See also

- Eeg for mapping details.
- https://en.wikipedia.org/wiki/Variance#Discrete_random_variable

8.33.2.43 VARIANCE_NOTCH_FILTERED_EEG

`com.choosemuse.libmuse.MuseDataPacketType.VARIANCE_NOTCH_FILTERED_EEG`

EEG derived value. Signal variance for notch filtered EEG. Variance is the numerical value that measures how widely a set of numbers within the interval are spread out from the average value. This packet contains the variance value of notch filtered EEG over the last second.

See also

- Eeg for mapping details.
- https://en.wikipedia.org/wiki/Variance#Discrete_random_variable

The documentation for this enum was generated from the following file:

- MuseDataPacketType.java

8.34 com.choosemuse.libmuse.MuseErrorListener Class Reference

Public Member Functions

- abstract void receiveError (Error error, Muse muse)

8.34.1 Detailed Description

Listens for Muse Error packets.

8.34.2 Member Function Documentation

8.34.2.1 receiveError()

```
abstract void com.choosemuse.libmuse.MuseErrorListener.receiveError (
            Error error,
            Muse muse )  [abstract]
```

Handler method for Muse error packets. Packet is sent iff muse receives an exception.

Generally if an error occurred, LogListener.receiveLog() will be called with a LogPacket of "error" severity. It is therefore not usually necessary or advisable to print log messages for errors received by this method. Rather, this handler should take actions like alerting users or deciding to end a session.

Warning

> It is important that you do not perform any computation intensive tasks in this callback. This would result in significant delays in all the listener callbacks from being called. You should delegate any intensive tasks to another thread or schedule it to run with a delay through handler/scheduler for the platform.

However, you can register/unregister listeners in this callback. All previously registered listeners would still receive callbacks for this current event. On subsequent events, the newly registered listeners will be called. For example, if you had 2 listeners 'A' and 'B' for this event. If, on the callback for listener A, listener A unregisters all listeners and registers a new listener 'C' and then in the callback for listener 'B', you unregister all listeners again and register a new listener 'D'. Then on the subsequent event callback, only listener D's callback will be invoked.

Parameters

error	The error encountered.
muse	The Muse that generated the error.

The documentation for this class was generated from the following file:

- MuseErrorListener.java

8.35 com.choosemuse.libmuse.MuseFile Class Reference

Public Member Functions

- abstract boolean open (boolean forWriting)
- abstract boolean write (byte[ ] buffer)
- abstract byte[ ] read (int length)
- abstract boolean close (boolean forWriting)

8.35.1 Detailed Description

A File interface. Normally you will not need to use this class directly as you can get a MuseFileReader or MuseFileWriter using MuseFileFactory .
If you provide your own implementation of this class, you should provide a way to specify the path to the file you want to access in a platform specific way.

Threading: MuseFile is not thread safe by default, but MuseFileWriter provides the necessary synchronization for its usage. If you are using MuseFile outside of that context, make sure the calls to the MuseFile.open() , MuseFile.write() and MuseFile.close() functions are thread-safe.

8.35.2 Member Function Documentation

8.35.2.1 close()

```
abstract boolean com.choosemuse.libmuse.MuseFile.close (
            boolean forWriting )  [abstract]
```

Closes opened file.
This method is called when MuseFileWriter or MuseFileReader is destroyed, or when close is called on either class.

Parameters

forWriting	`true` if the file should be closed for writing. `false` if for reading.

Returns

`true` if file is closed properly, `false` otherwise.

8.35.2.2 open()

```
abstract boolean com.choosemuse.libmuse.MuseFile.open (
            boolean forWriting )  [abstract]
```

Opens/creates the file. You can open same file for both writing and reading if you really (!) have to do so. Provide synchronization in this case. This method is called automatically when MuseFileWriter or MuseFileReader is created.

Parameters

forWriting	`true` if the file should be opened for writing. `false` if for reading.

Returns

`true` when file is opened successfully, `false` otherwise.

8.35.2.3 read()

```
abstract byte [] com.choosemuse.libmuse.MuseFile.read (
            int length )  [abstract]
```

Reads "length" bytes from file starting from current pointer position.

Parameters

length	The number of bytes to read.

Returns

A byte array containing the data read. An empty buffer is returned if the file is empty or if the end of file was reached.

8.35.2.4 write()

```
abstract boolean com.choosemuse.libmuse.MuseFile.write (
            byte[] buffer )  [abstract]
```

Writes buffer to file.
This method is called when you flush MuseFileWriter .

Parameters

buffer	The data to write.

Returns

`true` when buffer is written to successfully, `false` otherwise.

The documentation for this class was generated from the following file:

- MuseFile.java

8.36 com.choosemuse.libmuse.MuseFileFactory Class Reference

Static Public Member Functions

- static MuseFileWriter getMuseFileWriter (File file)
- static MuseFileReader getMuseFileReader (File file)
- static MuseFile getMuseFile (File file)

8.36.1 Detailed Description

Creates MuseFileWriter, MuseFileReader and MuseFile objects.

8.36.2 Member Function Documentation

8.36.2.1 getMuseFile()

```
static MuseFile com.choosemuse.libmuse.MuseFileFactory.getMuseFile (
            File file )  [inline], [static]
```

Constructs and returns MuseFile object, which uses Interaxon's implementation

Parameters

file	Reference to a File object, which will be used to read or store data.

Returns

MuseFile

8.36.2.2 getMuseFileReader()

```
static MuseFileReader com.choosemuse.libmuse.MuseFileFactory.getMuseFileReader (
            File file )  [inline], [static]
```

Constructs and returns MuseFileReader object

Parameters

file	Reference to a File object, which will be used to read data.

Returns

MuseFileReader

8.36.2.3 getMuseFileWriter()

```
static MuseFileWriter com.choosemuse.libmuse.MuseFileFactory.getMuseFileWriter (
            File file )  [inline], [static]
```

Constructs and returns MuseFileWriter object.

Note that upon creation of MuseFileWriter, an Annotation is automatically written out to the file. The annotation contains the app's name and version and libmuse version. If app's name and version can not be determined, they will be empty strings.

Parameters

file	Reference to a File object, which will be used to store data.

Returns

MuseFileWriter

The documentation for this class was generated from the following file:

- MuseFileFactory.java

8.37 com.choosemuse.libmuse.MuseFileReader Class Reference

Public Member Functions

- abstract boolean open ()
- abstract boolean close ()
- abstract Result gotoNextMessage ()
- abstract MessageType getMessageType ()
- abstract int getMessageId ()
- abstract long getMessageTimestamp ()
- abstract AnnotationData getAnnotation ()
- abstract MuseConfiguration getConfiguration ()
- abstract MuseVersion getVersion ()
- abstract ComputingDeviceConfiguration getComputingDeviceConfiguration ()
- abstract DspData getDsp ()
- abstract MuseDataPacket getDataPacket ()
- abstract MuseArtifactPacket getArtifactPacket ()

Static Public Member Functions

- static native MuseFileReader getFileReader (MuseFile file)

8.37.1 Detailed Description

Reads a .muse file formatted according to the Muse protobuf schema.

MuseFileReader is very similar to ReaderMuse as both classes allow you to parse a file and generate data packets from it. In most instances, ReaderMuse is the better choice at it abstracts the details of the parsing of the file and allows you to handle the data the same way that you would if you were connected to an actual Muse . Use MuseFileReader if you want finer control of the parsing of the file.

When the file reader is created it always starts reading from the beginning of the file to the end.

Threading: It is NOT thread safe, you have to provide your own synchronization mechanism if you plan to calls methods of this class from different threads.

8.37.2 Member Function Documentation

8.37.2.1 close()

```
abstract boolean com.choosemuse.libmuse.MuseFileReader.close ( )  [abstract]
```

Closes the file. Calls MuseFile.close() . You don't have to call close explicitly unless you want to close file immediately. close is called automatically, when the file reader object is destroyed.

8.37.2.2 getAnnotation()

```
abstract AnnotationData com.choosemuse.libmuse.MuseFileReader.getAnnotation ( )  [abstract]
```

Returns annotation data at the current position in the file.

Returns

The annotation data at the current position in the file.

Exceptions

IncorrectMessageType	If current message type is not MessageType.ANNOTATION

8.37.2.3 getArtifactPacket()

```
abstract MuseArtifactPacket com.choosemuse.libmuse.MuseFileReader.getArtifactPacket ( )  [abstract]
```

Returns the muse artifact packet data at the current position in the file.

Returns

The muse artifact packet at the current position in the file.

Exceptions

IncorrectMessageType	If current message type is not MessageType.ARTIFACT ,

8.37.2.4 getComputingDeviceConfiguration()

```
abstract ComputingDeviceConfiguration com.choosemuse.libmuse.MuseFileReader.getComputing↩
```

```
DeviceConfiguration ( )  [abstract]
```

Returns computing device configuration data at the current position in the file.

Returns

The device configuration data at the current position in the file.

Exceptions

IncorrectMessageType	If current message type is not MessageType.COMPUTING_DEVICE

8.37.2.5 getConfiguration()

```
abstract MuseConfiguration com.choosemuse.libmuse.MuseFileReader.getConfiguration ( )  [abstract]
```

Returns muse configuration data at the current position in the file.

Returns

The configuration data at the current position in the file.

Exceptions

IncorrectMessageType	If current message type is not MessageType.CONFIGURATION

8.37.2.6 getDataPacket()

```
abstract MuseDataPacket com.choosemuse.libmuse.MuseFileReader.getDataPacket ( )  [abstract]
```

Returns the muse data packet data at the current position in the file. Use this method to get EEG, Accelerometer, Battery and Quantization packets.

Returns

The muse data packet data at the current position in the file.

Exceptions

IncorrectMessageType	If current message type is not one of: MessageType.ACCELEROMETER , MessageType.ACC_DROPPED , MessageType.BATTERY , MessageType.EEG , MessageType.EEG_DROPPED , MessageType.GYRO , MessageType.MUSE_ELEMENTS or MessageType.QUANTIZATION

8.37.2.7 getDsp()

```
abstract DspData com.choosemuse.libmuse.MuseFileReader.getDsp ( )  [abstract]
```

Returns the dsp data at the current position in the file.

Returns

The dsp data at the current position in the file.

Exceptions

IncorrectMessageType	If current message type is not MessageType.DSP

8.37.2.8 getFileReader()

```
static native MuseFileReader com.choosemuse.libmuse.MuseFileReader.getFileReader (
            MuseFile file )  [static]
```

Returns an instance of file reader. Automatically opens a file.

Returns

a MuseFileReader instance.

8.37.2.9 getMessageId()

```
abstract int com.choosemuse.libmuse.MuseFileReader.getMessageId ( )  [abstract]
```

Returns the id of the message at the current position in the file or `-1` if the id isn't found in the protobuf specification.

Returns

The id of the message at the current position in the file or `-1` if the id isn't found in the protobuf specification.

8.37.2.10 getMessageTimestamp()

```
abstract long com.choosemuse.libmuse.MuseFileReader.getMessageTimestamp ( )  [abstract]
```

Returns the timestamp of the message at the current position in the file.

Returns

The timestamp of the message.

8.37.2.11 getMessageType()

```
abstract MessageType com.choosemuse.libmuse.MuseFileReader.getMessageType ( )  [abstract]
```

Returns the type of message at the current position in the file.

Returns

The type of message at the current position in the file.

8.37.2.12 getVersion()

```
abstract MuseVersion com.choosemuse.libmuse.MuseFileReader.getVersion ( )  [abstract]
```

Returns muse version data at the current position in the file.

Returns

The version data at the current position in the file.

Exceptions

IncorrectMessageType	If current message type is not MessageType.VERSION

8.37.2.13 gotoNextMessage()

```
abstract Result com.choosemuse.libmuse.MuseFileReader.gotoNextMessage ( )  [abstract]
```

Reads the next message in the protobuf stream.

Returns

The Result object with the status of the read operation.

8.37.2.14 open()

```
abstract boolean com.choosemuse.libmuse.MuseFileReader.open ( )  [abstract]
```

Opens an existing file Use this method if you explicitly closed file and want to open it again. Calls MuseFile.open() .

The documentation for this class was generated from the following file:

- MuseFileReader.java

8.38 com.choosemuse.libmuse.MuseFileWriter Class Reference

Public Member Functions

- abstract boolean open ()
- abstract boolean isOpen ()
- abstract boolean close ()
- abstract void discardBufferedPackets ()
- abstract boolean flush ()
- abstract int getBufferredMessagesCount ()
- abstract int getBufferedMessagesSize ()
- abstract long getTotalBytesWritten ()
- abstract void addArtifactPacket (int id, MuseArtifactPacket packet)
- abstract void addDataPacket (int id, MuseDataPacket packet)
- abstract void addAnnotationString (int id, String annotation)
- abstract void addAnnotation (int id, AnnotationData annotation)
- abstract void addConfiguration (int id, MuseConfiguration configuration)
- abstract void addVersion (int id, MuseVersion version)
- abstract void addComputingDeviceConfiguration (int id, ComputingDeviceConfiguration configuration)
- abstract void addDsp (int id, DspData dsp)
- abstract void setTimestampMode (TimestampMode mode)
- abstract void setTimestamp (long timestamp)

Static Public Member Functions

- static native MuseFileWriter getFileWriter (MuseFile file)

8.38.1 Detailed Description

This class manages saving Muse packets and other data into a file, which can later be read and replayed by MusePlayer. Google Protobuf is used to store the information. For better control, data is not written to the file immediately, but stored in a buffer instead. The buffer then can be flushed to a file or discarded.

Note that upon creation of a MuseFileWriter , an Annotation and the ComputingDeviceConfiguration are automatically written out to the file. The annotation contains the app's name and version and libmuse version. If app's name and version can not be determined, they will be empty strings. The ComputingDeviceConfiguration contains hardware information for the computing device. If any information fields can not be determined, they will be empty strings.

Threading: It is thread safe, so you can call write()/get()/flush()/discard() operations from different threads.

Warning

If you don't call flush() method, messages will be accumulated in memory and eventually you will get memory overflow error. It's library client responsibility to call flush() or discardBufferedPackets() and to clean the memory. You can use helper methods to get how many messages are currently stored in the buffer and how much memory is used.

8.38.2 Member Function Documentation

8.38.2.1 addAnnotation()

```
abstract void com.choosemuse.libmuse.MuseFileWriter.addAnnotation (
            int id,
            AnnotationData annotation )  [abstract]
```

Adds annotation data structure to the buffer. Similar to addAnnotationString(), but this method allows you to extra fields. Empty fields in annotation struct won't be added to protobuf. If annotation.data field is empty, method returns immediately.

Parameters

id	The id of the device generating the annotation. This is arbitrary and is used to differentiate data from multiple devices in a single file.
annotation	The annotation data to add.

8.38.2.2 addAnnotationString()

```
abstract void com.choosemuse.libmuse.MuseFileWriter.addAnnotationString (
            int id,
            String annotation )  [abstract]
```

Adds an annotation string to the buffer. It may be useful if you want to log specific events. This is a simplified version of addAnnotation(). Use this method if you want to add one string to protobuf. Use addAnnotation() for more advanced options. This function does nothing if annotation is empty.

Parameters

id	The id of the device generating the packet. This is arbitrary and is used to differentiate data from multiple devices in a single file.
annotation	The annotation string to add.

8.38.2.3 addArtifactPacket()

```
abstract void com.choosemuse.libmuse.MuseFileWriter.addArtifactPacket (
            int id,
            MuseArtifactPacket packet )  [abstract]
```

Adds MuseArtifactPacket to the buffer.

Parameters

id	The id of the device generating the packet. This is arbitrary and is used to differentiate data from multiple devices in a single file.
packet	The artifact packet to add.

8.38.2.4 addComputingDeviceConfiguration()

```
abstract void com.choosemuse.libmuse.MuseFileWriter.addComputingDeviceConfiguration (
            int id,
            ComputingDeviceConfiguration configuration )  [abstract]
```

Adds information about the running device to the buffer.

Parameters

id	The id of the device saving the configuration. This is arbitrary and is used to differentiate data from multiple devices in a single file.
configuration	The device configuration data to add.

8.38.2.5 addConfiguration()

```
abstract void com.choosemuse.libmuse.MuseFileWriter.addConfiguration (
            int id,
            MuseConfiguration configuration )  [abstract]
```

Adds MuseConfiguration to the buffer.

Parameters

id	The id of the device saving the configuration. This is arbitrary and is used to differentiate data from multiple devices in a single file.
configuration	The configuration data to add.

8.38.2.6 addDataPacket()

```
abstract void com.choosemuse.libmuse.MuseFileWriter.addDataPacket (
            int id,
            MuseDataPacket packet )  [abstract]
```

Adds MuseDataPacket to the buffer. All current packets are supported.

Parameters

id	The id of the device generating the packet. This is arbitrary and is used to differentiate data from multiple devices in a single file.
packet	The data packet to add.

8.38.2.7 addDsp()

```
abstract void com.choosemuse.libmuse.MuseFileWriter.addDsp (
            int id,
            DspData dsp )  [abstract]
```

With this method you can save your custom data.

Parameters

id	The id of the device saving the data. This is arbitrary and is used to differentiate data from multiple devices in a single file.
dsp	The data to store.

8.38.2.8 addVersion()

```
abstract void com.choosemuse.libmuse.MuseFileWriter.addVersion (
            int id,
            MuseVersion version )  [abstract]
```

Adds MuseVersion to the buffer.

Parameters

id	The id of the device saving the version. This is arbitrary and is used to differentiate data from multiple devices in a single file.
version	The version data to add.

8.38.2.9 close()

```
abstract boolean com.choosemuse.libmuse.MuseFileWriter.close ( )  [abstract]
```

Closes the file. Calls MuseFile.close() You don't have to call close explicitly unless you want to close file immediately. close is called automatically, when the file writer object is destroyed.

Returns

`false` if the file could not be closed for any reason. `true` otherwise.

8.38.2.10 discardBufferedPackets()

```
abstract void com.choosemuse.libmuse.MuseFileWriter.discardBufferedPackets ( )  [abstract]
```

Removes all saved messages from the memory.

8.38.2.11 flush()

```
abstract boolean com.choosemuse.libmuse.MuseFileWriter.flush ( )  [abstract]
```

Flashes saved messages to the disk. Calls MuseFile.write()

Returns

`false` if can not write to file for any reason. `true` otherwise.

8.38.2.12 getBufferedMessagesSize()

```
abstract int com.choosemuse.libmuse.MuseFileWriter.getBufferedMessagesSize ( )  [abstract]
```

Returns the size of saved messages in bytes.

Returns

the size of the saved messages in bytes.

8.38.2.13 getBufferredMessagesCount()

```
abstract int com.choosemuse.libmuse.MuseFileWriter.getBufferredMessagesCount ( )  [abstract]
```

Returns number of saved messages

Returns

the number of saved messages.

8.38.2.14 getFileWriter()

```
static native MuseFileWriter com.choosemuse.libmuse.MuseFileWriter.getFileWriter (
            MuseFile file )  [static]
```

Returns an instance of file writer. Automatically opens a file.

Returns

an instance of the file writer.

8.38.2.15 getTotalBytesWritten()

```
abstract long com.choosemuse.libmuse.MuseFileWriter.getTotalBytesWritten ( )  [abstract]
```

Returns the total number of bytes written to the file.

Returns

the total number of bytes written to the file.

8.38.2.16 isOpen()

```
abstract boolean com.choosemuse.libmuse.MuseFileWriter.isOpen ( )  [abstract]
```

Checks if the file writer is open.

Returns

`true` when file writer is open, `false` otherwise.

8.38.2.17 open()

```
abstract boolean com.choosemuse.libmuse.MuseFileWriter.open ( )  [abstract]
```

Opens a file if it exists or creates a new one. If file already exists, all new messages will be appended at the end. Use this method if you explicitly closed file and want to open it again. Calls MuseFile.open()

Returns

`false` if the file could not be opened for any reason. `true` otherwise.

8.38.2.18 setTimestamp()

```
abstract void com.choosemuse.libmuse.MuseFileWriter.setTimestamp (
            long timestamp )  [abstract]
```

Set the timestamp for the next message to be written.

This only does anything in explicit timestamp mode. In explicit mode, the timestamp set by set_timestamp applies to the next message and all subsequent messages. If set_timestamp is not called, then the timestamp is 0.

In all other modes, calling set_timestamp causes an assert failure – the app crashes in debug mode, and it is a noop in release mode.

The timestamp is in microseconds since some epoch (usually Jan 1, 1970).

Parameters

timestamp	The time to use for the timestamp in microseconds.

8.38.2.19 setTimestampMode()

```
abstract void com.choosemuse.libmuse.MuseFileWriter.setTimestampMode (
            TimestampMode mode )  [abstract]
```

Set the timestamp mode.

By default, the mode is TimestampMode.LEGACY

Parameters

mode	The timestamp mode to set.

The documentation for this class was generated from the following file:

- MuseFileWriter.java

8.39 com.choosemuse.libmuse.MuseListener Class Reference

Public Member Functions

- abstract void museListChanged ()

8.39.1 Detailed Description

Implement this interface to receive callbacks whenever a Muse headband is discovered after calling MuseManager.startListening()

8.39.2 Member Function Documentation

8.39.2.1 museListChanged()

```
abstract void com.choosemuse.libmuse.MuseListener.museListChanged ( )  [abstract]
```

Called when the list of Muses detected by MuseManager has changed.

- You will receive 1 callback for each headband discovered and added to the list. For Muse 2014 (MU_01) headbands, there will be a callback for each headband paired with the device even if the headband is powered off. MU_01) headbands, there will be a callback for each headband paired with the device only when the headband is powered on and connected to the device. MU_01) and Muse 2016 (MU_02) or later headbands, there will be a callback for each headband paired with the system even if the headband is powered off.
- You will receive 1 callback for each removal operation. If more than 1 headband is removed as the result of a single removal operation, you will only receive 1 callback instead of 1 callback for each headband.

The documentation for this class was generated from the following file:

- MuseListener.java

8.40 com.choosemuse.libmuse.MuseLog Class Reference

Static Public Member Functions

- static void r (String message)
- static void v (String message)
- static void i (String message)
- static void w (String message)
- static void e (String message)
- static void f (String message)
- static void d (String message)
- static void v (String message, Throwable t)
- static void i (String message, Throwable t)
- static void w (String message, Throwable t)
- static void e (String message, Throwable t)
- static void f (String message, Throwable t)
- static void d (String message, Throwable t)
- static void log (Severity sev, String message)
- static void log (Severity severity, String message, Throwable t)
- static void log (Severity severity, String message, boolean raw)

8.40.1 Detailed Description

A wrapper around android.util.Log which supports raw messages and uses the tag "MUSE".

8.40.2 Member Function Documentation

8.40.2.1 d() [1/2]

```
static void com.choosemuse.libmuse.MuseLog.d (
            String message )  [inline], [static]
```

Logs a debug message.

Parameters

message	The log message.

8.40.2.2 d() [2/2]

```
static void com.choosemuse.libmuse.MuseLog.d (
            String message,
            Throwable t )  [inline], [static]
```

Logs a debug message.

Parameters

message	The log message.
t	A Throwable object. The stack trace of the Throwable will be printed to the log.

8.40.2.3 e() [1/2]

```
static void com.choosemuse.libmuse.MuseLog.e (
            String message )  [inline], [static]
```

Logs an error message.

Parameters

message	The log message.

8.40.2.4 e() [2/2]

```
static void com.choosemuse.libmuse.MuseLog.e (
            String message,
            Throwable t )  [inline], [static]
```

Logs an error message.

Parameters

message	The log message.
t	A Throwable object. The stack trace of the Throwable will be printed to the log.

8.40.2.5 f() [1/2]

```
static void com.choosemuse.libmuse.MuseLog.f (
            String message )  [inline], [static]
```

Logs a fatal error message. The application will abort after the message is logged.

Parameters

message	The log message.

Exceptions

SIGABRT	After the message is logged.

8.40.2.6 f() [2/2]

```
static void com.choosemuse.libmuse.MuseLog.f (
            String message,
            Throwable t )  [inline], [static]
```

Logs a fatal error message. The application will abort after the message is logged.

Parameters

message	The log message.
t	A Throwable object. The stack trace of the Throwable will be printed to the log.

Exceptions

SIGABRT	After the message is logged.

8.40.2.7 i() [1/2]

```
static void com.choosemuse.libmuse.MuseLog.i (
            String message )  [inline], [static]
```

Logs an informational message.

Parameters

message	The log message.

8.40.2.8 i() [2/2]

```
static void com.choosemuse.libmuse.MuseLog.i (
            String message,
            Throwable t )  [inline], [static]
```

Logs an informational message.

Parameters

message	The log message.
t	A Throwable object. The stack trace of the Throwable will be printed to the log.

8.40.2.9 log() [1/3]

```
static void com.choosemuse.libmuse.MuseLog.log (
            Severity sev,
            String message )  [inline], [static]
```

Logs a message with the given severity.

Parameters

sev	The severity of the message.
message	The log message.

8.40.2.10 log() [2/3]

```
static void com.choosemuse.libmuse.MuseLog.log (
            Severity severity,
            String message,
            boolean raw )  [inline], [static]
```

Logs a message with the given severity.

Parameters

severity	The severity of the message.
message	The log message.
raw	`true` if this is a raw log message. `false` otherwise.

8.40.2.11 log() [3/3]

```
static void com.choosemuse.libmuse.MuseLog.log (
            Severity severity,
            String message,
            Throwable t )  [inline], [static]
```

Logs a message with the given severity.

Parameters

severity	The severity of the message.
message	The log message.
t	A Throwable object. The stack trace of the Throwable will be printed to the log.

8.40.2.12 r()

```
static void com.choosemuse.libmuse.MuseLog.r (
            String message )  [inline], [static]
```

Logs a raw message at verbose severity.

Parameters

message	The log message.

8.40.2.13 v() [1/2]

```
static void com.choosemuse.libmuse.MuseLog.v (
            String message )  [inline], [static]
```

Logs a verbose message.

Parameters

message	The log message.

8.40.2.14 v() [2/2]

```
static void com.choosemuse.libmuse.MuseLog.v (
            String message,
            Throwable t )  [inline], [static]
```

Logs a verbose message.

Parameters

message	The log message.
t	A Throwable object. The stack trace of the Throwable will be printed to the log.

8.40.2.15 w() [1/2]

```
static void com.choosemuse.libmuse.MuseLog.w (
            String message )  [inline], [static]
```

Logs a warning message.

Parameters

message	The log message.

8.40.2.16 w() [2/2]

```
static void com.choosemuse.libmuse.MuseLog.w (
            String message,
            Throwable t )  [inline], [static]
```

Logs a warning message.

Parameters

message	The log message.
t	A Throwable object. The stack trace of the Throwable will be printed to the log.

The documentation for this class was generated from the following file:

- MuseLog.java

8.41 com.choosemuse.libmuse.MuseManager Class Reference

Inheritance diagram for com.choosemuse.libmuse.MuseManager:

Public Member Functions

- abstract ArrayList< Muse > getMuses ()
- abstract void startListening ()
- abstract void stopListening ()
- abstract void setMuseListener (MuseListener listener)
- abstract AdvertisingStats getAdvertisingStats (Muse m)
- abstract void resetAdvertisingStats ()
- abstract void removeFromListAfter (long time)

Static Public Attributes

- static final long DEFAULT_REMOVE_FROM_LIST_AFTER = 30l

8.41.1 Detailed Description

Discovers the available Muse headbands that this device can connect to.

To connect to a headband, first call MuseManager.setMuseListener() to receive callbacks when a headband is discovered. Then call MuseManager.startListening() When MuseManager detects a headband you will receive a MuseListener.museListChanged() callback. You can then call MuseManager.getMuses() followed by the appropriate function on the Muse to connect to the headband.

See also

Muse

MuseListener

8.41.2 Member Function Documentation

8.41.2.1 getAdvertisingStats()

```
abstract AdvertisingStats com.choosemuse.libmuse.MuseManager.getAdvertisingStats (
            Muse m )  [abstract]
```

Returns information about the advertising packets seen by LibMuse. This is intended for internal usage at Interaxon and is currently only implemented on Android for Muse 2016 (MU_02) or later. This function does nothing on iOS or Windows.

Reimplemented in com.choosemuse.libmuse.MuseManagerAndroid.

8.41.2.2 getMuses()

```
abstract ArrayList<Muse> com.choosemuse.libmuse.MuseManager.getMuses ( )  [abstract]
```

Returns all currently available Muse headbands.

The returned list is sorted lexicographically by Muse name.

Note that until you start listening, this list will be empty.

Returns

The list of available muses.

Reimplemented in com.choosemuse.libmuse.MuseManagerAndroid.

8.41.2.3 removeFromListAfter()

```
abstract void com.choosemuse.libmuse.MuseManager.removeFromListAfter (
            long time )  [abstract]
```

MuseManager will automatically remove a Muse 2016 (MU_02) or later headband from the list of Muses if it has not received some type of communication from the headband within this time period.

By default this is MuseManager.DEFAULT_REMOVE_FROM_LIST_AFTER. To have Muse 2016 (MU_02) or later headbands remain in the list until the next call to MuseManager.startListening() set the time to 0.

Parameters

time	Specified in seconds, the time to wait before removing a headband from the list. If set to 0, headbands will not be removed automatically.

Reimplemented in com.choosemuse.libmuse.MuseManagerAndroid.

8.41.2.4 resetAdvertisingStats()

```
abstract void com.choosemuse.libmuse.MuseManager.resetAdvertisingStats ( )  [abstract]
```

Erases all previously stored information about advertising packets. This is intended for internal usage at Interaxon and is currently only implemented Muse 2016 (MU_02) or later. This function does nothing on iOS.

Reimplemented in com.choosemuse.libmuse.MuseManagerAndroid.

8.41.2.5 setMuseListener()

```
abstract void com.choosemuse.libmuse.MuseManager.setMuseListener (
            MuseListener listener )  [abstract]
```

Set the listener to be notified on changes to the list of Muses.

Parameters

listener	The listener that will receive the callback when a Muse is discovered.

Reimplemented in com.choosemuse.libmuse.MuseManagerAndroid.

8.41.2.6 startListening()

```
abstract void com.choosemuse.libmuse.MuseManager.startListening ( )  [abstract]
```

Start listening for any possible Muse devices.

As Muses are discovered, they are added to the list returned by MuseManager.getMuses() Listening is a fairly heavyweight operation, so it should be disabled when it is not needed, e.g. after a device has been found. Specifically, you *must* call stopListening() before you try to connect to a Muse you have previously discovered.

You must call MuseManagerAndroid.setContext() prior to calling startListening() or stopListening(), and the context should persist at least from before the call to startListening() till after the call to stopListening(). An application context is a good candidate for this.

Reimplemented in com.choosemuse.libmuse.MuseManagerAndroid.

8.41.2.7 stopListening()

```
abstract void com.choosemuse.libmuse.MuseManager.stopListening ( )  [abstract]
```

Stop listening for Muse devices.

Stops receiving advertising packets or modifying the list of Muses returned by MuseManager.getMuses() If this is called while not scanning, it is a no-op.

You must call MuseManagerAndroid.setContext() prior to calling startListening() or stopListening(), and the context should persist at least from before the call to startListening() till after the call to stopListening(). An application context is a good candidate for this.

Reimplemented in com.choosemuse.libmuse.MuseManagerAndroid.

8.41.3 Member Data Documentation

8.41.3.1 DEFAULT_REMOVE_FROM_LIST_AFTER

```
final long com.choosemuse.libmuse.MuseManager.DEFAULT_REMOVE_FROM_LIST_AFTER = 30l  [static]
```

The default time in seconds after which a headband will be removed from the list of muses if MuseManager has had no contact with it.

This has a value of 30 seconds.

The documentation for this class was generated from the following file:

- MuseManager.java

8.42 com.choosemuse.libmuse.MuseManagerAndroid Class Reference

Inheritance diagram for com.choosemuse.libmuse.MuseManagerAndroid:

Collaboration diagram for com.choosemuse.libmuse.MuseManagerAndroid:

Classes

- class **DiscoveryStats**

Public Member Functions

- BluetoothAdapter getDefaultAdapter ()
- ArrayList< Muse > getMuses ()
- void setContext (Context context)
- void setCompanionMode (boolean enable, String macAddress, byte[] manufacturerData, boolean singleDevice)
- void handleActivityResult (int requestCode, int resultCode, Intent data)
- void setMuseListener (MuseListener listener)
- void startListening ()
- void stopListening ()
- void removeFromListAfter (long time)
- AdvertisingStats getAdvertisingStats (Muse m)
- void resetAdvertisingStats ()

Static Public Member Functions

- static synchronized MuseManagerAndroid getInstance ()
- static UUID museUuid ()

Static Public Attributes

- static int SELECT_DEVICE_REQUEST_CODE = 100

8.42.1 Detailed Description

Provides access to all Muse devices paired to this device.

This class provides access to the Muse object associated with each Muse Headband the mobile device knows about. The recommended usage is to call getPairedMuses() and find the devices you want.

For applications that are designed only to work with a single headband, there are a few possibilities: you can display a list of names of all paired Muses in your application and have the user select the one they'd like to use; you may also save the MAC address of the last used Muse and use that as a default; or you can try connecting to each Muse in the list until you find one that works.

8.42.2 Member Function Documentation

8.42.2.1 getAdvertisingStats()

```
AdvertisingStats com.choosemuse.libmuse.MuseManagerAndroid.getAdvertisingStats (
            Muse m )  [inline]
```

Returns information about the advertising packets seen by LibMuse. This is intended for internal usage at Interaxon and is currently only implemented on Android for Muse 2016 (MU_02) or later. This function does nothing on iOS or Windows. each headband after the test is over. Know your device - if the device suffers from

Reimplemented from com.choosemuse.libmuse.MuseManager.

8.42.2.2 getDefaultAdapter()

```
BluetoothAdapter com.choosemuse.libmuse.MuseManagerAndroid.getDefaultAdapter ( )  [inline]
```

Returns the default Bluetooth Adapter.

Returns

the default Bluetooth Adapter.

8.42.2.3 getInstance()

```
static synchronized MuseManagerAndroid com.choosemuse.libmuse.MuseManagerAndroid.getInstance (
) [inline], [static]
```

Returns the instance of the MuseManager.

Returns

the instance of the MuseManager.

8.42.2.4 getMuses()

```
ArrayList<Muse> com.choosemuse.libmuse.MuseManagerAndroid.getMuses ( ) [inline]
```

Returns all currently available Muse headbands.

The returned list is sorted lexicographically by Muse name.

Note that until you start listening, this list will be empty.

Returns

The list of available muses.

Reimplemented from com.choosemuse.libmuse.MuseManager.

8.42.2.5 handleActivityResult()

```
void com.choosemuse.libmuse.MuseManagerAndroid.handleActivityResult (
            int requestCode,
            int resultCode,
            Intent data ) [inline]
```

Handle activity result from the CompanionDeviceManager. This is used to retrieve scan results when companion mode is enabled.

8.42.2.6 museUuid()

```
static UUID com.choosemuse.libmuse.MuseManagerAndroid.museUuid ( ) [inline], [static]
```

Returns the UUID of the Muse Headband.

Returns

the UUID of the Muse Headband.

8.42.2.7 removeFromListAfter()

```
void com.choosemuse.libmuse.MuseManagerAndroid.removeFromListAfter (
            long time ) [inline]
```

MuseManager will automatically remove a Muse 2016 (MU_02) or later headband from the list of Muses if it has not received some type of communication from the headband within this time period.

By default this is MuseManager.DEFAULT_REMOVE_FROM_LIST_AFTER. To have Muse 2016 (MU_02) or later headbands remain in the list until the next call to MuseManager.startListening() set the time to 0.

Parameters

time	Specified in seconds, the time to wait before removing a headband from the list. If set to 0, headbands will not be removed automatically.

Reimplemented from com.choosemuse.libmuse.MuseManager.

8.42.2.8 resetAdvertisingStats()

```
void com.choosemuse.libmuse.MuseManagerAndroid.resetAdvertisingStats ( )  [inline]
```

Erases all previously stored information about advertising packets. This is intended for internal usage at Interaxon and is currently only implemented Muse 2016 (MU_02) or later. This function does nothing on iOS.

Reimplemented from com.choosemuse.libmuse.MuseManager.

8.42.2.9 setCompanionMode()

```
void com.choosemuse.libmuse.MuseManagerAndroid.setCompanionMode (
            boolean enable,
            String macAddress,
            byte[] manufacturerData,
            boolean singleDevice )  [inline]
```

Companion mode allows device scanning using the CompanionDeviceManager. This mode removes the requirement for requesting location permissions.

If known device information is provided, a Muse will be constructed from this information and added to the muse list without any scanning.

Parameters

enable	Enable or disable companion device manager scanning
macAddress	Mac address of known device or null
manufacturerData	Manufacturer data of known device or null
singleDevice	When true, sets this device as the only device in the muse list

8.42.2.10 setContext()

```
void com.choosemuse.libmuse.MuseManagerAndroid.setContext (
            Context context )  [inline]
```

Sets the context object to use with this MuseManager.

Warning

Only call this method at or after onCreate().

Parameters

context	the context to use.

8.42.2.11 setMuseListener()

```
void com.choosemuse.libmuse.MuseManagerAndroid.setMuseListener (
            MuseListener listener )  [inline]
```

Set the listener to be notified on changes to the list of Muses.

Parameters

listener	The listener that will receive the callback when a Muse is discovered.

Reimplemented from com.choosemuse.libmuse.MuseManager.

8.42.2.12 startListening()

```
void com.choosemuse.libmuse.MuseManagerAndroid.startListening ( )  [inline]
```

Start listening for any possible Muse devices.

As Muses are discovered, they are added to the list returned by MuseManager.getMuses() Listening is a fairly heavyweight operation, so it should be disabled when it is not needed, e.g. after a device has been found. Specifically, you *must* call stopListening() before you try to connect to a Muse you have previously discovered.

You must call MuseManagerAndroid.setContext() prior to calling startListening() or stopListening(), and the context should persist at least from before the call to startListening() till after the call to stopListening(). An application context is a good candidate for this.

Reimplemented from com.choosemuse.libmuse.MuseManager.

8.42.2.13 stopListening()

```
void com.choosemuse.libmuse.MuseManagerAndroid.stopListening ( )  [inline]
```

Stop listening for Muse devices.

Stops receiving advertising packets or modifying the list of Muses returned by MuseManager.getMuses() If this is called while not scanning, it is a no-op.

You must call MuseManagerAndroid.setContext() prior to calling startListening() or stopListening(), and the context should persist at least from before the call to startListening() till after the call to stopListening(). An application context is a good candidate for this.

Reimplemented from com.choosemuse.libmuse.MuseManager.

8.42.3 Member Data Documentation

8.42.3.1 SELECT_DEVICE_REQUEST_CODE

```
int com.choosemuse.libmuse.MuseManagerAndroid.SELECT_DEVICE_REQUEST_CODE = 100  [static]
```

The activity result request code used during companion device scanning.

The documentation for this class was generated from the following file:

- MuseManagerAndroid.java

8.43 com.choosemuse.libmuse.MuseModel Enum Reference

Public Attributes

- MU_01
- MU_02
- MU_03
- MU_04
- MU_05
- MU_06
- MS_03

8.43.1 Detailed Description

The model identifier of the headbands. The model is laser printed on the inside of the left pod of the headband as either "MU-01", "MU-02", "MU-03" or "MU-06". MuseS 2019 (MU_04) MuseS 2021 (MU_05) and MuseS 2025 (MS_03) models are marked on the underside as "MS-01", "MS-02" or "MS-03".

8.43.2 Member Data Documentation

8.43.2.1 MS_03

```
com.choosemuse.libmuse.MuseModel.MS_03
```

MuseS 2025 softband with USB-C, Bluetooth 5.3, improved EEG and Optics (MS-03).

8.43.2.2 MU_01

```
com.choosemuse.libmuse.MuseModel.MU_01
```

First model of Muse, Muse 2014

8.43.2.3 MU_02

`com.choosemuse.libmuse.MuseModel.MU_02`

Muse 2016 with Bluetooth Low Energy support.

8.43.2.4 MU_03

`com.choosemuse.libmuse.MuseModel.MU_03`

Muse2 2018 adding PPG sensor, Accelerometer and Gyroscope.

8.43.2.5 MU_04

`com.choosemuse.libmuse.MuseModel.MU_04`

MuseS 2019 softband (MS-01).

8.43.2.6 MU_05

`com.choosemuse.libmuse.MuseModel.MU_05`

MuseS 2021 softband refresh (MS-02).

8.43.2.7 MU_06

`com.choosemuse.libmuse.MuseModel.MU_06`

Muse2 2024 with USB-C connector.

The documentation for this enum was generated from the following file:

- MuseModel.java

8.44 com.choosemuse.libmuse.MusePreset Enum Reference

Public Attributes

- PRESET_10
- PRESET_12
- PRESET_14
- PRESET_20
- PRESET_21
- PRESET_22
- PRESET_23
- PRESET_AB
- PRESET_AD
- PRESET_31
- PRESET_32
- PRESET_50
- PRESET_51
- PRESET_52
- PRESET_53
- PRESET_54
- PRESET_55
- PRESET_60
- PRESET_61
- PRESET_63
- PRESET_1021
- PRESET_1022
- PRESET_1023
- PRESET_1024
- PRESET_1025
- PRESET_1026
- PRESET_1027
- PRESET_1028
- PRESET_1029
- PRESET_102A
- PRESET_1031
- PRESET_1032
- PRESET_1033
- PRESET_1034
- PRESET_1035
- PRESET_1036
- PRESET_1041
- PRESET_1042
- PRESET_1043
- PRESET_1044
- PRESET_1045
- PRESET_1046

8.44.1 Detailed Description

Defines all possible Muse presets.

Presets are specific to the model of headband. For example, setting a preset that is intended for a Muse 2014 (MU_01) headband on a Muse 2016 (MU_02) headband will not work. Attempting to set an incorrect preset on a headband will result in a warning in the log and the incorrect preset will be ignored. To avoid this, you should first connect to the headband without setting a preset and check the MuseConfiguration to get the model of the headband and then set the appropriate preset. You can also obtain the current preset of the headband from the MuseConfiguration

If you change the preset while the headband is connected, the headband will disconnect. If the preset is valid for that headband model, then the headband will automatically reconnect. If the preset is invalid for that headband model, then the headband will remain disconnected.

See also

MuseConfiguration

8.44.2 Member Data Documentation

8.44.2.1 PRESET_10

`com.choosemuse.libmuse.MusePreset.PRESET_10`

4 CH EEG, 10 bit @ 220 Hz, compression ON, no accelerometer, no battery data, no error data, no DRL/REF data.

Availability: Muse 2014 (MU_01) only

8.44.2.2 PRESET_1021

`com.choosemuse.libmuse.MusePreset.PRESET_1021`

4 CH EEG, 14 bit @ 256 Hz, 52 Hz accelerometer/gyro, 1 Hz battery, 32 Hz DRL/REF

Availability: MuseS 2025 (MS_03)

8.44.2.3 PRESET_1022

`com.choosemuse.libmuse.MusePreset.PRESET_1022`

8 CH EEG, 14 bit @ 256 Hz, 1 Hz battery, 32 Hz DRL/REF

Availability: MuseS 2025 (MS_03)

8.44.2.4 PRESET_1023

`com.choosemuse.libmuse.MusePreset.PRESET_1023`

Battery only @ 50 Hz

Availability: MuseS 2025 (MS_03)

8.44.2.5 PRESET_1024

`com.choosemuse.libmuse.MusePreset.PRESET_1024`

52 Hz accelerometer/gyro

Availability: MuseS 2025 (MS_03)

8.44.2.6 PRESET_1025

`com.choosemuse.libmuse.MusePreset.PRESET_1025`

16 CH Optics @ 64 Hz, low power

Availability: MuseS 2025 (MS_03)

8.44.2.7 PRESET_1026

`com.choosemuse.libmuse.MusePreset.PRESET_1026`

16 CH Optics @ 64 Hz, high power

Availability: MuseS 2025 (MS_03)

8.44.2.8 PRESET_1027

`com.choosemuse.libmuse.MusePreset.PRESET_1027`

8 CH Optics @ 64 Hz, low power

Availability: MuseS 2025 (MS_03)

8.44.2.9 PRESET_1028

`com.choosemuse.libmuse.MusePreset.PRESET_1028`

8 CH Optics @ 64 Hz, high power

Availability: MuseS 2025 (MS_03)

8.44.2.10 PRESET_1029

`com.choosemuse.libmuse.MusePreset.PRESET_1029`

4 CH Optics @ 64 Hz, low power

Availability: MuseS 2025 (MS_03)

8.44.2.11 PRESET_102A

`com.choosemuse.libmuse.MusePreset.PRESET_102A`

4 CH Optics @ 64 Hz, high power

Availability: MuseS 2025 (MS_03)

8.44.2.12 PRESET_1031

`com.choosemuse.libmuse.MusePreset.PRESET_1031`

4 CH EEG, 14 bit @ 256 Hz, 52 Hz accelerometer/gyro, 1 Hz battery, 32 Hz DRL/REF, 16 CH Optics @ 64 Hz, low power

Availability: MuseS 2025 (MS_03)

8.44.2.13 PRESET_1032

`com.choosemuse.libmuse.MusePreset.PRESET_1032`

4 CH EEG, 14 bit @ 256 Hz, 52 Hz accelerometer/gyro, 1 Hz battery, 32 Hz DRL/REF, 16 CH Optics @ 64 Hz, high power

Availability: MuseS 2025 (MS_03)

8.44.2.14 PRESET_1033

`com.choosemuse.libmuse.MusePreset.PRESET_1033`

4 CH EEG, 14 bit @ 256 Hz, 52 Hz accelerometer/gyro, 1 Hz battery, 32 Hz DRL/REF, 8 CH Optics @ 64 Hz, low power

Availability: MuseS 2025 (MS_03)

8.44.2.15 PRESET_1034

`com.choosemuse.libmuse.MusePreset.PRESET_1034`

4 CH EEG, 14 bit @ 256 Hz, 52 Hz accelerometer/gyro, 1 Hz battery, 32 Hz DRL/REF, 8 CH Optics @ 64 Hz, high power

Availability: MuseS 2025 (MS_03)

8.44.2.16 PRESET_1035

`com.choosemuse.libmuse.MusePreset.PRESET_1035`

4 CH EEG, 14 bit @ 256 Hz, 52 Hz accelerometer/gyro, 1 Hz battery, 32 Hz DRL/REF, 4 CH Optics @ 64 Hz, low power

Availability: MuseS 2025 (MS_03)

8.44.2.17 PRESET_1036

`com.choosemuse.libmuse.MusePreset.PRESET_1036`

4 CH EEG, 14 bit @ 256 Hz, 52 Hz accelerometer/gyro, 1 Hz battery, 32 Hz DRL/REF, 4 CH Optics @ 64 Hz, high power

Availability: MuseS 2025 (MS_03)

8.44.2.18 PRESET_1041

`com.choosemuse.libmuse.MusePreset.PRESET_1041`

8 CH EEG, 14 bit @ 256 Hz, 52 Hz accelerometer/gyro, 1 Hz battery, 32 Hz DRL/REF, 16 CH Optics @ 64 Hz, low power

Availability: MuseS 2025 (MS_03)

8.44.2.19 PRESET_1042

`com.choosemuse.libmuse.MusePreset.PRESET_1042`

8 CH EEG, 14 bit @ 256 Hz, 52 Hz accelerometer/gyro, 1 Hz battery, 32 Hz DRL/REF, 16 CH Optics @ 64 Hz, high power

Availability: MuseS 2025 (MS_03)

8.44.2.20 PRESET_1043

`com.choosemuse.libmuse.MusePreset.PRESET_1043`

8 CH EEG, 14 bit @ 256 Hz, 52 Hz accelerometer/gyro, 1 Hz battery, 32 Hz DRL/REF, 8 CH Optics @ 64 Hz, low power

Availability: MuseS 2025 (MS_03)

8.44.2.21 PRESET_1044

`com.choosemuse.libmuse.MusePreset.PRESET_1044`

8 CH EEG, 14 bit @ 256 Hz, 52 Hz accelerometer/gyro, 1 Hz battery, 32 Hz DRL/REF, 8 CH Optics @ 64 Hz, high power

Availability: MuseS 2025 (MS_03)

8.44.2.22 PRESET_1045

`com.choosemuse.libmuse.MusePreset.PRESET_1045`

8 CH EEG, 14 bit @ 256 Hz, 52 Hz accelerometer/gyro, 1 Hz battery, 32 Hz DRL/REF, 4 CH Optics @ 64 Hz, low power

Availability: MuseS 2025 (MS_03)

8.44.2.23 PRESET_1046

`com.choosemuse.libmuse.MusePreset.PRESET_1046`

8 CH EEG, 14 bit @ 256 Hz, 52 Hz accelerometer/gyro, 1 Hz battery, 32 Hz DRL/REF, 4 CH Optics @ 64 Hz, high power

Availability: MuseS 2025 (MS_03)

8.44.2.24 PRESET_12

`com.choosemuse.libmuse.MusePreset.PRESET_12`

4 CH EEG, 10 bit @ 220 Hz, compression ON, 50 Hz accelerometer data, 0.1 Hz battery data, no error data, no DRL/REF data

Availability: Muse 2014 (MU_01) only

8.44.2.25 PRESET_14

`com.choosemuse.libmuse.MusePreset.PRESET_14`

4 CH EEG, 10 bit @ 220 Hz, compression ON, 50 Hz accelerometer data, 0.1 Hz battery data, real-time error data, 10 bit @ 10 Hz DRL/REF data

This is the default for Muse 2014 (MU_01).

Availability: Muse 2014 (MU_01) only

8.44.2.26 PRESET_20

`com.choosemuse.libmuse.MusePreset.PRESET_20`

5 CH EEG, 12 bit @ 256 Hz, 52 Hz accelerometer/gyro, 0.1 Hz battery, 32 Hz DRL/REF.

Availability: Muse 2016 (MU_02), Muse2 2018 (MU_03), MuseS 2019 (MU_04), MuseS 2021 (MU_05), Muse2 2024 (MU_06)

8.44.2.27 PRESET_21

`com.choosemuse.libmuse.MusePreset.PRESET_21`

4 CH EEG, 12 bit @ 256 Hz, 52 Hz accelerometer/gyro, 0.1 Hz battery, 32 Hz DRL/REF.

This is the default for Muse 2016 (MU_02), Muse2 2018 (MU_03), MuseS 2019 (MU_04), MuseS 2021 (MU_05), Muse2 2024 (MU_06).

Availability: Muse 2016 (MU_02), Muse2 2018 (MU_03), MuseS 2019 (MU_04), MuseS 2021 (MU_05), Muse2 2024 (MU_06)

8.44.2.28 PRESET_22

`com.choosemuse.libmuse.MusePreset.PRESET_22`

4 CH EEG, 12 bit @ 256 Hz, 0.1 Hz battery, 32 Hz DRL/REF.

Availability: Muse 2016 (MU_02), Muse2 2018 (MU_03), MuseS 2019 (MU_04), MuseS 2021 (MU_05), Muse2 2024 (MU_06)

8.44.2.29 PRESET_23

`com.choosemuse.libmuse.MusePreset.PRESET_23`

5 CH EEG, 12 bit @ 256 Hz, 0.1 Hz battery, 32 Hz DRL/REF.

Availability: Muse2 2018 (MU_03), MuseS 2019 (MU_04), MuseS 2021 (MU_05), Muse2 2024 (MU_06)

8.44.2.30 PRESET_31

`com.choosemuse.libmuse.MusePreset.PRESET_31`

4 CH EEG, 12 bit @ 256 Hz, 0.1 Hz battery, 32 Hz DRL/REF, 52 Hz Acc, Gyro and Magnetometer, 0.1 Hz for UV and Pressure.

Availability: glasses only

8.44.2.31 PRESET_32

`com.choosemuse.libmuse.MusePreset.PRESET_32`

No EEG data, only other sensors. 32 Hz DRL/REF, 52 Hz Acc, Gyro and Magnetometer, 0.1 Hz for battery, UV and Pressure.

Availability: glasses only

8.44.2.32 PRESET_50

`com.choosemuse.libmuse.MusePreset.PRESET_50`

5 CH EEG, 12 bit @ 256 Hz, 52 Hz accelerometer/gyro, 0.1 Hz battery, 32 Hz DRL/REF, PPG @ 64 Hz

Availability: Muse2 2018 (MU_03), MuseS 2019 (MU_04), MuseS 2021 (MU_05), Muse2 2024 (MU_06)

8.44.2.33 PRESET_51

`com.choosemuse.libmuse.MusePreset.PRESET_51`

4 CH EEG, 12 bit @ 256 Hz, 52 Hz accelerometer/gyro, 0.1 Hz battery, 32 Hz DRL/REF, PPG @ 64 Hz

Availability: Muse2 2018 (MU_03), MuseS 2019 (MU_04), MuseS 2021 (MU_05), Muse2 2024 (MU_06)

8.44.2.34 PRESET_52

`com.choosemuse.libmuse.MusePreset.PRESET_52`

4 CH EEG, 12 bit @ 256 Hz, 0.1 Hz battery, 32 Hz DRL/REF, PPG @ 64 Hz

Availability: Muse2 2018 (MU_03), MuseS 2019 (MU_04), MuseS 2021 (MU_05), Muse2 2024 (MU_06)

8.44.2.35 PRESET_53

`com.choosemuse.libmuse.MusePreset.PRESET_53`

6 CH EEG, 12 bit @ 256 Hz, 52 Hz accelerometer/gyro, 0.1 Hz battery, 32 Hz DRL/REF, PPG @ 64 Hz

Availability: MuseS 2019 (MU_04), MuseS 2021 (MU_05), Muse2 2024 (MU_06)

8.44.2.36 PRESET_54

`com.choosemuse.libmuse.MusePreset.PRESET_54`

6 CH EEG, 12 bit @ 128 Hz, 52 Hz accelerometer/gyro, 0.1 Hz battery, 16 Hz DRL/REF

Availability: MuseS 2021 (MU_05)

8.44.2.37 PRESET_55

`com.choosemuse.libmuse.MusePreset.PRESET_55`

4 CH EEG, 12 bit @ 128 Hz, 0.1 Hz battery, 16 Hz DRL/REF, PPG @ 64 Hz

Availability: MuseS 2021 (MU_05)

8.44.2.38 PRESET_60

`com.choosemuse.libmuse.MusePreset.PRESET_60`

5 CH EEG, 12 bit @ 256 Hz, 52 Hz accelerometer/gyro, 0.1 Hz battery, 32 Hz DRL/REF, PPG @ 64 Hz, THERM↩ ISTOR @ 16 Hz

Availability: MuseS 2019 (MU_04), MuseS 2021 (MU_05)

8.44.2.39 PRESET_61

`com.choosemuse.libmuse.MusePreset.PRESET_61`

4 CH EEG, 12 bit @ 256 Hz, 52 Hz accelerometer/gyro, 0.1 Hz battery, 32 Hz DRL/REF, PPG @ 64 Hz, THERM↩ ISTOR @ 16 Hz

Availability: MuseS 2019 (MU_04), MuseS 2021 (MU_05)

8.44.2.40 PRESET_63

`com.choosemuse.libmuse.MusePreset.PRESET_63`

6 CH EEG, 12 bit @ 256 Hz, 52 Hz accelerometer/gyro, 0.1 Hz battery, 32 Hz DRL/REF, PPG @ 64 Hz, THERM↩ ISTOR @ 16 Hz

Availability: MuseS 2019 (MU_04), MuseS 2021 (MU_05)

8.44.2.41 PRESET_AB

`com.choosemuse.libmuse.MusePreset.PRESET_AB`

6 CH EEG, 16 bit @ 500 Hz, 50 Hz accelerometer, 0.1 Hz battery, compression OFF, notch filter OFF, no error data, no DRL/REF data

Research preset:
Only the following data packets are generated with this preset: MuseDataPacketType.EEG , MuseDataPacketType.ACCELEROMETER , MuseDataPacketType.BATTERY
Artifacts are not generated with this preset.

Availability: Muse 2014 (MU_01) only

8.44.2.42 PRESET_AD

`com.choosemuse.libmuse.MusePreset.PRESET_AD`

4 CH EEG, 16 bit @ 500 Hz, 50 Hz accelerometer, 0.1 Hz battery, compression OFF, notch filter OFF, no error data, no DRL/REF data

Research preset:
Only the following data packets are generated with this preset: MuseDataPacketType.EEG , MuseDataPacketType.ACCELEROMETER , MuseDataPacketType.BATTERY
Artifacts are not generated with this preset.

Availability: Muse 2014 (MU_01) only

The documentation for this enum was generated from the following file:

- MusePreset.java

8.45 com.choosemuse.libmuse.MuseVersion Class Reference

Public Member Functions

- abstract String getRunningState ()
- abstract String getHardwareVersion ()
- abstract String getBspVersion ()
- abstract String getFirmwareVersion ()
- abstract String getBootloaderVersion ()
- abstract String getFirmwareBuildNumber ()
- abstract String getFirmwareType ()
- abstract int getProtocolVersion ()
- abstract String getBleFirmwareVersion ()

Static Public Member Functions

- static native MuseVersion makeDefaultVersion ()
- static native MuseVersion makeVersion (String json)

8.45.1 Detailed Description

Provides access to Muse firmware and hardware versions.

You must connect to the headband at least once to before this information is available. Once you have connected once the information will remain available, even after you disconnect.

8.45.2 Member Function Documentation

8.45.2.1 getBleFirmwareVersion()

```
abstract String com.choosemuse.libmuse.MuseVersion.getBleFirmwareVersion ( )  [abstract]
```

Provides access to the Bluetooth firmware version.

Returns

The BLE firmware version.

8.45.2.2 getBootloaderVersion()

```
abstract String com.choosemuse.libmuse.MuseVersion.getBootloaderVersion ( )  [abstract]
```

Provides access to Muse bootloader version.

Returns

The bootloader version.

8.45.2.3 getBspVersion()

```
abstract String com.choosemuse.libmuse.MuseVersion.getBspVersion ( )  [abstract]
```

BSP (board support package) version. This is only available on Muse 2016 (MU_02) or later.

Returns

The BSP version.

8.45.2.4 getFirmwareBuildNumber()

```
abstract String com.choosemuse.libmuse.MuseVersion.getFirmwareBuildNumber ( )  [abstract]
```

Provides access to Muse firmware build number. This is only available on Muse 2014 (MU_01).

Returns

The firmware build number.

8.45.2.5 getFirmwareType()

```
abstract String com.choosemuse.libmuse.MuseVersion.getFirmwareType ( )  [abstract]
```

Type of firmware. One of `consumer`, `research` or `test`.

Returns

The type of firmware.

8.45.2.6 getFirmwareVersion()

```
abstract String com.choosemuse.libmuse.MuseVersion.getFirmwareVersion ( )  [abstract]
```

Provides access to the firmware version.

Returns

The firmware version.

8.45.2.7 getHardwareVersion()

```
abstract String com.choosemuse.libmuse.MuseVersion.getHardwareVersion ( )  [abstract]
```

Provides access to hardware version.

Returns

The hardware version.

8.45.2.8 getProtocolVersion()

```
abstract int com.choosemuse.libmuse.MuseVersion.getProtocolVersion ( )  [abstract]
```

Provides access to Muse communication protocol version.

Returns

The communication protocol version.

8.45.2.9 getRunningState()

```
abstract String com.choosemuse.libmuse.MuseVersion.getRunningState ( )  [abstract]
```

Provides access to the running state.

- For Muse 2014 (MU_01) this is one of `app`, `bootloader` or `test`.
- For Muse 2016 (MU_02) or later this is one of `headband` or `bootloader`.

Returns

The current running state of the headband.

8.45.2.10 makeDefaultVersion()

```
static native MuseVersion com.choosemuse.libmuse.MuseVersion.makeDefaultVersion ( )  [static]
```

Internal use only. Create a default version.

8.45.2.11 makeVersion()

```
static native MuseVersion com.choosemuse.libmuse.MuseVersion.makeVersion (
            String json )  [static]
```

Internal use only. Create version from JSON string on Muse 2016 (MU_02) or later.

The documentation for this class was generated from the following file:

- MuseVersion.java

8.46 com.choosemuse.libmuse.NotchFrequency Enum Reference

Public Attributes

- NOTCH_NONE
- NOTCH_50HZ
- NOTCH_60HZ

8.46.1 Detailed Description

Notch Frequencies refer to the possible power line frequencies in a geographic area that may create noise in raw EEG data.

Muse 2014 (MU_01) headbands are equipped with a hardware filter to remove this noise in certain preset configurations.

For Muse 2016 (MU_02) or later headbands or Muse 2014 (MU_01) headbands using a research preset there is no hardware filtering.

See also

- MusePreset
- http://en.wikipedia.org/wiki/Utility_frequency

8.46.2 Member Data Documentation

8.46.2.1 NOTCH_50HZ

`com.choosemuse.libmuse.NotchFrequency.NOTCH_50HZ`

50 Hz frequency is used in most parts of the world: Europe, Russia, Africa

8.46.2.2 NOTCH_60HZ

`com.choosemuse.libmuse.NotchFrequency.NOTCH_60HZ`

60 Hz frequency is used in North America and Asia. This is the default setting.

8.46.2.3 NOTCH_NONE

`com.choosemuse.libmuse.NotchFrequency.NOTCH_NONE`

The notch filter is not available on some hardware versions.

The documentation for this enum was generated from the following file:

- NotchFrequency.java

8.47 com.choosemuse.libmuse.Optics Enum Reference

Public Attributes

- OPTICS1
- OPTICS2
- OPTICS3
- OPTICS4
- OPTICS5
- OPTICS6
- OPTICS7
- OPTICS8
- OPTICS9
- OPTICS10
- OPTICS11
- OPTICS12
- OPTICS13
- OPTICS14
- OPTICS15
- OPTICS16

8.47.1 Detailed Description

Represents the data mapping in an Optics packet.

Functional near-infrared spectroscopy (fNIRS) is a non-invasive imaging technique that measures brain activity by detecting changes in blood oxygenation.

The light absorbtion of the blood varies by wavelength. The 750nm wavelength is sensitive to deoxygenated hemoglobin (HbR), while the 850nm wavelength is sensitive to oxygenated hemoglobin (HbO). By analyzing the absorption of these wavelengths, we can determine changes in blood oxygenation levels.

For more information on fNIRS, see the following Wikipedia article:
https://en.wikipedia.org/wiki/Functional_near-infrared_spectroscopy

The Optics data is also used to measure PPG.

Optics data values are given in microamps (uA).

See also

MuseDataPacketType.OPTICS

MuseDataPacket.getOpticsChannelValue()

8.47.2 Member Data Documentation

8.47.2.1 OPTICS1

`com.choosemuse.libmuse.Optics.OPTICS1`

730nm left outer (8 and 16 channel modes) or 730nm left inner (4 channel mode)

8.47.2.2 OPTICS10

`com.choosemuse.libmuse.Optics.OPTICS10`

Red right outer (16 channel mode)

8.47.2.3 OPTICS11

`com.choosemuse.libmuse.Optics.OPTICS11`

Ambient left outer (16 channel mode)

8.47.2.4 OPTICS12

`com.choosemuse.libmuse.Optics.OPTICS12`

Ambient right outer (16 channel mode)

8.47.2.5 OPTICS13

`com.choosemuse.libmuse.Optics.OPTICS13`

Red left inner (16 channel mode)

8.47.2.6 OPTICS14

`com.choosemuse.libmuse.Optics.OPTICS14`

Red right inner (16 channel mode)

8.47.2.7 OPTICS15

`com.choosemuse.libmuse.Optics.OPTICS15`

Ambient left inner (16 channel mode)

8.47.2.8 OPTICS16

`com.choosemuse.libmuse.Optics.OPTICS16`

Ambient right inner (16 channel mode)

8.47.2.9 OPTICS2

`com.choosemuse.libmuse.Optics.OPTICS2`

730nm right outer (8 and 16 channel modes) or 730nm right inner (4 channel mode)

8.47.2.10 OPTICS3

`com.choosemuse.libmuse.Optics.OPTICS3`

850nm left outer (8 and 16 channel modes) or 850nm left inner (4 channel mode)

8.47.2.11 OPTICS4

`com.choosemuse.libmuse.Optics.OPTICS4`

850nm right outer (8 and 16 channel modes) or 850nm right inner (4 channel mode)

8.47.2.12 OPTICS5

`com.choosemuse.libmuse.Optics.OPTICS5`

730nm left inner (8 and 16 channel modes)

8.47.2.13 OPTICS6

`com.choosemuse.libmuse.Optics.OPTICS6`

730nm right inner (8 and 16 channel modes)

8.47.2.14 OPTICS7

`com.choosemuse.libmuse.Optics.OPTICS7`

850nm left inner (8 and 16 channel modes)

8.47.2.15 OPTICS8

`com.choosemuse.libmuse.Optics.OPTICS8`

850nm right inner (8 and 16 channel modes)

8.47.2.16 OPTICS9

`com.choosemuse.libmuse.Optics.OPTICS9`

Red left outer (16 channel mode)

The documentation for this enum was generated from the following file:

- Optics.java

8.48 com.choosemuse.libmuse.Ppg Enum Reference

Public Attributes

- AMBIENT
- IR
- RED

8.48.1 Detailed Description

Represents the data mapping in a PPG packet. Enum values for IR and RED correspond to the Infrared and Red values. The AMBIENT enum value is repurposed on models MuseS 2019 (MU_04) and MuseS 2021 (MU_05).

Muse2 2018 (MU_03) AMBIENT represents the Ambient value.
MuseS 2019 (MU_04) AMBIENT represents the Green value. There is no Ambient available and Red is not used.
MuseS 2021 (MU_05) AMBIENT represents the IR-H16 value. There is no Ambient available.

MuseS 2025 (MS_03) uses Optics data for PPG.

Raw PPG values are given in arbitrary units.

See also

MuseDataPacketType.PPG

MuseDataPacket.getPpgChannelValue()

8.48.2 Member Data Documentation

8.48.2.1 AMBIENT

`com.choosemuse.libmuse.Ppg.AMBIENT`

Ambient, Green or IR-H16.

8.48.2.2 IR

`com.choosemuse.libmuse.Ppg.IR`

IR.

8.48.2.3 RED

`com.choosemuse.libmuse.Ppg.RED`

Red.

The documentation for this enum was generated from the following file:

- Ppg.java

8.49 com.choosemuse.libmuse.Pressure Enum Reference

Public Attributes

- RAW
- AVERAGED

8.49.1 Detailed Description

Represents the data mapping in a Pressure packet.

See also

MuseDataPacketType.PRESSURE

MuseDataPacket.getPressureValue()

8.49.2 Member Data Documentation

8.49.2.1 AVERAGED

`com.choosemuse.libmuse.Pressure.AVERAGED`

The averaged pressure value in mBar based on the last 10 readings. This provides a smoother curve for the pressure values.

8.49.2.2 RAW

`com.choosemuse.libmuse.Pressure.RAW`

The raw pressure value returned by the pressure sensor in mBar

The documentation for this enum was generated from the following file:

- Pressure.java

8.50 com.choosemuse.libmuse.ReaderListener Class Reference

Public Member Functions

- abstract void receiveAnnotation (AnnotationData annotation)
- abstract void receiveVersion (MuseVersion version)
- abstract void receiveConfiguration (MuseConfiguration configuration)
- abstract void receiveComputingDeviceConfiguration (ComputingDeviceConfiguration computingDevice↩ Configuration)

8.50.1 Detailed Description

Implement this interface to receive annotations and configuration/version info from a ReaderMuse

8.50.2 Member Function Documentation

8.50.2.1 receiveAnnotation()

```
abstract void com.choosemuse.libmuse.ReaderListener.receiveAnnotation (
            AnnotationData annotation )  [abstract]
```

Called once for each annotation in the file.

Parameters

annotation	The annotation data that was read.

8.50.2.2 receiveComputingDeviceConfiguration()

```
abstract void com.choosemuse.libmuse.ReaderListener.receiveComputingDeviceConfiguration (
            ComputingDeviceConfiguration computingDeviceConfiguration )  [abstract]
```

Called each time a computing device packet is encountered.

Parameters

computingDeviceConfiguration	The device configuration that was read.

8.50.2.3 receiveConfiguration()

```
abstract void com.choosemuse.libmuse.ReaderListener.receiveConfiguration (
            MuseConfiguration configuration )  [abstract]
```

Called each time a configuration packet is encountered.

Parameters

configuration	The configuration data that was read.

8.50.2.4 receiveVersion()

```
abstract void com.choosemuse.libmuse.ReaderListener.receiveVersion (
            MuseVersion version )  [abstract]
```

Called each time a version packet is encountered.

Parameters

version	The version data that was read.

The documentation for this class was generated from the following file:

- ReaderListener.java

8.51 com.choosemuse.libmuse.ReaderMuse Class Reference

Public Member Functions

- abstract void run ()

- abstract void runInRealTimespan ()
- abstract long currentTime ()
- abstract void playback ()
- abstract void stopPlayback ()
- abstract void setPlaybackSettings (ReaderMusePlaybackSettings settings)
- abstract ReaderMusePlaybackSettings getPlaybackSettings ()
- abstract void setReaderListener (ReaderListener listener)
- abstract void setPlaybackListener (ReaderPlaybackListener listener)
- abstract Muse asMuse ()

8.51.1 Detailed Description

A Muse associated with a MuseFileReader .

This interface can be used to play back packets from a .muse file so that they can be handled the same way that you would handle data from an actual Muse . Details of the parsing are abstracted and you will simply receive data packets with the data that is read.

When you use this class, register MuseDataListener for the data packets you would like to receive by calling ReaderMuse.asMuse() then Muse.registerDataListener().

See also

ReaderMuseBuilder

ReaderMusePlaybackSettings

ReaderPlaybackListener

8.51.2 Member Function Documentation

8.51.2.1 asMuse()

```
abstract Muse com.choosemuse.libmuse.ReaderMuse.asMuse ( )  [abstract]
```

Downcast to Muse Note that most of the API of the returned Muse is stubbed and will cause assert failures if called in debug mode. The methods that are implemented are:

- isLowEnergy / is_low_energy
- enableDataTransmission / enable_data_transmission
- register∗
- unregister∗

Note that the only listener that will ever receive packets from a ReaderMuse is MuseDataListener .

8.51.2.2 currentTime()

```
abstract long com.choosemuse.libmuse.ReaderMuse.currentTime ( )  [abstract]
```

Corresponds to the timestamp field for the last packet read.

This does not change within a given listener's body.

Returns

The timestamp of the last packet read.

8.51.2.3 getPlaybackSettings()

```
abstract ReaderMusePlaybackSettings com.choosemuse.libmuse.ReaderMuse.getPlaybackSettings ( )
[abstract]
```

Returns the settings that will be used when playing back a file with this ReaderMuse .

Returns

The settings that will be used to play back the file.

8.51.2.4 playback()

```
abstract void com.choosemuse.libmuse.ReaderMuse.playback ( )  [abstract]
```

Play back the MuseFile and call each listener for each packet as appropriate.

The speed of the playback is controlled by the playback settings. Simulated playback settings require the ReaderMuse to be constructed with an EventLoop to simulate the time between packets. Calling playback without an EventLoop will not playback the file and will log a warning.

This only works once; the file is not rewound on subsequent calls. If you want to play a file again, create another ReaderMuse

See also

ReaderMusePlaybackSettings

8.51.2.5 run()

```
abstract void com.choosemuse.libmuse.ReaderMuse.run ( )  [abstract]
```

Run through all packets in the file, calling each listener for each packet as appropriate.

This only works once; the file is not rewound on subsequent calls. If you want to play a file again, create another ReaderMuse

Deprecated Use playback() with the playback setting ReaderMusePlaybackSettings.AS_FAST_AS_POSSIBLE_WITH_SAVED_TIM to replicate this behaviour.

8.51.2.6 runInRealTimespan()

```
abstract void com.choosemuse.libmuse.ReaderMuse.runInRealTimespan ( )  [abstract]
```

Run through all packets in the file, calling each listener for each packet as appropriate in 1:1 timespan

This only works once; the file is not rewound on subsequent calls. If you want to play a file again, create another ReaderMuse

Deprecated Use playback()with the playback setting ReaderMusePlaybackSettings.SIMULATED_WITH_SAVED_TIMESTAMP to replicate this behaviour.

8.51.2.7 setPlaybackListener()

```
abstract void com.choosemuse.libmuse.ReaderMuse.setPlaybackListener (
            ReaderPlaybackListener listener )  [abstract]
```

Set a listener to receive events related to playback. For example when playback has finished successfully or was interrupted due to error.

Parameters

listener	The playback listener to receive callbacks.

8.51.2.8 setPlaybackSettings()

```
abstract void com.choosemuse.libmuse.ReaderMuse.setPlaybackSettings (
            ReaderMusePlaybackSettings settings )  [abstract]
```

Sets the settings to use when playing back a file with this ReaderMuse .

Parameters

settings	The settings to use for playback.

8.51.2.9 setReaderListener()

```
abstract void com.choosemuse.libmuse.ReaderMuse.setReaderListener (
            ReaderListener listener )  [abstract]
```

Set a listener to receive annotations, version, and configuration packets.

Parameters

listener	The listener for annotation, version and configuration packets.

8.51.2.10 stopPlayback()

```
abstract void com.choosemuse.libmuse.ReaderMuse.stopPlayback ( )  [abstract]
```

Stop playback of the MuseFile

The documentation for this class was generated from the following file:

- ReaderMuse.java

8.52 com.choosemuse.libmuse.ReaderMuseBuilder Class Reference

Public Member Functions

- abstract ReaderMuseBuilder withPacketTypes (HashSet< MuseDataPacketType > types)
- abstract ReaderMuseBuilder skipPacketTypes (HashSet< MuseDataPacketType > types)
- abstract ReaderMuseBuilder withModel (MuseModel model)
- abstract ReaderMuseBuilder withPlaybackSettings (ReaderMusePlaybackSettings settings)
- abstract ReaderMuseBuilder withEventLoop (EventLoop loop)
- abstract ReaderMuse build (MuseFileReader reader)
- abstract ReaderMuse buildWithAsync (MuseFileReader reader, EventLoop asyncLoop)

Static Public Member Functions

- static native ReaderMuseBuilder get ()

8.52.1 Detailed Description

Builds a ReaderMuse that plays back the packets in the given MuseFileReader .

It exposes a way to set the configuration with chainable method calls, e.g.:

```
ReaderMuse muse = ReaderMuseBuilder.get()
    .withPacketTypes(myPacketTypes)
    .build(myMuseFileReader);
```

8.52.2 Member Function Documentation

8.52.2.1 build()

```
abstract ReaderMuse com.choosemuse.libmuse.ReaderMuseBuilder.build (
            MuseFileReader reader )  [abstract]
```

Construct a ReaderMuse

Parameters

reader	The MuseFileReader to use to read the file.

Returns

A ReaderMuse configured with the parameters that were set.

8.52.2.2 buildWithAsync()

```
abstract ReaderMuse com.choosemuse.libmuse.ReaderMuseBuilder.buildWithAsync (
            MuseFileReader reader,
            EventLoop asyncLoop )  [abstract]
```

Construct a ReaderMuse

Parameters

reader	The MuseFileReader to use to read the file.
asyncLoop	The EventLoop to use for the call to Muse::runAsynchronously()

Returns

A ReaderMuse configured with the parameters that were set.

Deprecated Set the EventLoop with ReaderMuseBuilder.withEventLoop() and then call ReaderMuseBuilder.build() instead.

8.52.2.3 get()

```
static native ReaderMuseBuilder com.choosemuse.libmuse.ReaderMuseBuilder.get ( )  [static]
```

Returns a reference to a ReaderMuseBuilder

Returns

A reference to a ReaderMuseBuilder

8.52.2.4 skipPacketTypes()

```
abstract ReaderMuseBuilder com.choosemuse.libmuse.ReaderMuseBuilder.skipPacketTypes (
            HashSet< MuseDataPacketType > types )  [abstract]
```

The set of packet types to explicitly ignore in the file.

The default is the empty set.

The eventual set used is computed by the last call to withPacketTypes() minus the last call to skipPacketTypes()

Parameters

types	The set of packet types to skip.

Returns

A reference to the same ReaderMuseBuilder

8.52.2.5 withEventLoop()

```
abstract ReaderMuseBuilder com.choosemuse.libmuse.ReaderMuseBuilder.withEventLoop (
            EventLoop loop )  [abstract]
```

The EventLoop to use to handle simulated playback.

The default is a null pointer (no event loop).

Parameters

loop	The EventLoop to use.

Returns

A reference to the same ReaderMuseBuilder

8.52.2.6 withModel()

```
abstract ReaderMuseBuilder com.choosemuse.libmuse.ReaderMuseBuilder.withModel (
            MuseModel model )  [abstract]
```

The model that this Muse should say it is.

The default is Muse 2014 (MU_01).

Parameters

model	The model to use.

Returns

A reference to the same ReaderMuseBuilder

8.52.2.7 withPacketTypes()

```
abstract ReaderMuseBuilder com.choosemuse.libmuse.ReaderMuseBuilder.withPacketTypes (
            HashSet< MuseDataPacketType > types )  [abstract]
```

The set of packet types to pass through from the file.

The default set contains the following all packet types in MuseDataPacketType .

The eventual set used is computed by the last call to withPacketTypes() minus the last call to skipPacketTypes()

Parameters

types	The set of packet types to read.

Returns

A reference to the same ReaderMuseBuilder

8.52.2.8 withPlaybackSettings()

```
abstract ReaderMuseBuilder com.choosemuse.libmuse.ReaderMuseBuilder.withPlaybackSettings (
            ReaderMusePlaybackSettings settings )  [abstract]
```

The playback settings to use with playing back the file.

The default is ReaderMusePlaybackSettings.AS_FAST_AS_POSSIBLE_WITH_SAVED_TIMESTAMP

Parameters

settings	The playback settings to use.

Returns

A reference to the same ReaderMuseBuilder

The documentation for this class was generated from the following file:

- ReaderMuseBuilder.java

8.53 com.choosemuse.libmuse.ReaderMusePlaybackSettings Enum Reference

Public Attributes

- AS_FAST_AS_POSSIBLE_WITH_SAVED_TIMESTAMP
- SIMULATED_WITH_SAVED_TIMESTAMP
- SIMULATED_WITH_SYSTEM_CLOCK_TIMESTAMP

8.53.1 Detailed Description

Configures the playback speed and timestamps of a ReaderMuse

See also

ReaderMuse

ReaderMuseBuilder

8.53.2 Member Data Documentation

8.53.2.1 AS_FAST_AS_POSSIBLE_WITH_SAVED_TIMESTAMP

`com.choosemuse.libmuse.ReaderMusePlaybackSettings.AS_FAST_AS_POSSIBLE_WITH_SAVED_TIMESTAMP`

The ReaderMuse should playback the data as fast as it can be read from the file. There will be no simulated time between packets and the timestamps of the packets created during playback will match the timestamps that are saved in the file.

8.53.2.2 SIMULATED_WITH_SAVED_TIMESTAMP

`com.choosemuse.libmuse.ReaderMusePlaybackSettings.SIMULATED_WITH_SAVED_TIMESTAMP`

The ReaderMuse should playback the data simulating the time between packets. For example, if the second packet in the file has a timestamp that is 20ms later than the timestamp of the first packet, then LibMuse will wait for 20ms after the first packet is sent before sending the second packet. The timestamps of the packets created during playback will match the timestamps that are saved in the file.

Using this setting requires that the ReaderMuse be constructed with an EventLoop that will be used to simulate the time between packets. If this setting is used, but the ReaderMuse does not have an EventLoop then playback will not read the file and a warning will be logged.

8.53.2.3 SIMULATED_WITH_SYSTEM_CLOCK_TIMESTAMP

`com.choosemuse.libmuse.ReaderMusePlaybackSettings.SIMULATED_WITH_SYSTEM_CLOCK_TIMESTAMP`

The ReaderMuse should playback the data simulating the time between packets. For example, if the second packet in the file has a timestamp that is 20ms later than the timestamp of the first packet, then LibMuse will wait for 20ms after the first packet is sent before sending the second packet. The timestamps of the packets created during playback will use the current times of the system clock rather than the timestamp of the packet saved in the file.

Using this setting requires that the ReaderMuse be constructed with an EventLoop that will be used to simulate the time between packets. If this setting is used, but the ReaderMuse does not have an EventLoop then playback will not read the file and a warning will be logged.

The documentation for this enum was generated from the following file:

- ReaderMusePlaybackSettings.java

8.54 com.choosemuse.libmuse.ReaderPlaybackListener Class Reference

Public Member Functions

- abstract void receivePlaybackDone ()
- abstract void receivePlaybackInterrupted ()

8.54.1 Detailed Description

Listens for ReaderMuse events related to the playback of the file.

8.54.2 Member Function Documentation

8.54.2.1 receivePlaybackDone()

```
abstract void com.choosemuse.libmuse.ReaderPlaybackListener.receivePlaybackDone ( )  [abstract]
```

Called when playback of muse file has finished (i.e., reach EOF).

8.54.2.2 receivePlaybackInterrupted()

```
abstract void com.choosemuse.libmuse.ReaderPlaybackListener.receivePlaybackInterrupted ( )
[abstract]
```

Called when playback was interrupted/error occurred when reading the file.

The documentation for this class was generated from the following file:

- ReaderPlaybackListener.java

8.55 com.choosemuse.libmuse.Result Class Reference

Collaboration diagram for com.choosemuse.libmuse.Result:

Public Member Functions

- Result (ResultLevel level, String type, int code, String info)
- ResultLevel getLevel ()
- String getType ()
- int getCode ()
- String getInfo ()
- String toString ()

8.55.1 Detailed Description

Represents the result of a synchronous operation.

Results can be at different levels: eg, success, info, warning, error

8.55.2 Constructor & Destructor Documentation

8.55.2.1 Result()

```
com.choosemuse.libmuse.Result.Result (
            ResultLevel level,
            String type,
            int code,
            String info )  [inline]
```

8.55.3 Member Function Documentation

8.55.3.1 getCode()

```
int com.choosemuse.libmuse.Result.getCode ( )  [inline]
```

A machine-parseable error code

Returns

0 if the result is a success, non-zero otherwise.

8.55.3.2 getInfo()

```
String com.choosemuse.libmuse.Result.getInfo ( )  [inline]
```

Human-readable description

Returns

A verbose description of the result (error message or status)

8.55.3.3 getLevel()

```
ResultLevel com.choosemuse.libmuse.Result.getLevel ( )  [inline]
```

The level of the result.

Returns

The ResultLevel

8.55.3.4 getType()

```
String com.choosemuse.libmuse.Result.getType ( )  [inline]
```

The type of operation that this result applies to as a string.

Returns

The type of this result.

8.55.3.5 toString()

```
String com.choosemuse.libmuse.Result.toString ( )  [inline]
```

Returns a string representation of this object.

Returns

a string representation of this object.

The documentation for this class was generated from the following file:

- Result.java

8.56 com.choosemuse.libmuse.ResultLevel Enum Reference

Public Attributes

- R_NONE
- R_SUCCESS
- R_INFO
- R_WARN
- R_ERROR
- R_DEBUG

8.56.1 Detailed Description

Represents the level of a result

8.56.2 Member Data Documentation

8.56.2.1 R_DEBUG

```
com.choosemuse.libmuse.ResultLevel.R_DEBUG
```

debug

8.56.2.2 R_ERROR

```
com.choosemuse.libmuse.ResultLevel.R_ERROR
```

error

8.56.2.3 R_INFO

`com.choosemuse.libmuse.ResultLevel.R_INFO`

info

8.56.2.4 R_NONE

`com.choosemuse.libmuse.ResultLevel.R_NONE`

default

8.56.2.5 R_SUCCESS

`com.choosemuse.libmuse.ResultLevel.R_SUCCESS`

success

8.56.2.6 R_WARN

`com.choosemuse.libmuse.ResultLevel.R_WARN`

warn

The documentation for this enum was generated from the following file:

- ResultLevel.java

8.57 com.choosemuse.libmuse.Severity Enum Reference

Public Attributes

- SEV_VERBOSE
- SEV_INFO
- SEV_WARN
- SEV_ERROR
- SEV_FATAL
- SEV_DEBUG
- TOTAL

8.57.1 Detailed Description

The log message severity.

8.57.2 Member Data Documentation

8.57.2.1 SEV_DEBUG

`com.choosemuse.libmuse.Severity.SEV_DEBUG`

Debug-only logs. These are only interesting to developers trying to track down problems in the library.

8.57.2.2 SEV_ERROR

`com.choosemuse.libmuse.Severity.SEV_ERROR`

Error logs. These messages indicate that something has gone wrong – for instance, a connection terminated unexpectedly or a corrupted packet was received.

8.57.2.3 SEV_FATAL

`com.choosemuse.libmuse.Severity.SEV_FATAL`

Fatal logs. These are sent when the library is in a completely unrecoverable state from which the only reasonable outcome is termination of the running process. The process will be aborted as soon as the log handler returns.

8.57.2.4 SEV_INFO

`com.choosemuse.libmuse.Severity.SEV_INFO`

Informational logs. These messages are sent when significant but expected events happen (e.g. a connection started or was completed successfully.)

8.57.2.5 SEV_VERBOSE

`com.choosemuse.libmuse.Severity.SEV_VERBOSE`

Verbose logs. These provide lots of details that are probably irrelevant except for tracing or debugging problems with the headband or library.

8.57.2.6 SEV_WARN

`com.choosemuse.libmuse.Severity.SEV_WARN`

Warning logs. These messages indicate that something out of the ordinary but recoverable happened (e.g. a connection attempt failed but will be retried automatically.)

8.57.2.7 TOTAL

```
com.choosemuse.libmuse.Severity.TOTAL
```

Enum size sentinel; not an actual severity.

The documentation for this enum was generated from the following file:

- Severity.java

8.58 com.choosemuse.libmuse.Stringify Class Reference

Public Member Functions

- abstract String packetType (MuseDataPacketType type)
- abstract String connectionState (ConnectionState state)

Static Public Member Functions

- static native Stringify instance ()

8.58.1 Detailed Description

Converts some LibMuse constants to strings.

This is primarily used for LibMuse internal logging purposes. Don't use this to generate UI elements; the results are not guaranteed to be human-friendly, nor localized. You may, however, use this for your own logging or debugging purposes.

8.58.2 Member Function Documentation

8.58.2.1 connectionState()

```
abstract String com.choosemuse.libmuse.Stringify.connectionState (
            ConnectionState state )  [abstract]
```

String connection state.

Parameters

state	The connection state.

Returns

The string representation of the connection state.

8.58.2.2 instance()

```
static native Stringify com.choosemuse.libmuse.Stringify.instance ( )  [static]
```

Returns the singleton instance.

Returns

the singleton instance.

8.58.2.3 packetType()

```
abstract String com.choosemuse.libmuse.Stringify.packetType (
            MuseDataPacketType type )  [abstract]
```

String packet type.

Only the primitive packet types (eeg, accelerometer, gyro, battery, drl_ref) are human-readable. The rest show up as "type_<n>" where n is the integer corresponding to the constant.

Parameters

type	The packet type.

Returns

The string representation of the packet type.

The documentation for this class was generated from the following file:

- Stringify.java

8.59 com.choosemuse.libmuse.TimestampMode Enum Reference

Public Attributes

- LEGACY
- CURRENT
- EXPLICIT

8.59.1 Detailed Description

File writer timestamp mode.

8.59.2 Member Data Documentation

8.59.2.1 CURRENT

`com.choosemuse.libmuse.TimestampMode.CURRENT`

Use the current time for timestamps.

8.59.2.2 EXPLICIT

`com.choosemuse.libmuse.TimestampMode.EXPLICIT`

Use set_timestamp for timestamps.

8.59.2.3 LEGACY

`com.choosemuse.libmuse.TimestampMode.LEGACY`

Legacy mode.

Use the current time for everything except data packets. Use the data packet's timestamp field for those.

The documentation for this enum was generated from the following file:

- TimestampMode.java

8.60 com.choosemuse.libmuse.UltraViolet Enum Reference

Public Attributes

- UV_INDEX
- UV_A
- UV_B

8.60.1 Detailed Description

Represents the data mapping in a UltraViolet data packet

The ultra violet sensor will measure UVA and UVB wavelength. The UV index is the average of UVA and UVB calculated index values from raw sensor data.

See also

MuseDataPacketType.ULTRA_VIOLET

MuseDataPacket.getUvValue()

8.60.2 Member Data Documentation

8.60.2.1 UV_A

`com.choosemuse.libmuse.UltraViolet.UV_A`

8.60.2.2 UV_B

`com.choosemuse.libmuse.UltraViolet.UV_B`

8.60.2.3 UV_INDEX

`com.choosemuse.libmuse.UltraViolet.UV_INDEX`

The documentation for this enum was generated from the following file:

- UltraViolet.java

Chapter 9

File Documentation

9.1 Accelerometer.java File Reference

Classes

- enum com.choosemuse.libmuse.Accelerometer

Packages

- package com.choosemuse.libmuse

9.2 Action.java File Reference

Classes

- class com.choosemuse.libmuse.Action

Packages

- package com.choosemuse.libmuse

9.3 AdvertisingStats.java File Reference

Classes

- class com.choosemuse.libmuse.AdvertisingStats

Packages

- package com.choosemuse.libmuse

9.4 AndroidLogListener.java File Reference

Classes

- class com.choosemuse.libmuse.AndroidLogListener

Packages

- package com.choosemuse.libmuse

9.5 AnnotationData.java File Reference

Classes

- class com.choosemuse.libmuse.AnnotationData

Packages

- package com.choosemuse.libmuse

9.6 AnnotationFormat.java File Reference

Classes

- enum com.choosemuse.libmuse.AnnotationFormat

Packages

- package com.choosemuse.libmuse

9.7 ApiVersion.java File Reference

Classes

- class com.choosemuse.libmuse.ApiVersion

Packages

- package com.choosemuse.libmuse

9.8 Battery.java File Reference

Classes

- enum com.choosemuse.libmuse.Battery

Packages

- package com.choosemuse.libmuse

9.9 ComputingDeviceConfiguration.java File Reference

Classes

- class com.choosemuse.libmuse.ComputingDeviceConfiguration

Packages

- package com.choosemuse.libmuse

9.10 ComputingDeviceConfigurationFactory.java File Reference

Classes

- class com.choosemuse.libmuse.ComputingDeviceConfigurationFactory

Packages

- package com.choosemuse.libmuse

9.11 ConnectionState.java File Reference

Classes

- enum com.choosemuse.libmuse.ConnectionState

Packages

- package com.choosemuse.libmuse

9.12 DrlRef.java File Reference

Classes

- enum com.choosemuse.libmuse.DrlRef

Packages

- package com.choosemuse.libmuse

9.13 DspData.java File Reference

Classes

- class com.choosemuse.libmuse.DspData

Packages

- package com.choosemuse.libmuse

9.14 Eeg.java File Reference

Classes

- enum com.choosemuse.libmuse.Eeg

Packages

- package com.choosemuse.libmuse

9.15 Error.java File Reference

Classes

- class com.choosemuse.libmuse.Error

Packages

- package com.choosemuse.libmuse

9.16 ErrorType.java File Reference

Classes

- enum com.choosemuse.libmuse.ErrorType

Packages

- package com.choosemuse.libmuse

9.17 EventLoop.java File Reference

Classes

- class com.choosemuse.libmuse.EventLoop

Packages

- package com.choosemuse.libmuse

9.18 Gyro.java File Reference

Classes

- enum com.choosemuse.libmuse.Gyro

Packages

- package com.choosemuse.libmuse

9.19 HandlerEventLoop.java File Reference

Classes

- class com.choosemuse.libmuse.HandlerEventLoop

Packages

- package com.choosemuse.libmuse

9.20 LibmuseVersion.java File Reference

Classes

- class com.choosemuse.libmuse.LibmuseVersion

Packages

- package com.choosemuse.libmuse

9.21 LogListener.java File Reference

Classes

- class com.choosemuse.libmuse.LogListener

Packages

- package com.choosemuse.libmuse

9.22 LogManager.java File Reference

Classes

- class com.choosemuse.libmuse.LogManager

Packages

- package com.choosemuse.libmuse

9.23 LogPacket.java File Reference

Classes

- class com.choosemuse.libmuse.LogPacket

Packages

- package com.choosemuse.libmuse

9.24 Magnetometer.java File Reference

Classes

- enum com.choosemuse.libmuse.Magnetometer

Packages

- package com.choosemuse.libmuse

9.25 mainpage.dox File Reference

9.26 MessageType.java File Reference

Classes

- enum com.choosemuse.libmuse.MessageType

Packages

- package com.choosemuse.libmuse

9.27 Muse.java File Reference

Classes

- class com.choosemuse.libmuse.Muse

Packages

- package com.choosemuse.libmuse

9.28 MuseArtifactPacket.java File Reference

Classes

- class com.choosemuse.libmuse.MuseArtifactPacket

Packages

- package com.choosemuse.libmuse

9.29 MuseConfiguration.java File Reference

Classes

- class com.choosemuse.libmuse.MuseConfiguration

Packages

- package com.choosemuse.libmuse

9.30 MuseConnectionListener.java File Reference

Classes

- class com.choosemuse.libmuse.MuseConnectionListener

Packages

- package com.choosemuse.libmuse

9.31 MuseConnectionPacket.java File Reference

Classes

- class com.choosemuse.libmuse.MuseConnectionPacket

Packages

- package com.choosemuse.libmuse

9.32 MuseDataListener.java File Reference

Classes

- class com.choosemuse.libmuse.MuseDataListener

Packages

- package com.choosemuse.libmuse

9.33 MuseDataPacket.java File Reference

Classes

- class com.choosemuse.libmuse.MuseDataPacket

Packages

- package com.choosemuse.libmuse

9.34 MuseDataPacketType.java File Reference

Classes

- enum com.choosemuse.libmuse.MuseDataPacketType

Packages

- package com.choosemuse.libmuse

9.35 MuseErrorListener.java File Reference

Classes

- class com.choosemuse.libmuse.MuseErrorListener

Packages

- package com.choosemuse.libmuse

9.36 MuseFile.java File Reference

Classes

- class com.choosemuse.libmuse.MuseFile

Packages

- package com.choosemuse.libmuse

9.37 MuseFileFactory.java File Reference

Classes

- class com.choosemuse.libmuse.MuseFileFactory

Packages

- package com.choosemuse.libmuse

9.38 MuseFileReader.java File Reference

Classes

- class com.choosemuse.libmuse.MuseFileReader

Packages

- package com.choosemuse.libmuse

9.39 MuseFileWriter.java File Reference

Classes

- class com.choosemuse.libmuse.MuseFileWriter

Packages

- package com.choosemuse.libmuse

9.40 MuseListener.java File Reference

Classes

- class com.choosemuse.libmuse.MuseListener

Packages

- package com.choosemuse.libmuse

9.41 MuseLog.java File Reference

Classes

- class com.choosemuse.libmuse.MuseLog

Packages

- package com.choosemuse.libmuse

9.42 MuseManager.java File Reference

Classes

- class com.choosemuse.libmuse.MuseManager

Packages

- package com.choosemuse.libmuse

9.43 MuseManagerAndroid.java File Reference

Classes

- class com.choosemuse.libmuse.MuseManagerAndroid
- class **com.choosemuse.libmuse.MuseManagerAndroid.DiscoveryStats**

Packages

- package com.choosemuse.libmuse

9.44 MuseModel.java File Reference

Classes

- enum com.choosemuse.libmuse.MuseModel

Packages

- package com.choosemuse.libmuse

9.45 MusePreset.java File Reference

Classes

- enum com.choosemuse.libmuse.MusePreset

Packages

- package com.choosemuse.libmuse

9.46 MuseVersion.java File Reference

Classes

- class com.choosemuse.libmuse.MuseVersion

Packages

- package com.choosemuse.libmuse

9.47 NotchFrequency.java File Reference

Classes

- enum com.choosemuse.libmuse.NotchFrequency

Packages

- package com.choosemuse.libmuse

9.48 Optics.java File Reference

Classes

- enum com.choosemuse.libmuse.Optics

Packages

- package com.choosemuse.libmuse

9.49 Ppg.java File Reference

Classes

- enum com.choosemuse.libmuse.Ppg

Packages

- package com.choosemuse.libmuse

9.50 Pressure.java File Reference

Classes

- enum com.choosemuse.libmuse.Pressure

Packages

- package com.choosemuse.libmuse

9.51 ReaderListener.java File Reference

Classes

- class com.choosemuse.libmuse.ReaderListener

Packages

- package com.choosemuse.libmuse

9.52 ReaderMuse.java File Reference

Classes

- class com.choosemuse.libmuse.ReaderMuse

Packages

- package com.choosemuse.libmuse

9.53 ReaderMuseBuilder.java File Reference

Classes

- class com.choosemuse.libmuse.ReaderMuseBuilder

Packages

- package com.choosemuse.libmuse

9.54 ReaderMusePlaybackSettings.java File Reference

Classes

- enum com.choosemuse.libmuse.ReaderMusePlaybackSettings

Packages

- package com.choosemuse.libmuse

9.55 ReaderPlaybackListener.java File Reference

Classes

- class com.choosemuse.libmuse.ReaderPlaybackListener

Packages

- package com.choosemuse.libmuse

9.56 Result.java File Reference

Classes

- class com.choosemuse.libmuse.Result

Packages

- package com.choosemuse.libmuse

9.57 ResultLevel.java File Reference

Classes

- enum com.choosemuse.libmuse.ResultLevel

Packages

- package com.choosemuse.libmuse

9.58 Severity.java File Reference

Classes

- enum com.choosemuse.libmuse.Severity

Packages

- package com.choosemuse.libmuse

9.59 Stringify.java File Reference

Classes

- class com.choosemuse.libmuse.Stringify

Packages

- package com.choosemuse.libmuse

9.60 TimestampMode.java File Reference

Classes

- enum com.choosemuse.libmuse.TimestampMode

Packages

- package com.choosemuse.libmuse

9.61 UltraViolet.java File Reference

Classes

- enum com.choosemuse.libmuse.UltraViolet

Packages

- package com.choosemuse.libmuse